

From Linear Models to Neural Networks

Linear & logistic regression, neurons, MLPs, universal approximation

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ES 667 · Deep Learning · IIT Gandhinagar

Lecture 1 chose the loss; Lecture 2 expands the prediction rule

Keep from Lecture 1

Regression output $\rightarrow$ Gaussian model
 $\rightarrow$ MSE

Class probabilities $\rightarrow$ Bernoulli /
categorical model $\rightarrow$ cross-entropy

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We will not re-derive the losses; today is about making f_{θ} more expressive.

Outline

1. **Linear prediction** — weighted sums, batches, and output shapes
2. **One neuron** — add an activation to create a nonlinear feature
3. **The limitation** — see exactly why a linear model cannot solve XOR
4. **An MLP** — compose learned features across width and depth
5. **Practice** — trace one forward pass, count parameters, and make one update

One notation covers every prediction rule

Write $f_{\theta}(x)$ for the model's prediction at input x .

x : the input features

θ : all trainable weights and biases

f_{θ} : the rule that maps input to a score
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Starting point — one weighted sum plus a bias:

$$f_{\theta}(x) = w^{\top} x + b$$

We will build the network gradually; the full nested formula comes only after each part is concrete.

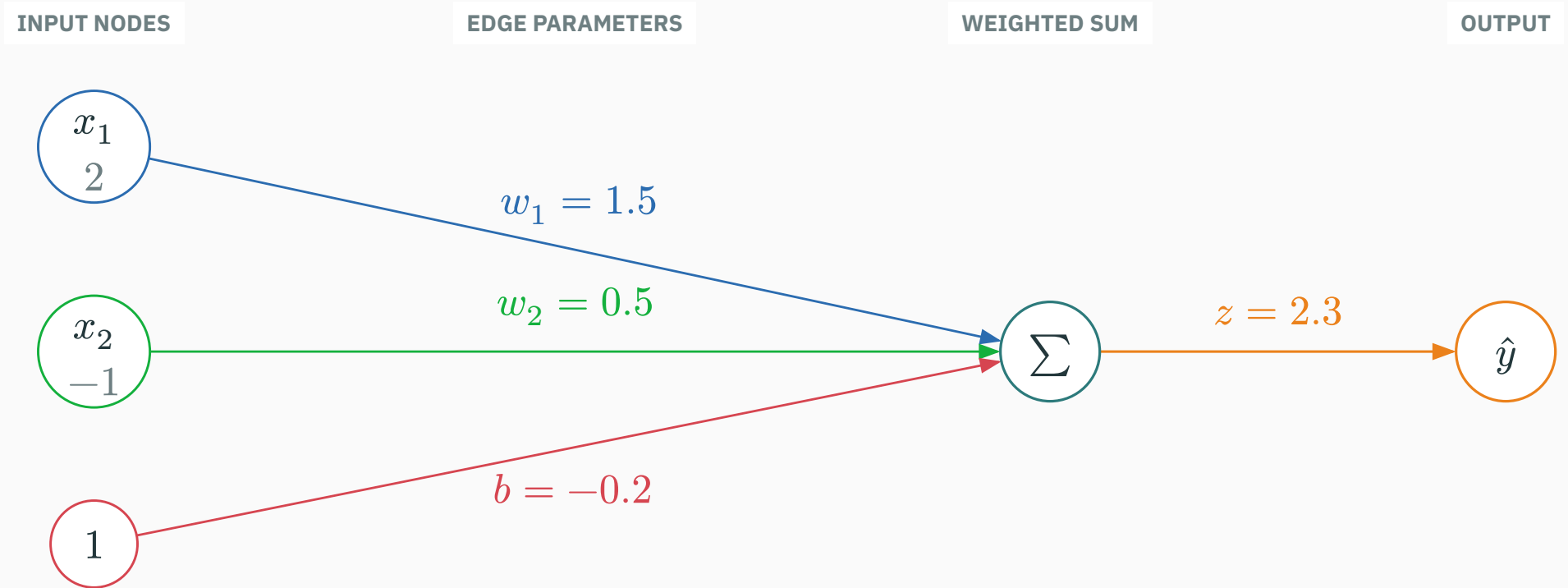
The linear starting point

Start with one weighted sum

For one output, every input node sends a weighted value into one summation node.

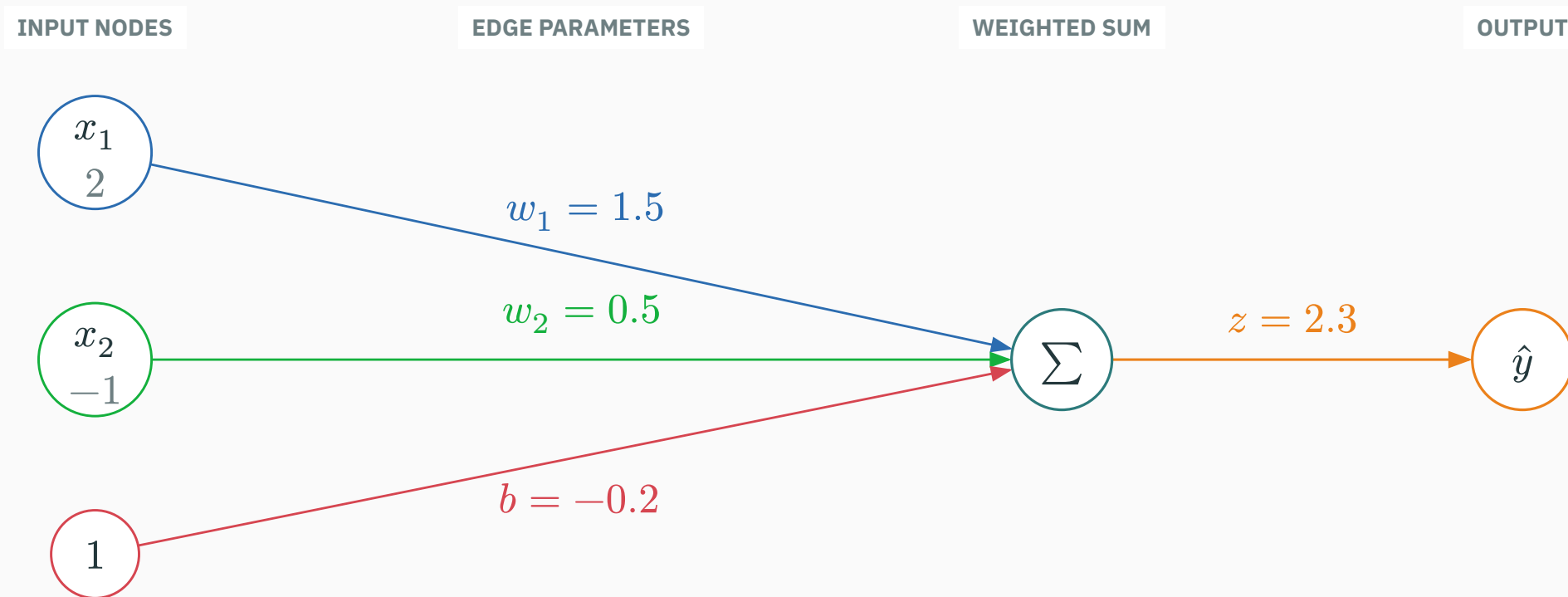
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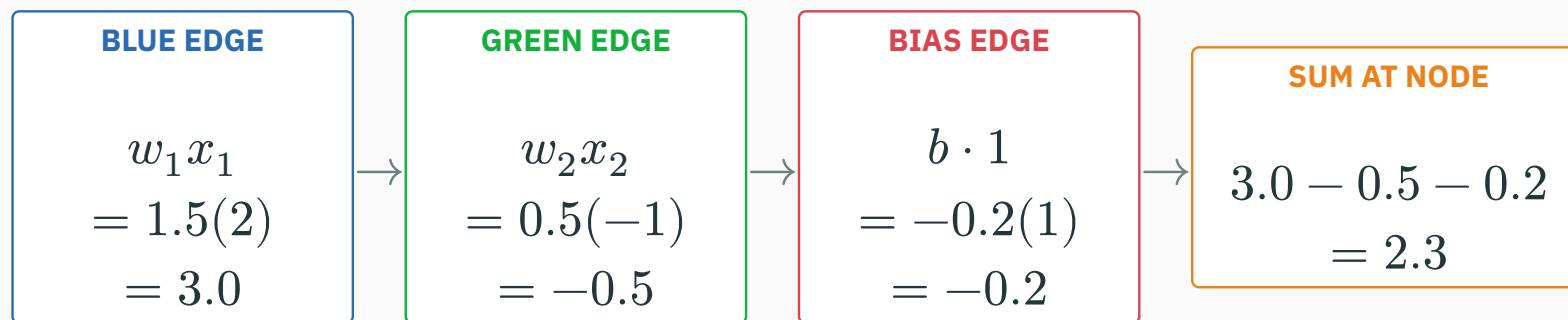
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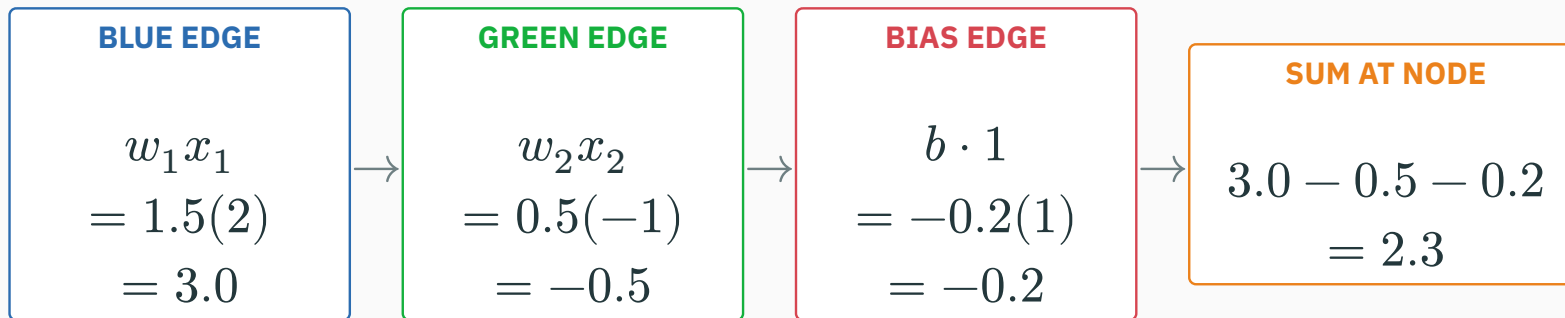


The constant node 1 turns the bias into one more incoming edge: $b \cdot 1$.

Calculate every contribution entering the node

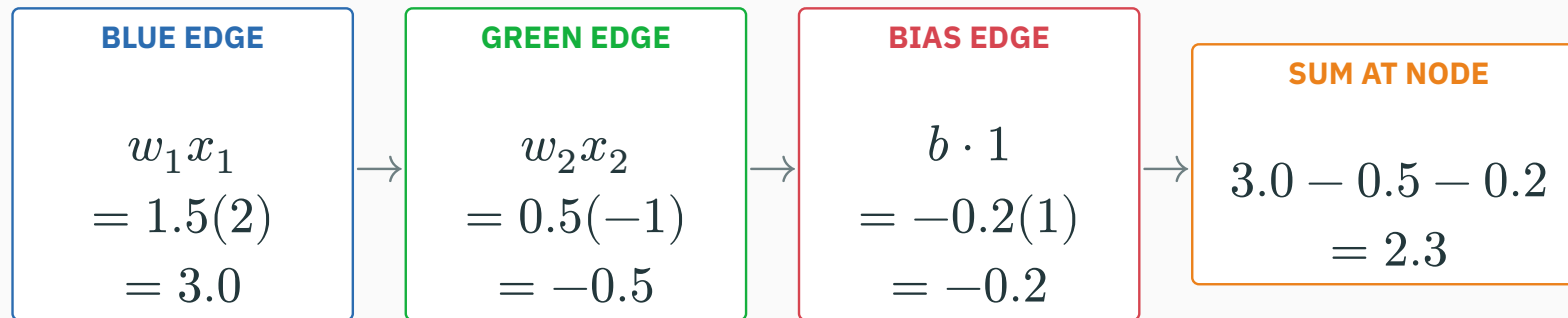


Calculate every contribution entering the node



$$\hat{y} = \underbrace{1.5(2)}_{3.0} + \underbrace{0.5(-1)}_{-0.5} + \underbrace{(-0.2)(1)}_{-0.2} = 2.3.$$

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Each edge contributes “weight × incoming value”; the node adds all contributions.

A batch is a labelled data matrix

example	feature x_1	feature x_2	target y
1	2	-1	2.4
2	0	3	1.0
3	-1	1	-1.1

DATA MATRIX X

$$X = \begin{pmatrix} 2 & -1 \\ 0 & 3 \\ -1 & 1 \end{pmatrix}$$

3 rows = examples
2 columns = features

TARGET VECTOR y

$$y = \begin{pmatrix} 2.4 \\ 1.0 \\ -1.1 \end{pmatrix}$$

one observed target
per example

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Here $n = 3$ examples and $d = 2$ input features, so $X \in \mathbb{R}^{3 \times 2}$ and $y \in \mathbb{R}^3$.

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Row i contains the input $x^{(i)}$ and its observed answer $y^{(i)}$.

One matrix multiplication predicts all three rows

D DERIVATION

Reuse $w = \begin{pmatrix} 1.5 \\ 0.5 \end{pmatrix}$ and $b = -0.2$ for every example.

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Reuse $\mathbf{w} = \begin{pmatrix} 1.5 \\ 0.5 \end{pmatrix}$ and $b = -0.2$ for every example.

$$\mathbf{X}\mathbf{w} + b\mathbf{1} = \begin{pmatrix} 2 & -1 \\ 0 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1.5 \\ 0.5 \end{pmatrix} + (-0.2) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

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$$1.5(0) + 0.5(3) - 0.2 = 1.3$$

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$\hat{y}$ contains predictions; y contains observed targets. They are not the same object.

AUGMENT THE DATA MATRIX

$$\tilde{X} = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 3 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

 $\tilde{X}$

original feature columns
+ one constant column

STACK THE PARAMETERS

$$\theta = \begin{pmatrix} w_1 \\ w_2 \\ b \end{pmatrix} = \begin{pmatrix} 1.5 \\ 0.5 \\ -0.2 \end{pmatrix}$$

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all weights
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$$\hat{y} = \tilde{X}\theta, \quad (n \times (d + 1))((d + 1) \times 1) \rightarrow (n \times 1).$$

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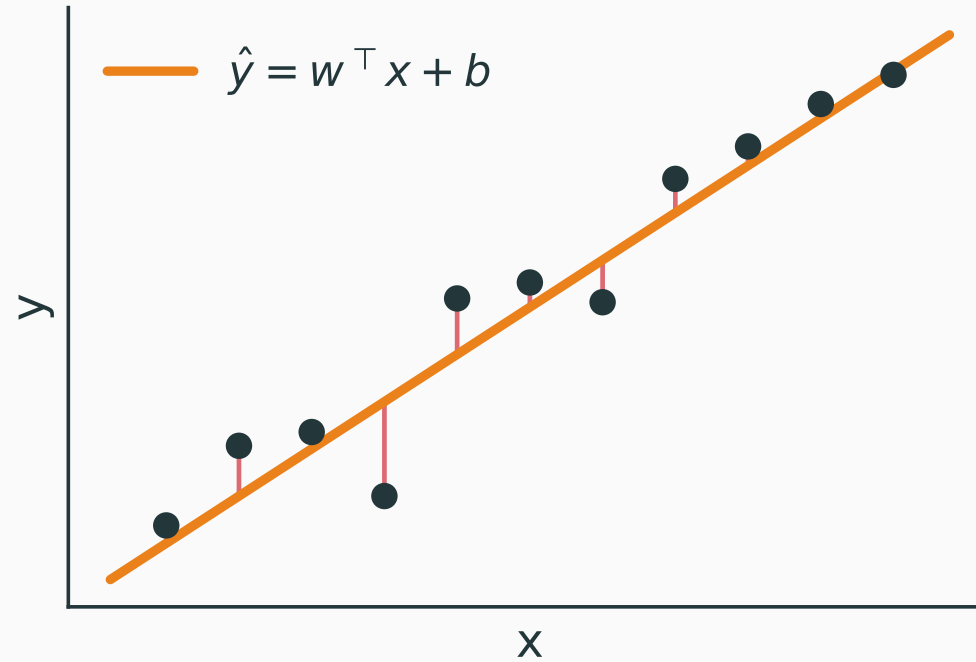
The column of ones is the batch version of the constant bias node on the previous network diagram.

For regression, one weighted sum gives one flat trend

$$\hat{y} = w^{\top} x + b$$

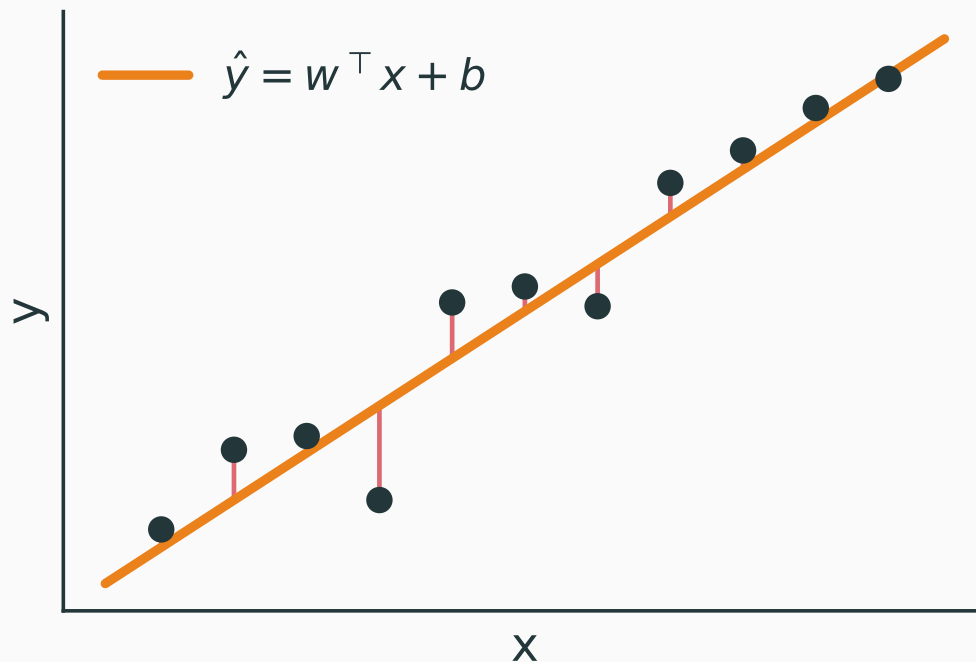
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The model can tilt and shift a line or plane, but it cannot bend to follow nonlinear structure.

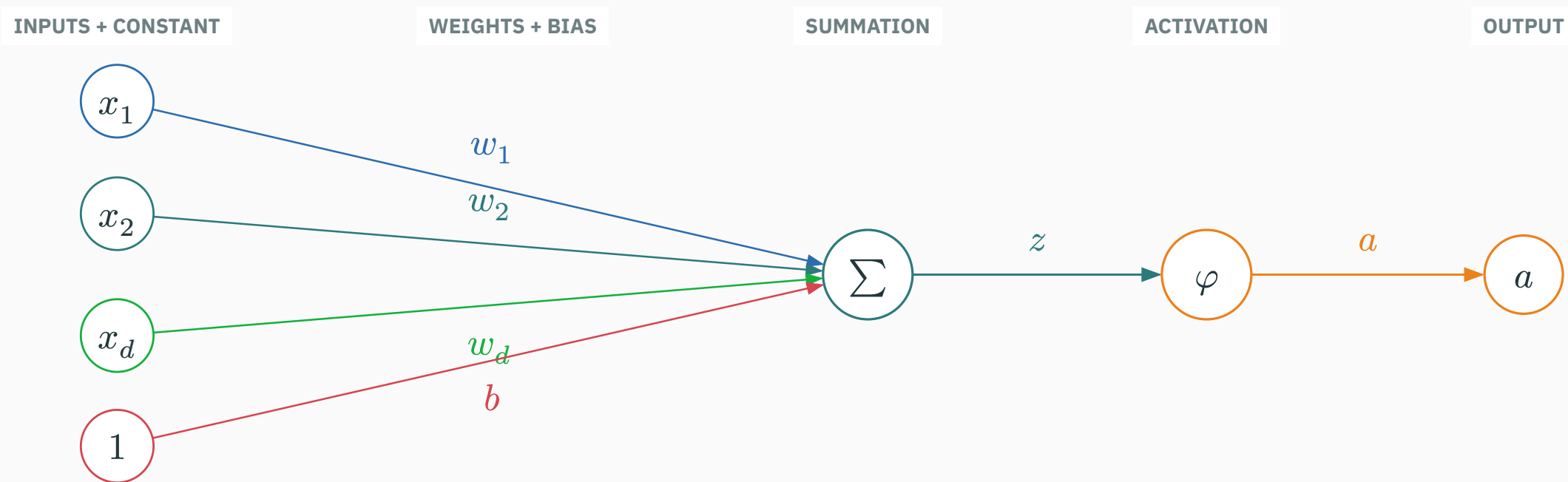
From a linear score to a neuron

A neuron has two stages

First compute a weighted sum plus bias; then pass that value through an activation.

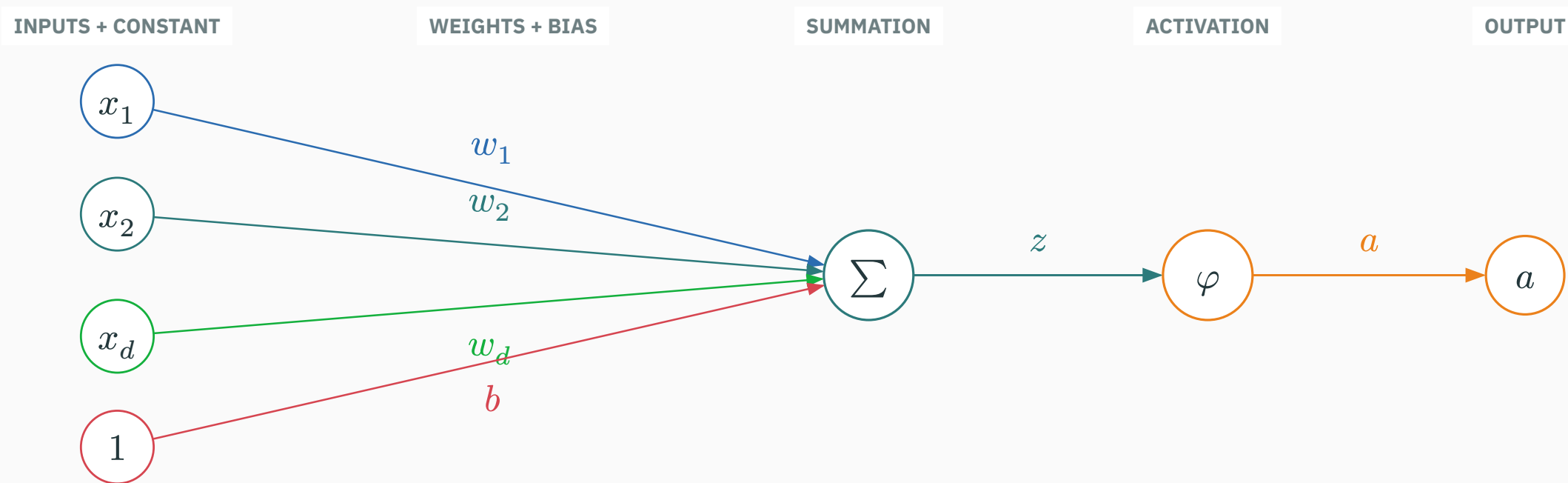
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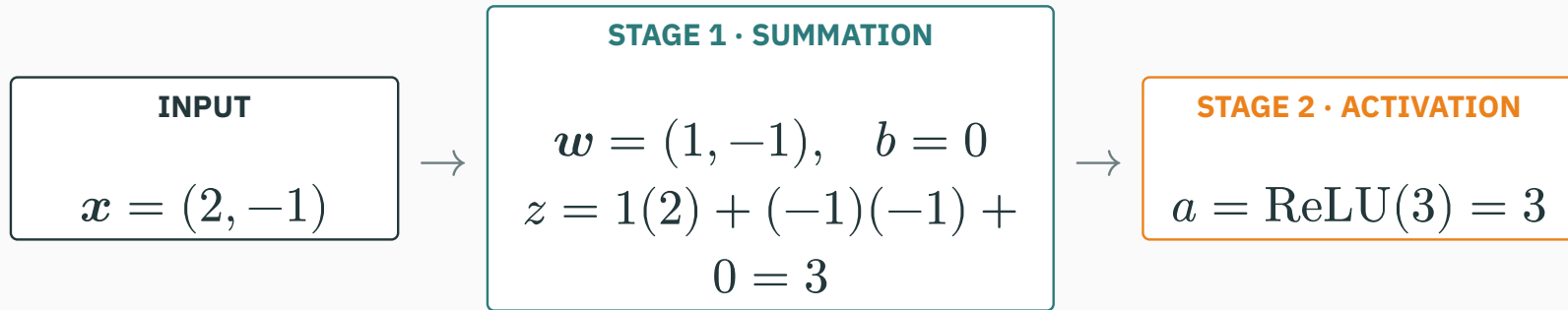
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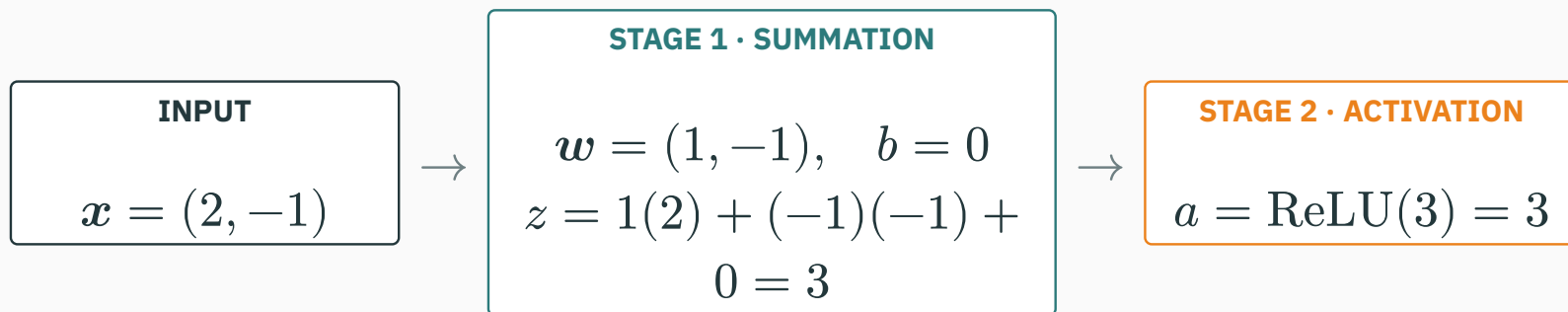


Weights and bias create z at the Σ node; the separate φ node reshapes z into a .

Trace one concrete neuron through the two stages



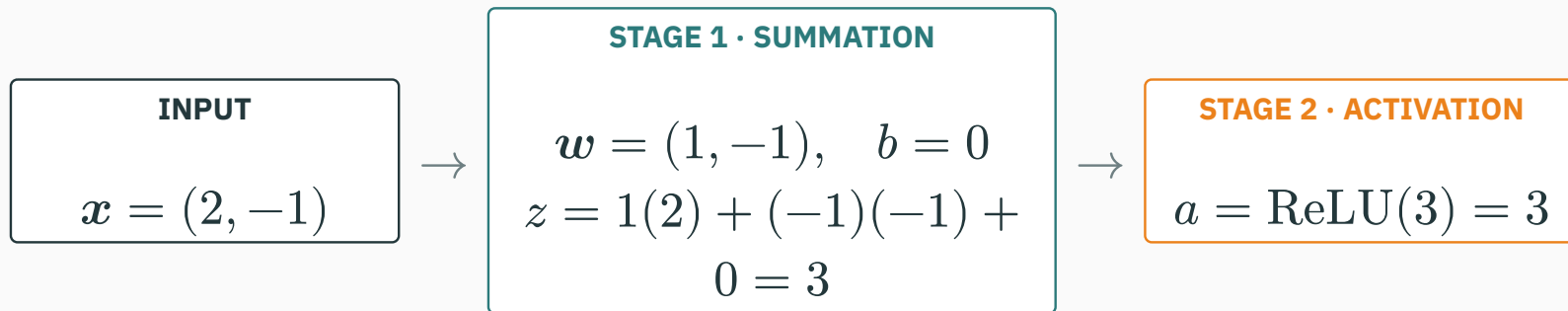
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Changing w or b changes which input patterns make z large.

Changing φ changes how the same pre-activation z is gated or reshaped.

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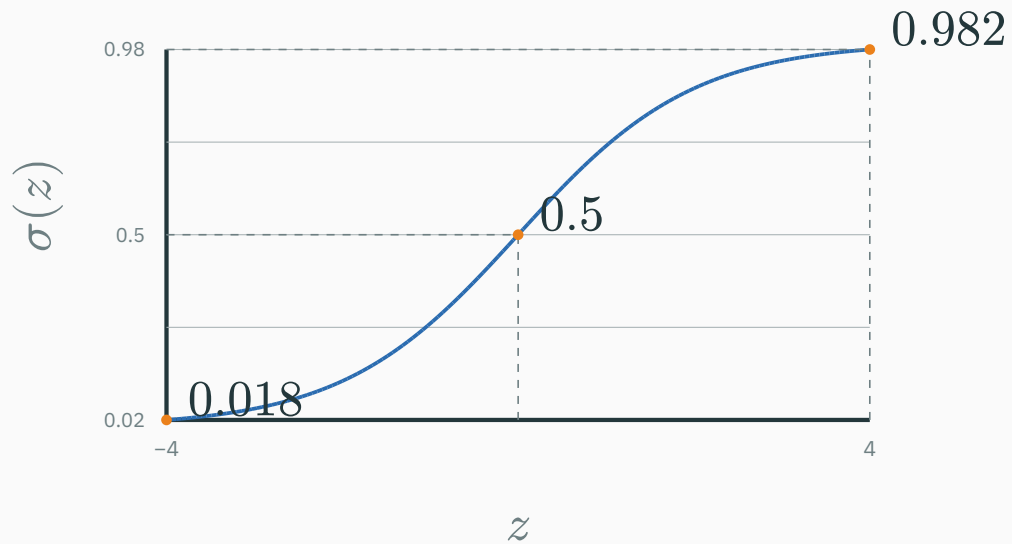
Changing φ changes how the same pre-activation z is gated or reshaped.

Topology tells us where values flow; this calculation tells us the number carried along each stage.

Bounded activations behave like smooth switches

$$\sigma(z) = \frac{1}{1 + \exp(-z)}$$

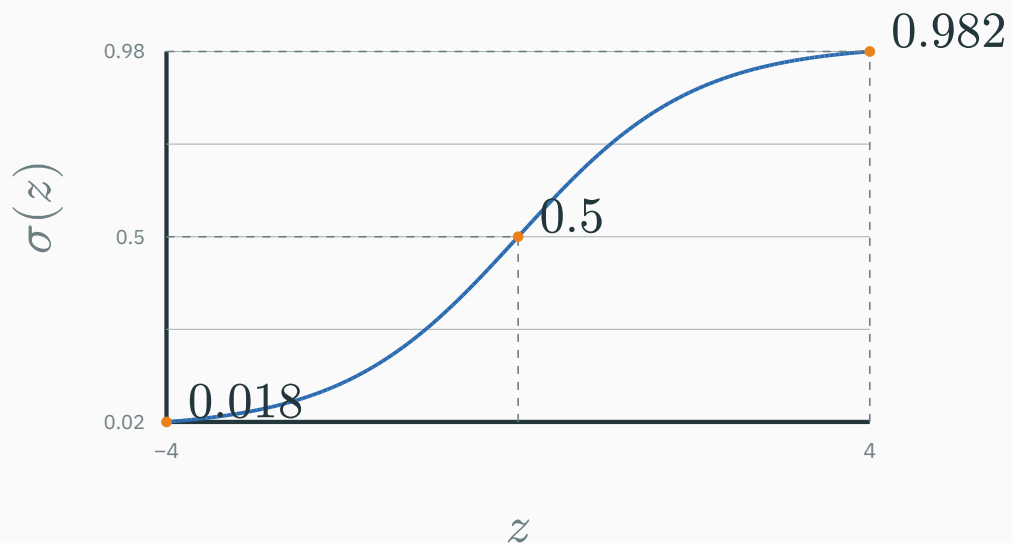
sigmoid: off $\rightarrow$ on



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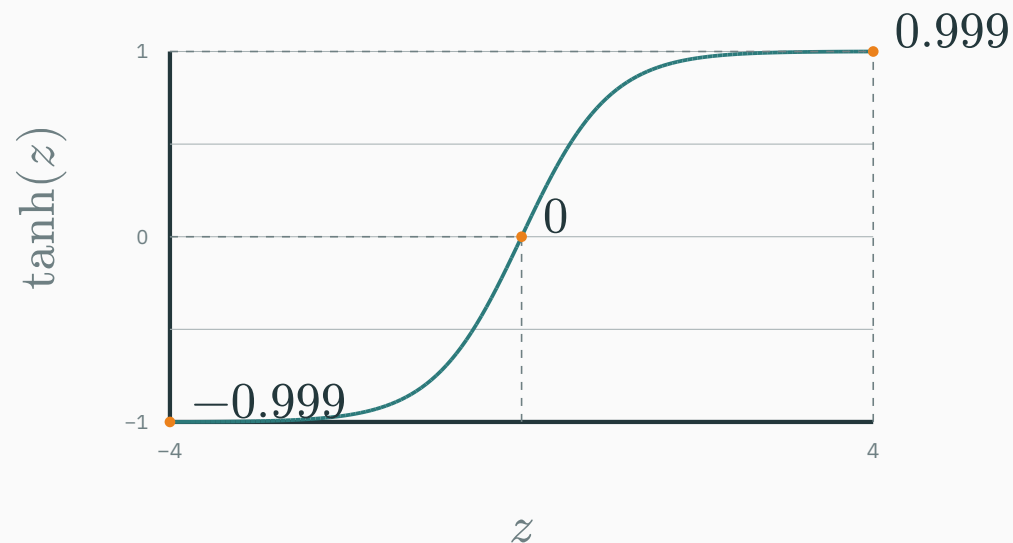
$$\sigma(z) = \frac{1}{1 + \exp(-z)}$$

sigmoid: off $\rightarrow$ on



$$\tanh(z) = \frac{\exp(z) - \exp(-z)}{\exp(z) + \exp(-z)}$$

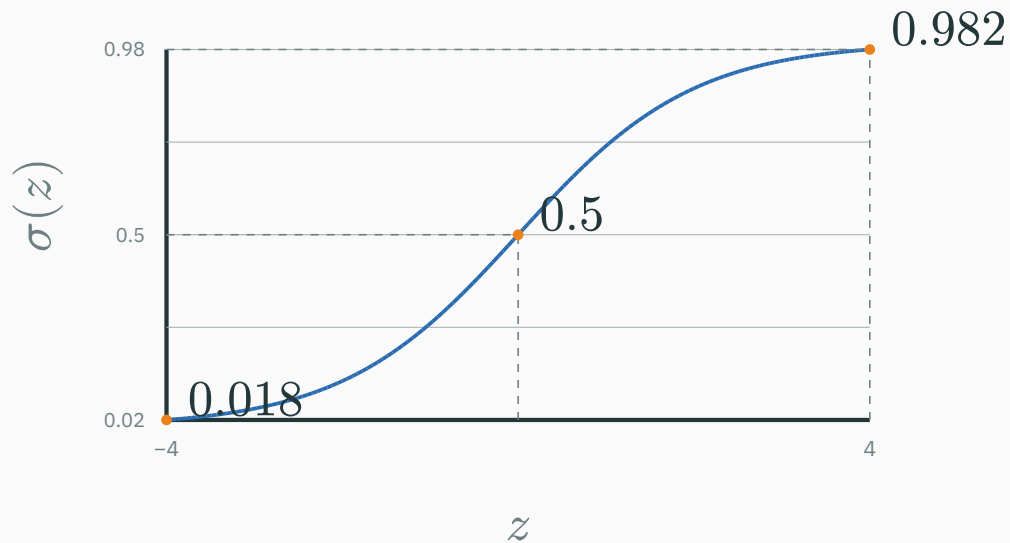
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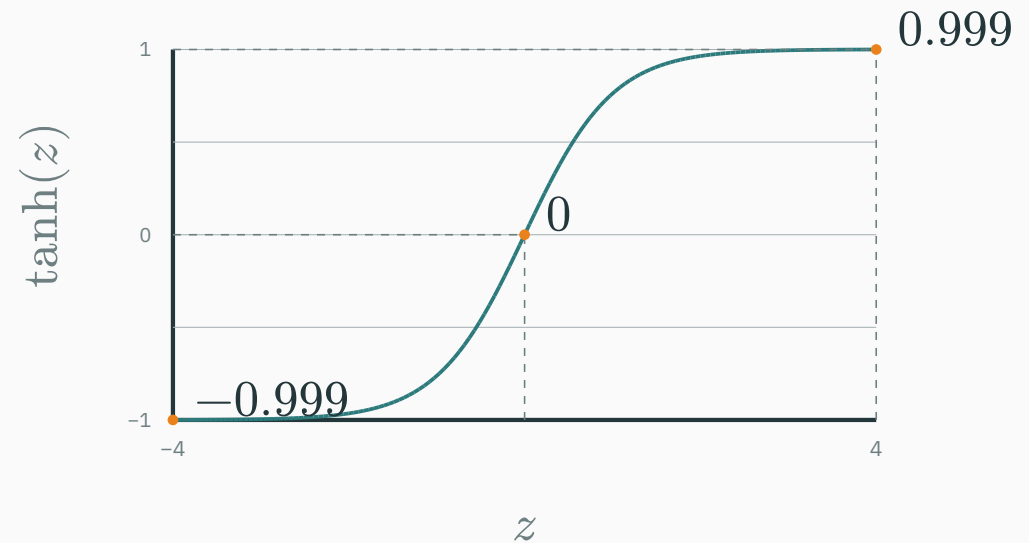
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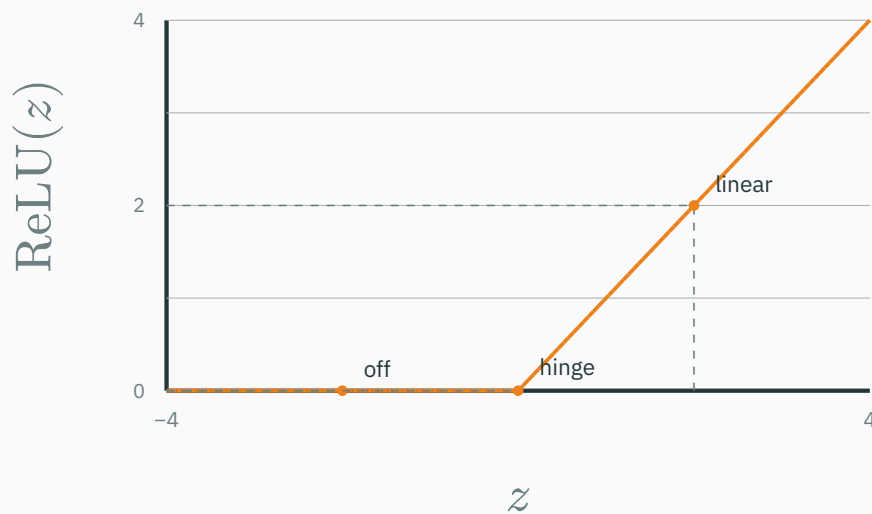


Sigmoid answers “how on?”; tanh answers “negative or positive, and how strongly?”

ReLU and GELU behave like ramps

$$\text{ReLU}(z) = \max(0, z)$$

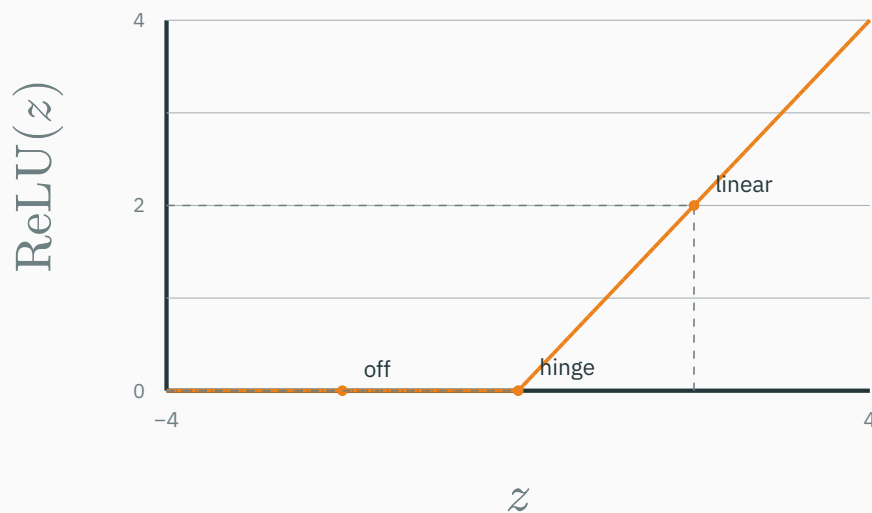
ReLU: hard gate + ramp



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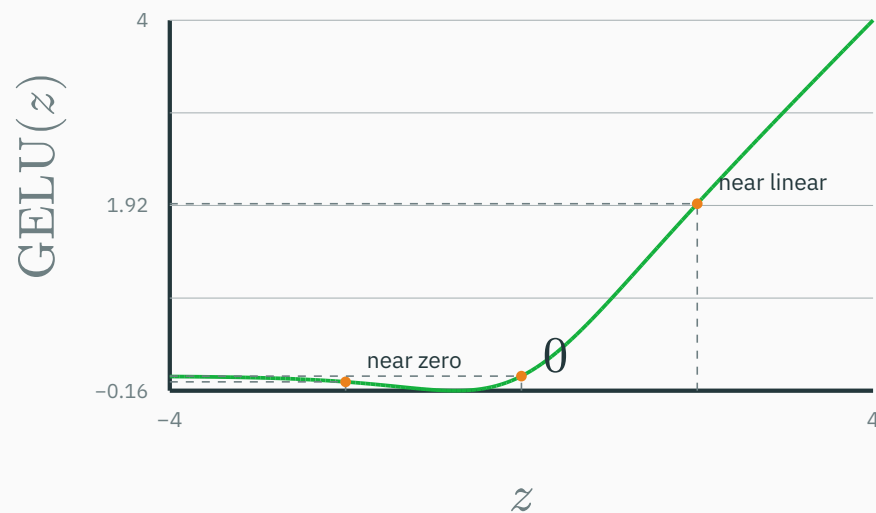
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ReLU: hard gate + ramp



$$\text{GELU}(z) \approx z\sigma(1.702z)$$

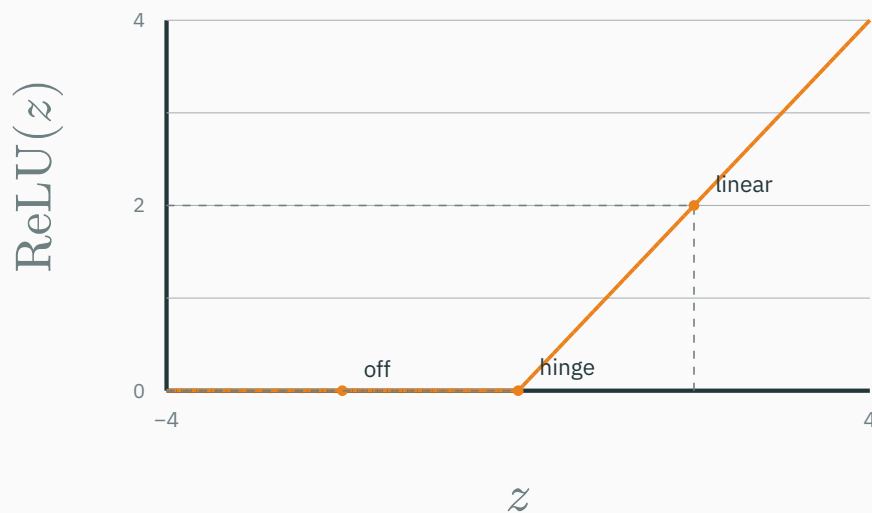
GELU: smooth gate + ramp



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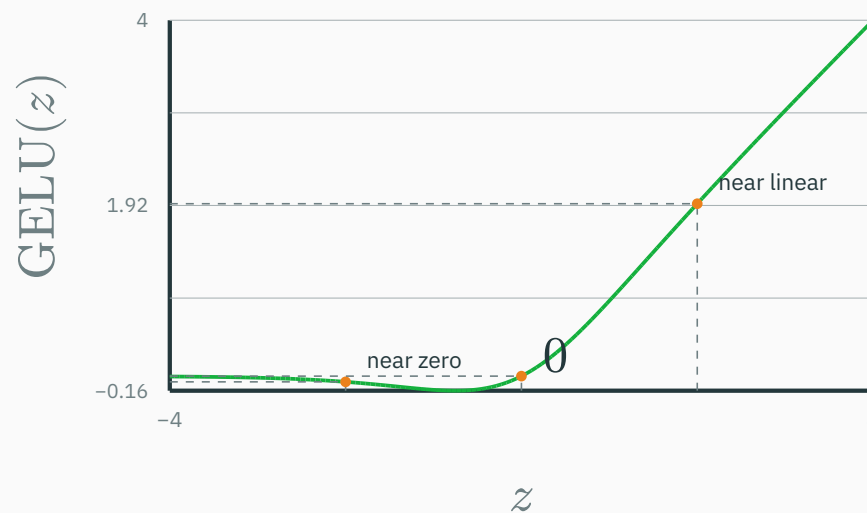
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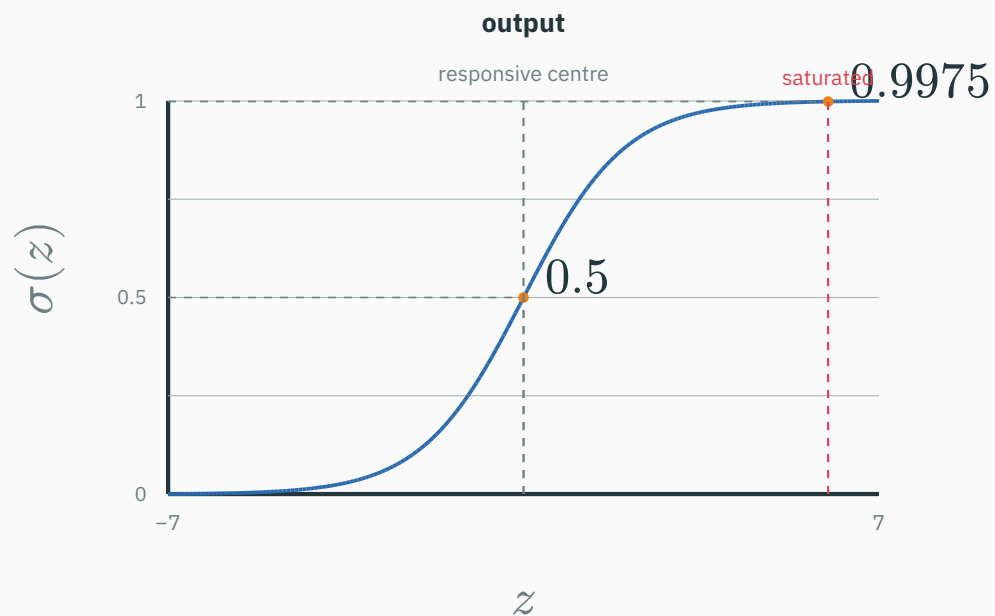
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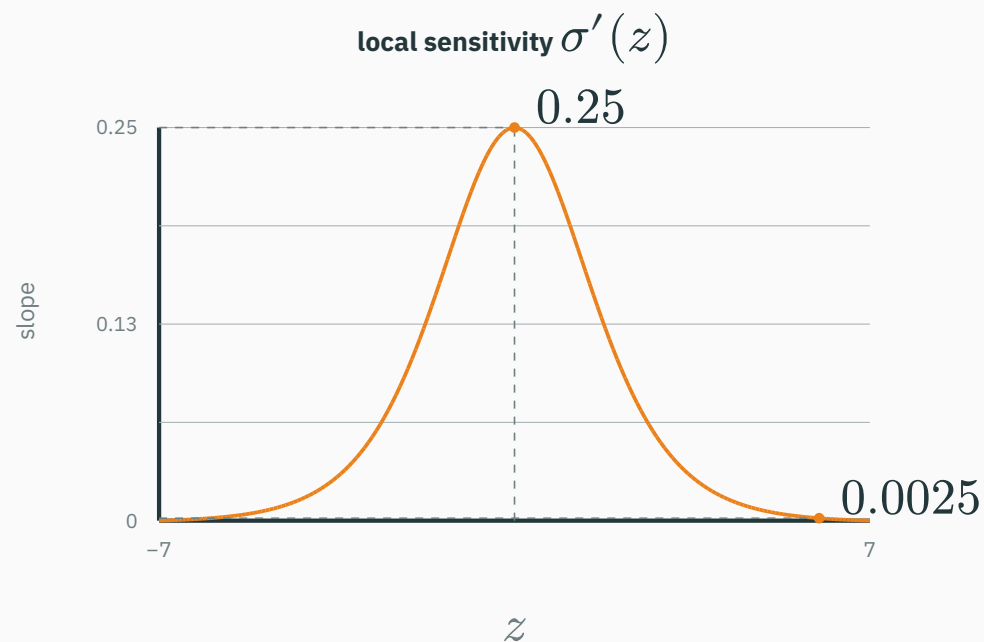
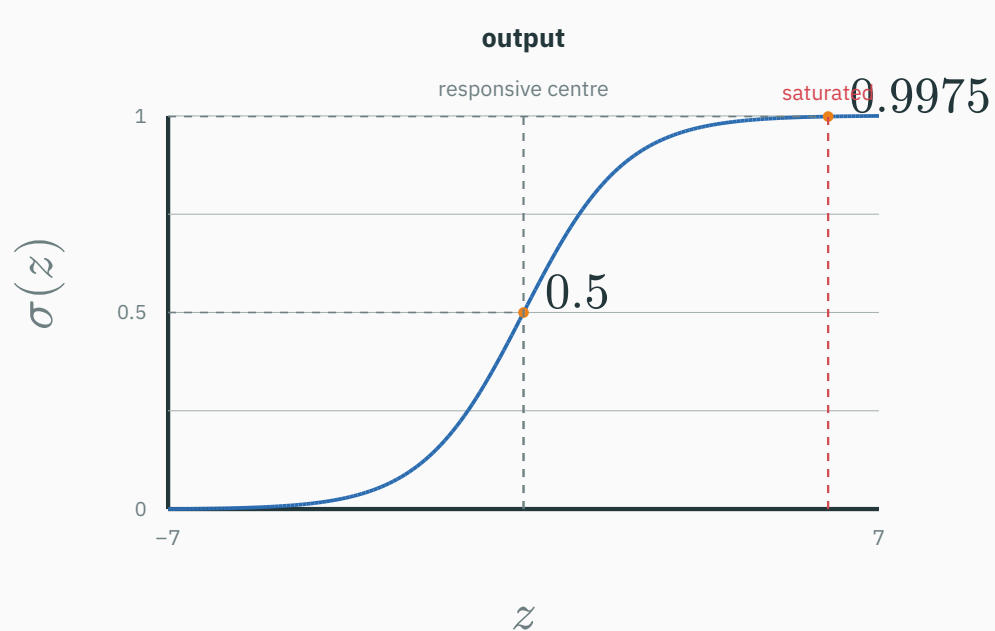


Both build ramp-like hidden features; ReLU has a sharp hinge, while GELU eases through the gate.

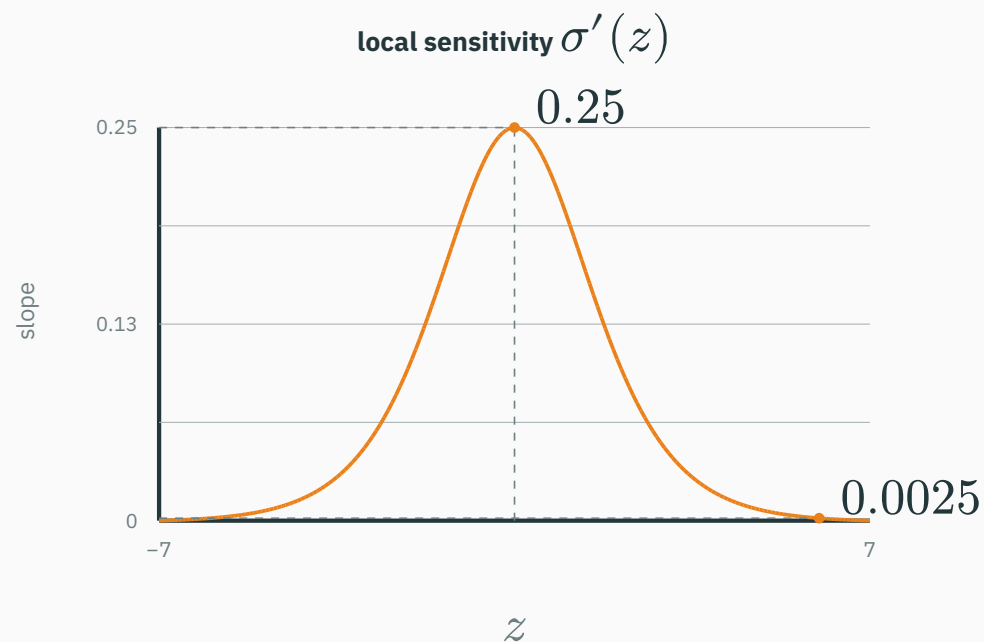
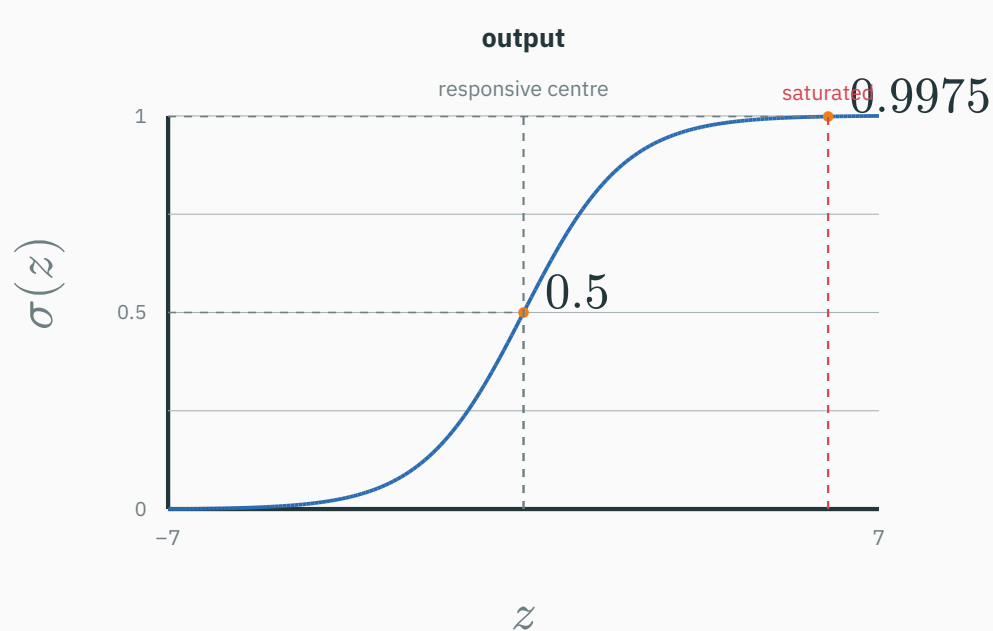
Saturation means the curve has become locally flat



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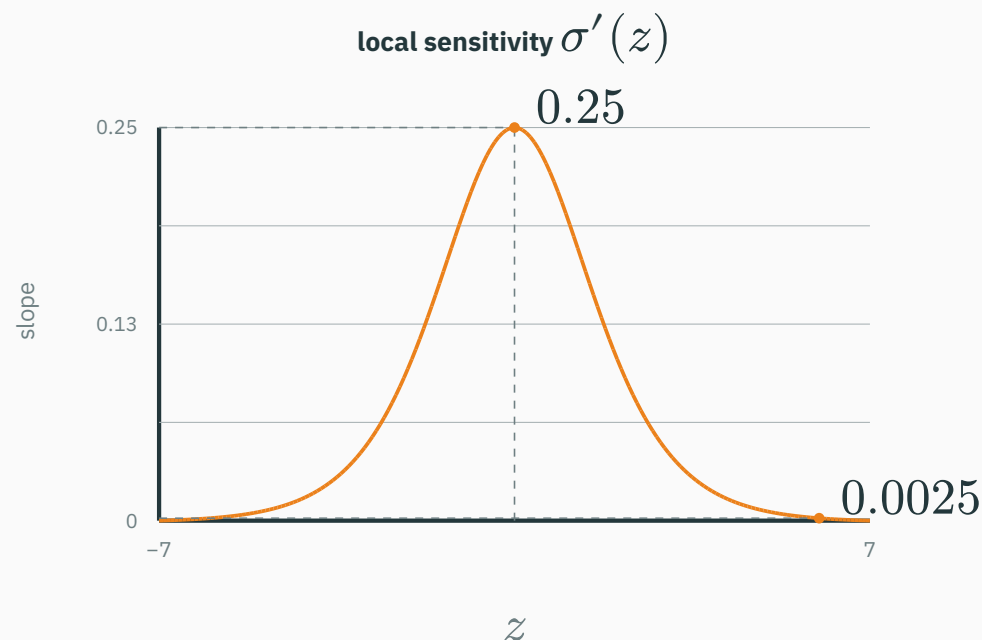
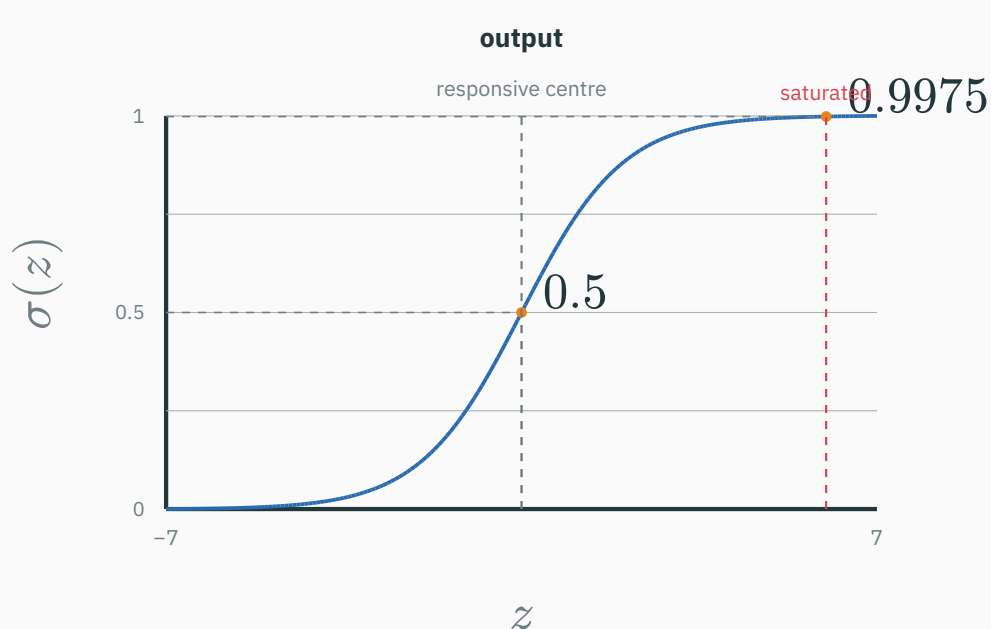


Saturation means the curve has become locally flat



Near $z = 0$, the curve is steep; near $z = 6$, it is almost horizontal.

Saturation means the curve has become locally flat



Near $z = 0$, the curve is steep; near $z = 6$, it is almost horizontal.

A saturated unit still outputs a number, but changing its input barely changes that output.

Calculate the effect of saturation

CENTRE: $z = 0$

$$\begin{aligned}\sigma'(0) &= 0.25 \\ \text{for } \Delta z &= 0.1: \\ \Delta h &\approx 0.25(0.1) = 0.025\end{aligned}$$

same
 Δz
 $\rightarrow$

SATURATED: $z = 6$

$$\begin{aligned}\sigma'(6) &\approx 0.0025 \\ \text{for } \Delta z &= 0.1: \\ \Delta h &\approx 0.0025(0.1) = \\ &0.00025\end{aligned}$$

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$$\frac{0.025}{0.00025} = 100 \rightarrow \text{the saturated unit responds } \mathbf{100\times \text{ less.}}$$

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For a positive ReLU input, $\text{ReLU}'(z) = 1$, so the same $\Delta z = 0.1$ gives $\Delta h \approx 0.1$.

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The derivative is a local responsiveness meter; Lecture 3 will connect it to the learning signal.

Remove every activation. What can a stack of “weighted sum + bias” layers represent?

A. One equivalent weighted sum + bias

B. Any curved decision boundary

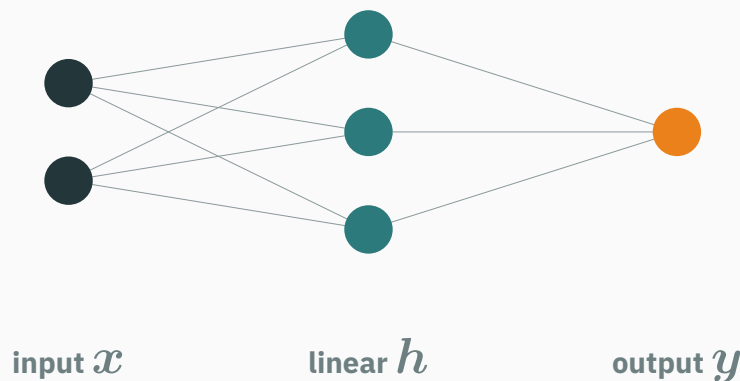
C. Only a constant function

D. A probability distribution automatically

Answer in nodes: two linear layers collapse into one

A ANSWER

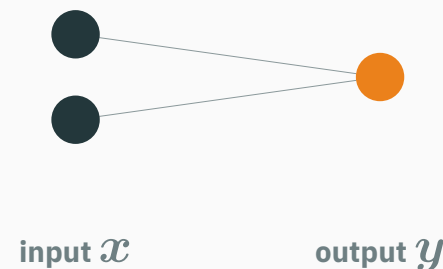
TWO LINEAR LAYERS



$$h = W_1 x + b_1$$

$$y = W_2 h + b_2$$

ONE EQUIVALENT LAYER



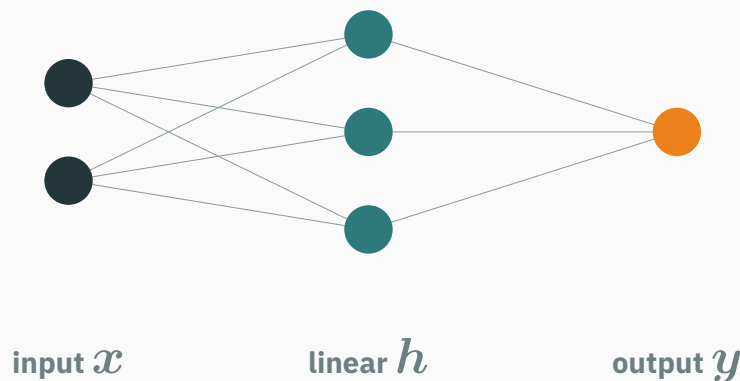
$$y = W_{\text{eq}} x + b_{\text{eq}}$$

no hidden representation is needed

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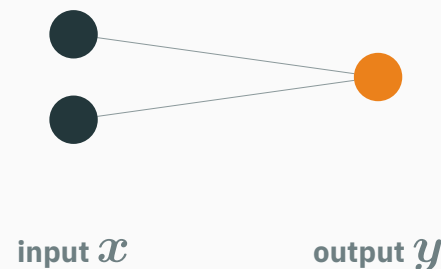


$$h = W_1 x + b_1$$

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$$W_{\text{eq}} = W_2 W_1, \quad b_{\text{eq}} = W_2 b_1 + b_2$$

ONE EQUIVALENT LAYER



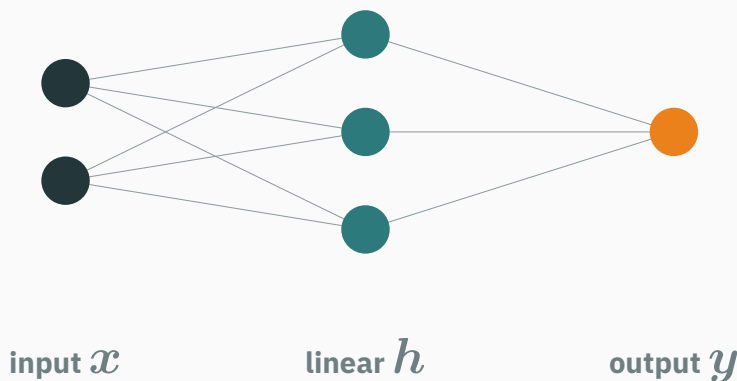
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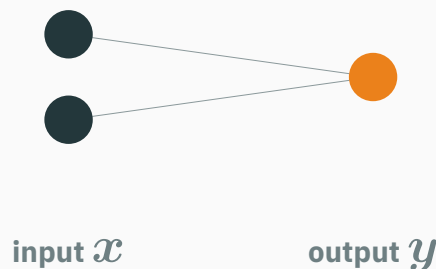
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no hidden representation is needed

$$W_{\text{eq}} = W_2 W_1, \quad b_{\text{eq}} = W_2 b_1 + b_2$$

Without a φ node, the middle column adds parameters but no new bend or feature shape.

INTERACTIVE

↗ nipunbatra.github.io/dl-teaching/interactives/activation-explorer.html

Compare activations and their **derivatives**, and see where gradients vanish.
Controls: activation type · input z · show derivative · compare saturation.

INTERACTIVE

↗ nipunbatra.github.io/dl-teaching/interactives/activation-explorer.html

Compare activations and their **derivatives**, and see where gradients vanish.

Controls: activation type · input z · show derivative · compare saturation.

Explore large positive and negative logits, where sigmoid's derivative becomes tiny.

**Where a single weighted sum
fails**

A binary classifier draws one straight boundary

LINEAR SCORE

$$z = \mathbf{w}^\top \mathbf{x} + b$$

THRESHOLD

$$\hat{y} = 1 \text{ if } z \geq 0$$

$$\hat{y} = 0 \text{ if } z < 0$$

The boundary is where $z = 0$.

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LINEAR SCORE

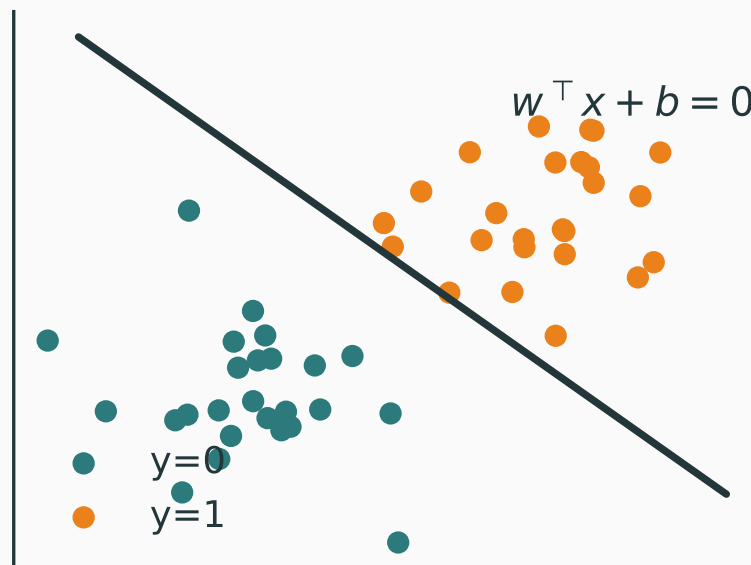
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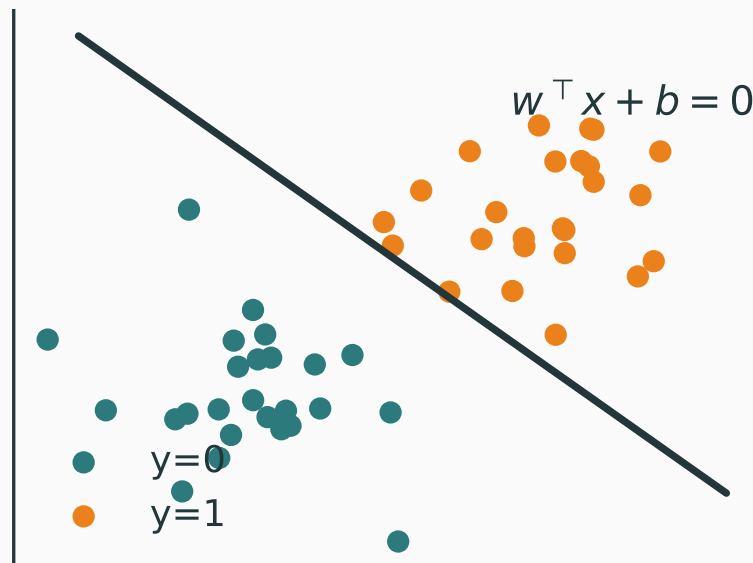
$$z = w^T x + b$$

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The boundary is where $z = 0$.



$w^T x + b = 0$ is one line in 2-D, so binary linear classification works only when one line can separate the classes.

XOR is the smallest pattern that defeats one line

CLASS A

$(0, 0)$ and $(1, 1)$

CLASS B

$(1, 0)$ and $(0, 1)$

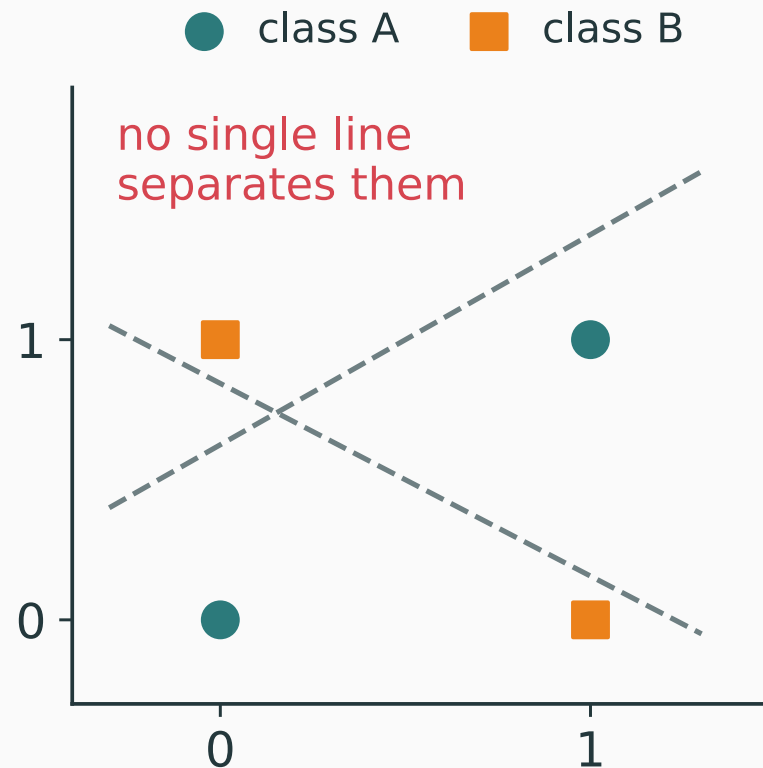
Every proposed line leaves one same-class corner on the wrong side.

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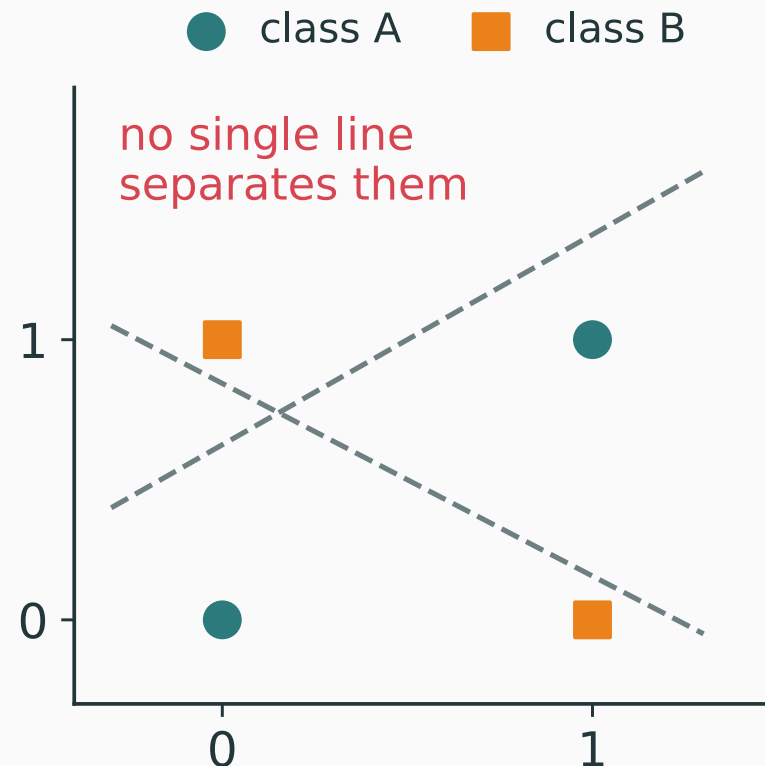


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Every proposed line leaves one same-class corner on the wrong side.



XOR requires a nonlinear feature transformation before a final straight separator can work.

For each class k , compute one linear score $z_{k(\mathbf{x})} = \mathbf{w}_k^\top \mathbf{x} + b_k$.

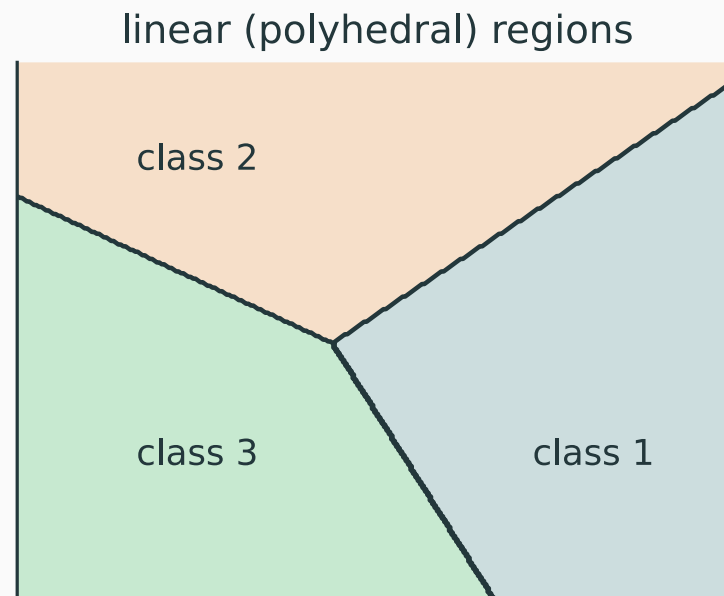
For each class k , compute one linear score $z_{k(\mathbf{x})} = \mathbf{w}_k^\top \mathbf{x} + b_k$. Predict the largest score: $\hat{y}(\mathbf{x}) = \arg \max_k z_{k(\mathbf{x})}$.

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Each pair of classes can exchange the lead only where their two scores tie.

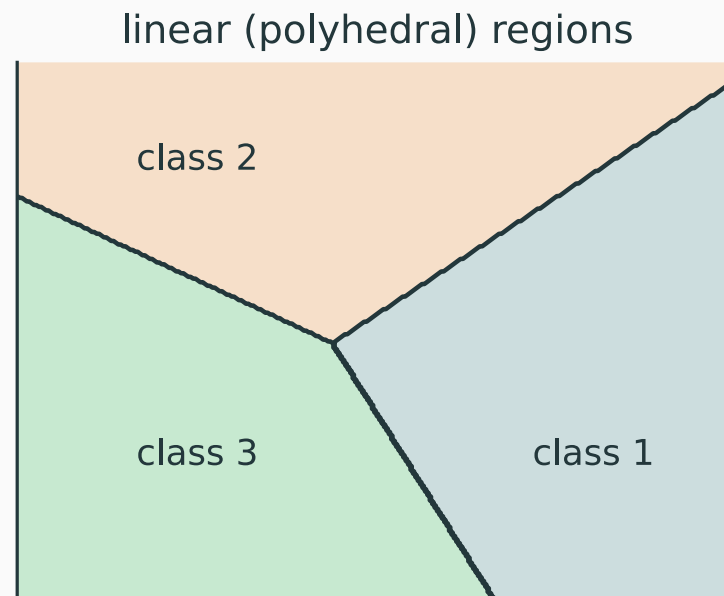
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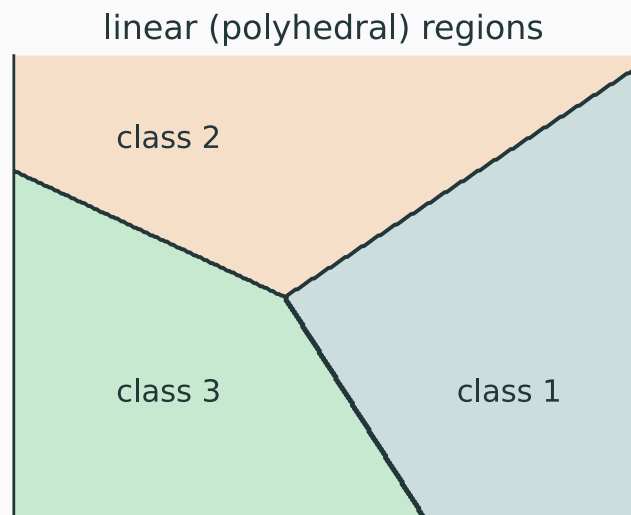
Each pair of classes can exchange the lead only where their two scores tie.



Multi-class regions can form polygons, but every edge still comes from equality between two linear scores.

Question: why are the multi-class boundaries straight?

QUESTION



three linear scores divide the input plane

The classifier predicts

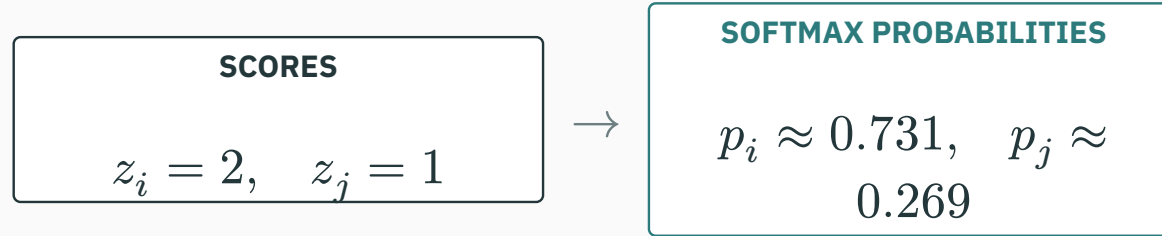
$$\hat{y}(\mathbf{x}) = \arg \max_k p_k(\mathbf{x}).$$

When is a point on the boundary between predicted classes i and j ?

- A.** Whenever $p_i = p_j$
- B.** When $p_i = p_j$ and both are maximal
- C.** Whenever $p_i + p_j = 1$
- D.** Only when all K probabilities are equal

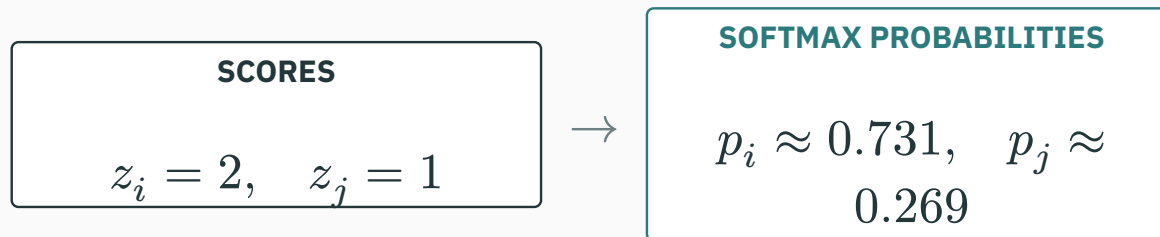
Softmax preserves the winner

A ANSWER



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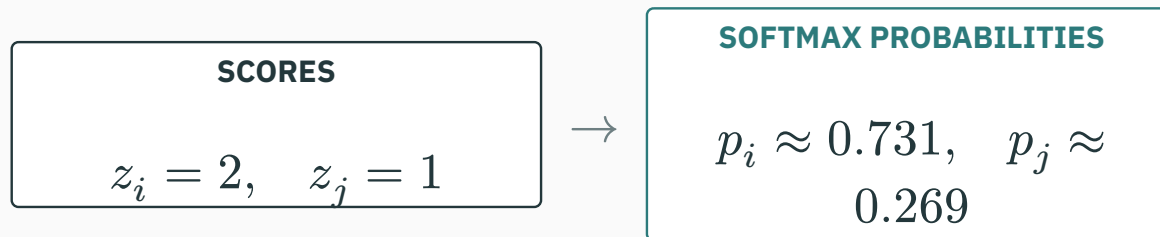
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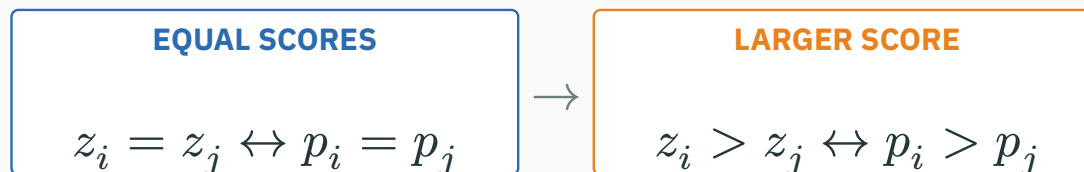
$$\frac{p_i}{p_j} = \frac{\exp(z_i)}{\exp(z_j)} = \exp(z_i - z_j)$$

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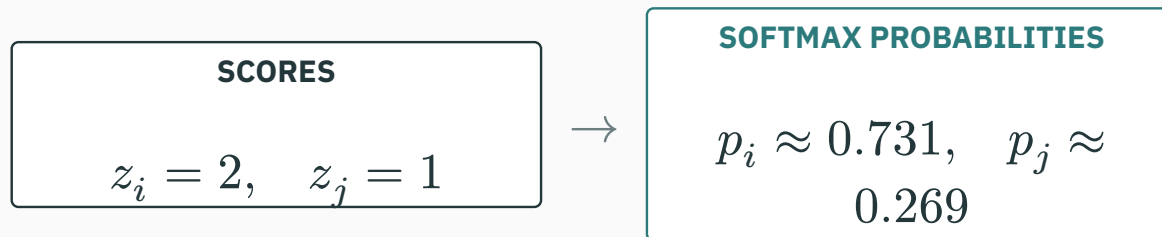


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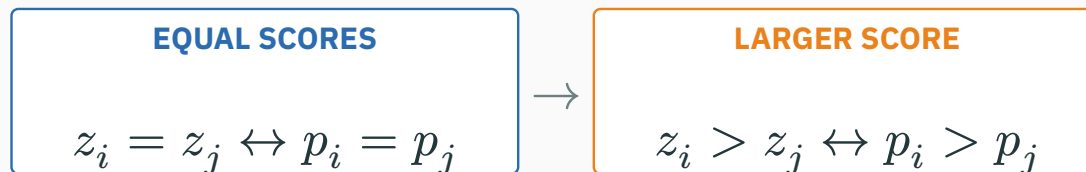


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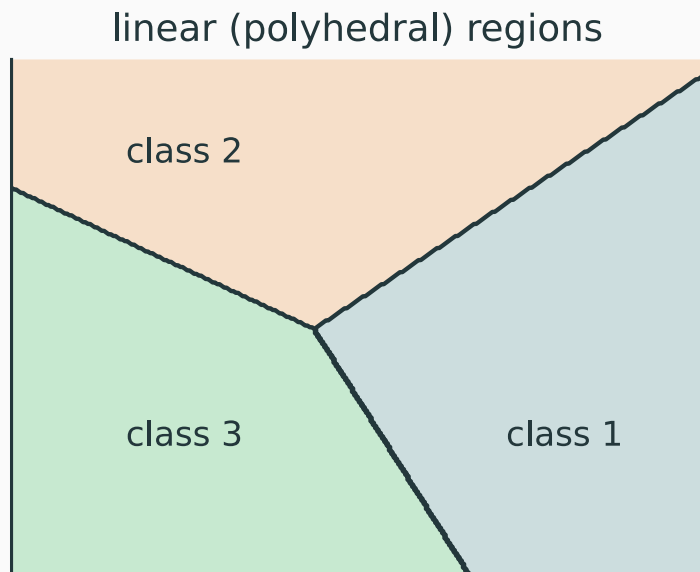
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Softmax changes score scale, not score order: $\arg \max_k p_k = \arg \max_k z_k$.

Linear scores can tie only on a flat set

A ANSWER

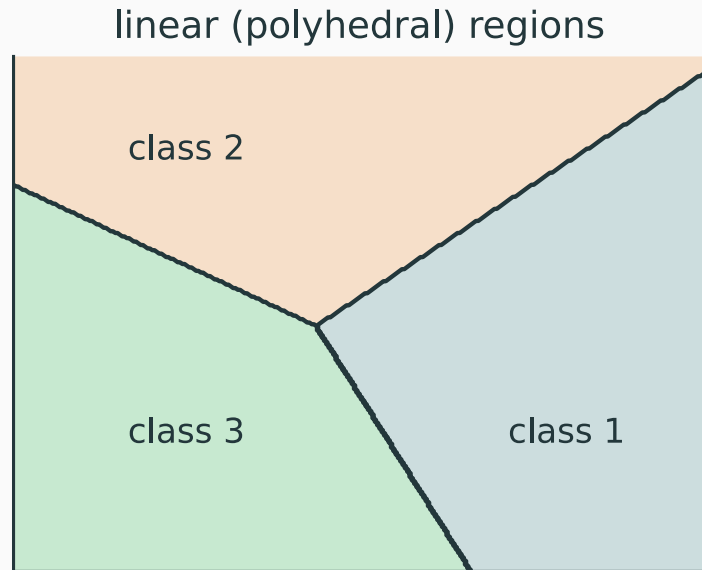


For class k , $z_{k(x)} = \mathbf{w}_k^\top \mathbf{x} + b_k$.

Each coloured region is where one linear score is largest.

Linear scores can tie only on a flat set

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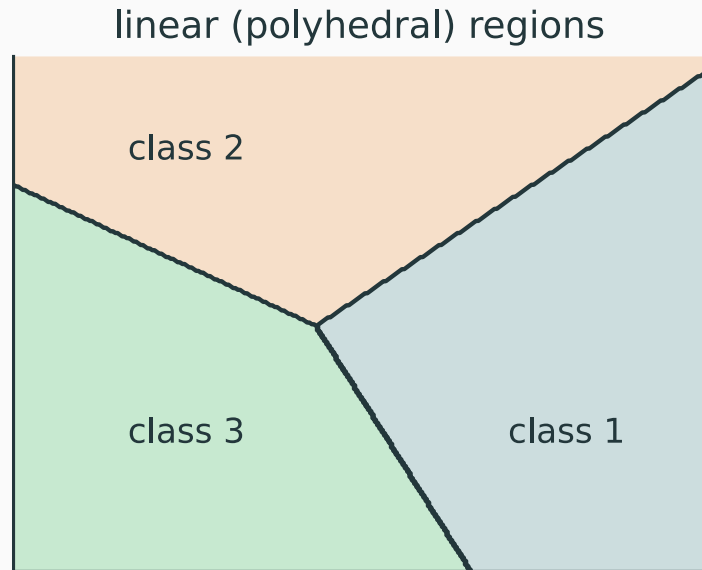


For class k , $z_{k(\mathbf{x})} = \mathbf{w}_k^\top \mathbf{x} + b_k$. Two classes tie when $z_{i(\mathbf{x})} = z_{j(\mathbf{x})}$, so $(\mathbf{w}_i - \mathbf{w}_j)^\top \mathbf{x} + (b_i - b_j) = 0$.

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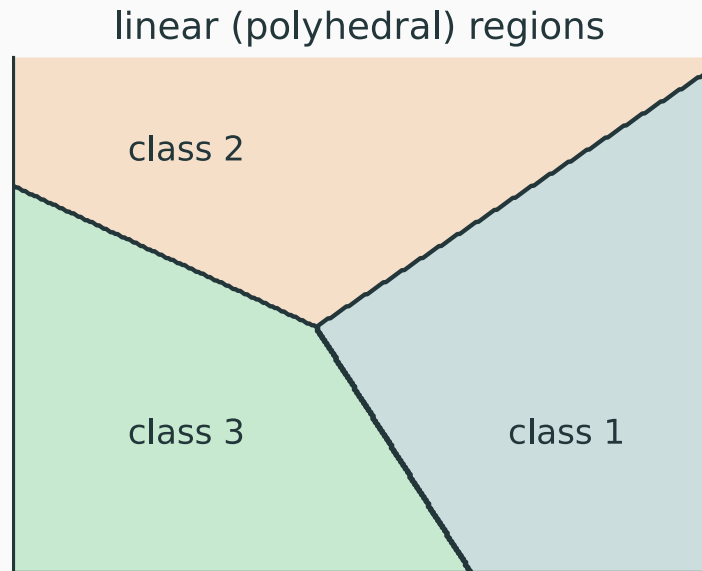
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This is a line in 2-D, a plane in 3-D, and a hyperplane in d dimensions.

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Equal linear scores create a candidate straight boundary; softmax cannot bend it.

A concrete example: three linear scores compete

CLASS 1 SCORE

$$z_1(x) = x$$

CLASS 2 SCORE

$$z_2(x) = -x$$

CLASS 3 SCORE

$$z_3(x) = 0.5$$

A concrete example: three linear scores compete

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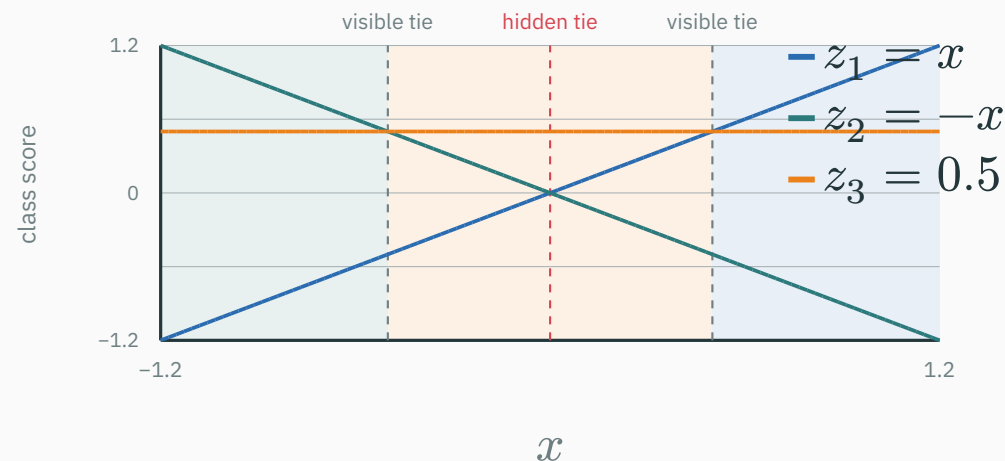
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PREDICTED CLASS = $\arg \max_k z_k(x)$ • boundaries at $x = \pm 0.5$

$$2: x < -.5$$

$$3: |x| < .5$$

$$1: x > .5$$

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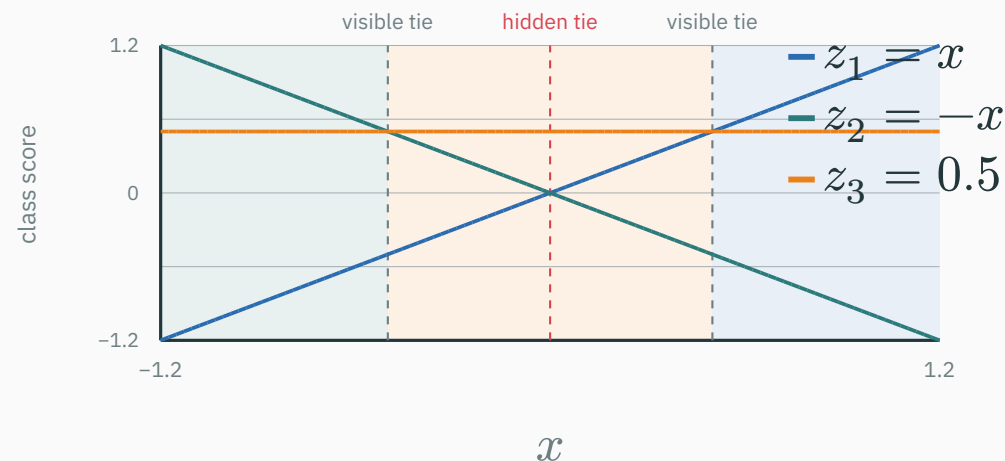
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$$2: x < -.5$$

$$3: |x| < .5$$

$$1: x > .5$$

The learned boundaries are $x = -0.5$ and $x = 0.5$; the class-1/class-2 tie at $x = 0$ remains hidden.

Evaluate five explicit inputs to see which ties are visible

Prediction = $\arg \max_k z_{k(x)}$; a boundary appears only when the tied scores are maximal.

x	$z_1 = x$	$z_2 = -x$	$z_3 = 0.5$	what is visible?
-1	-1	1	0.5	class 2 wins
-0.5	-0.5	0.5	0.5	visible boundary 2/3
0	0	0	0.5	class 3 wins; 1/2 tie hidden
0.5	0.5	-0.5	0.5	visible boundary 1/3
1	1	-1	0.5	class 1 wins

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The candidate **1/2 tie** occurs at $x = 0$, but the visible boundaries occur at $x = -0.5$ **and** $x = 0.5$.

Candidate boundary versus visible boundary

A ANSWER

CANDIDATE PAIRWISE TIE

$\mathcal{T}_{ij} = \{x : z_i = z_j\}$
an entire hyperplane

keep
only
winners
→

VISIBLE CLASS BOUNDARY

$\mathcal{B}_{ij} = \{x : z_i = z_j \geq z_l \text{ for every } l\}$
the top-scoring portion

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In probability language: $p_i = p_j$ **and both are maximal.**

Candidate boundary versus visible boundary

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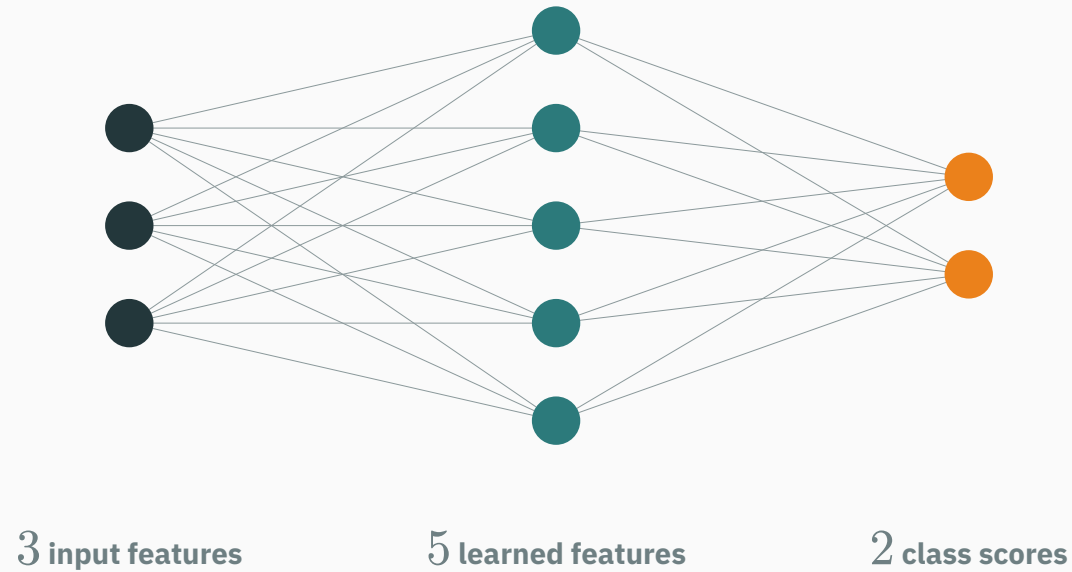
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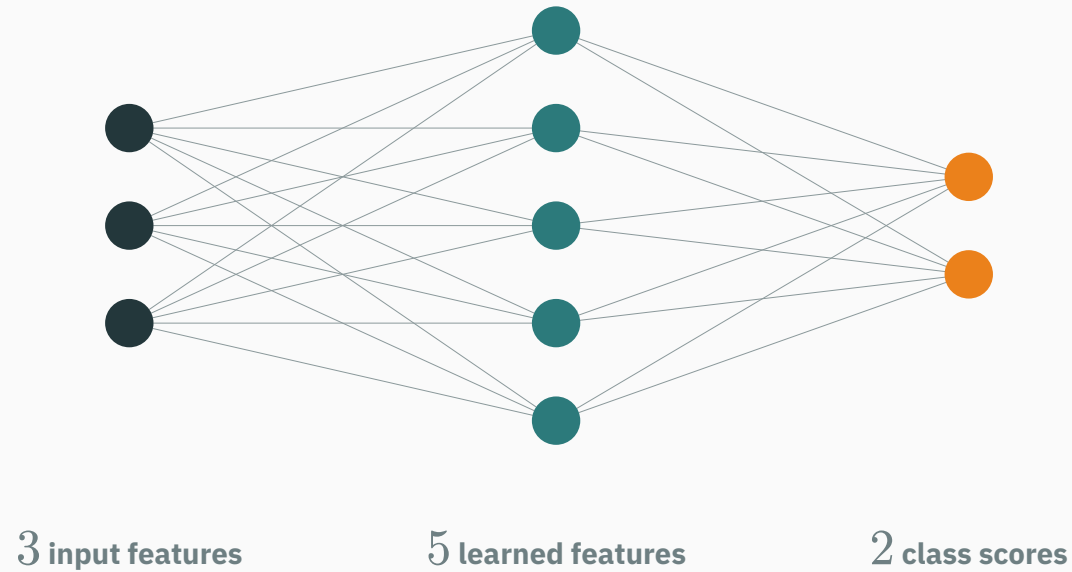
Correct: B. Multi-class linear classifiers have piecewise-straight visible boundaries.

The multilayer perceptron

A hidden layer is a new column of learned features

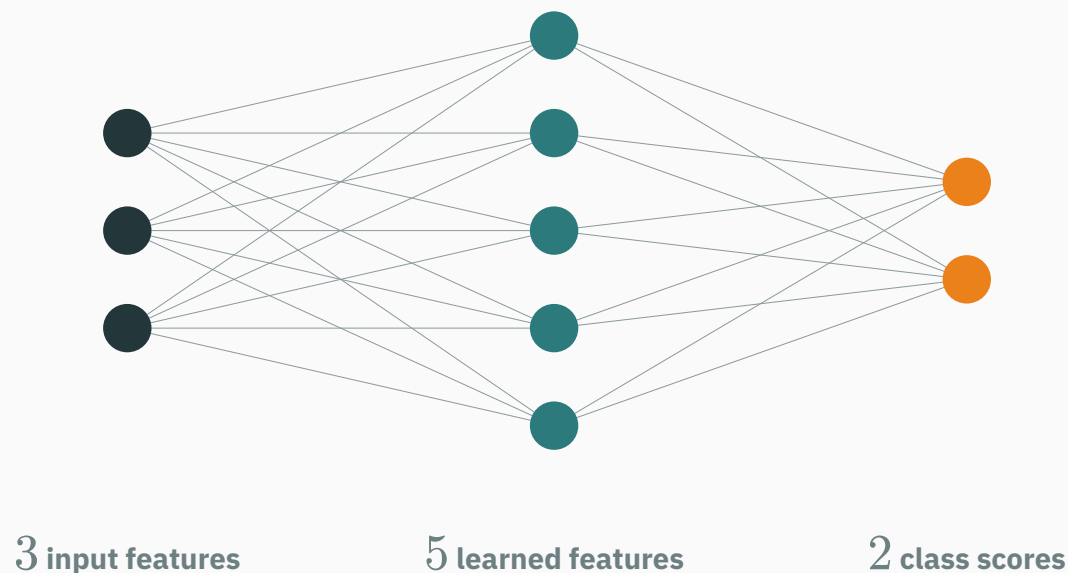


A hidden layer is a new column of learned features



$$h = \varphi(W_1 x + b_1), \quad z = W_2 h + b_2.$$

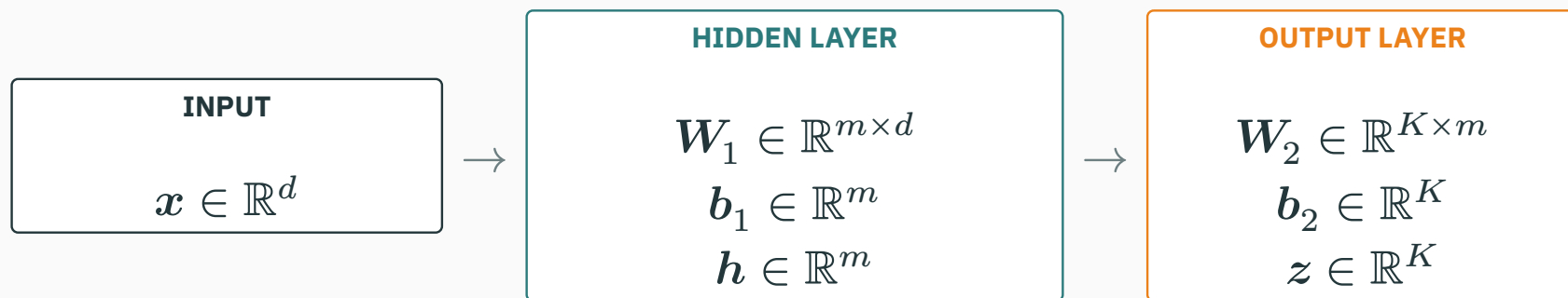
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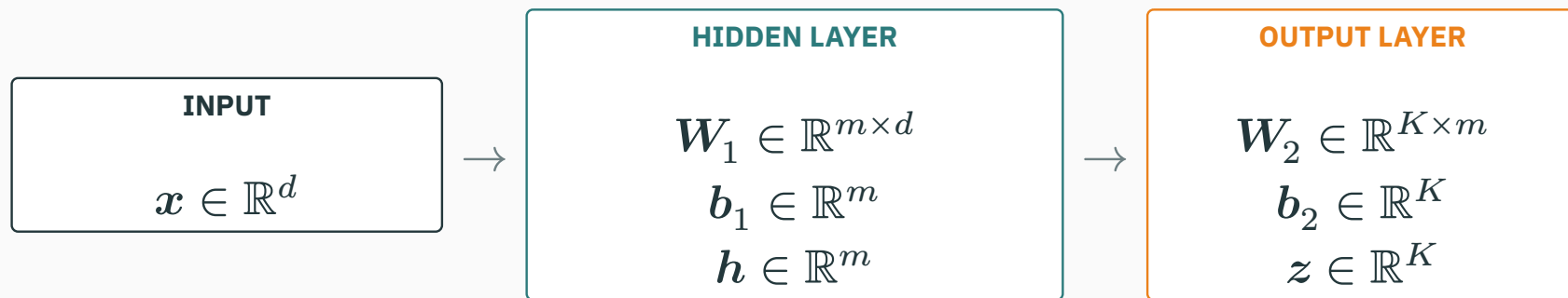
$$h = \varphi(W_1 x + b_1), \quad z = W_2 h + b_2.$$

W_1 stores the $3 \times 5 = 15$ blue-to-teal edges; W_2 stores the $5 \times 2 = 10$ teal-to-orange edges.

General layer dimensions must agree

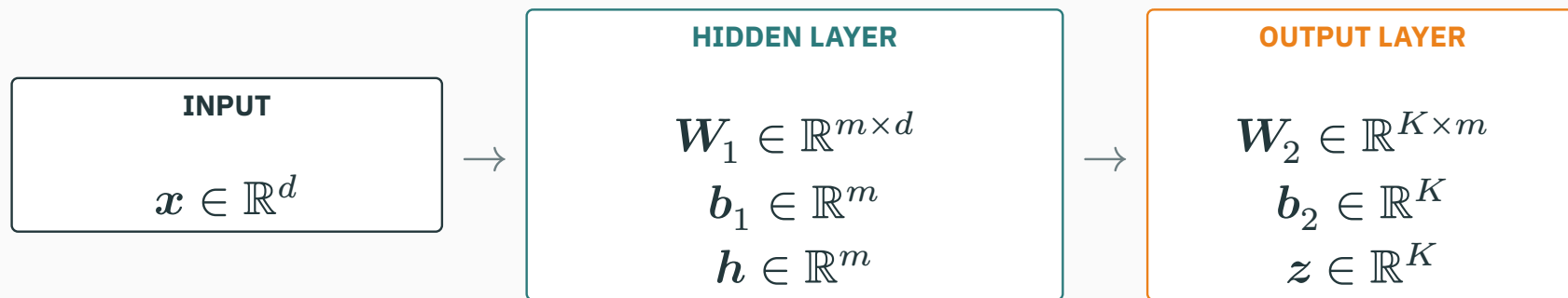


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$$(m \times d)(d \times 1) \rightarrow (m \times 1), \quad (K \times m)(m \times 1) \rightarrow (K \times 1)$$

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The inner dimensions cancel; the number of rows determines the next representation size.

One XOR network: keep these two features throughout

Both hidden units receive the same pair $x = (x_1, x_2)$:

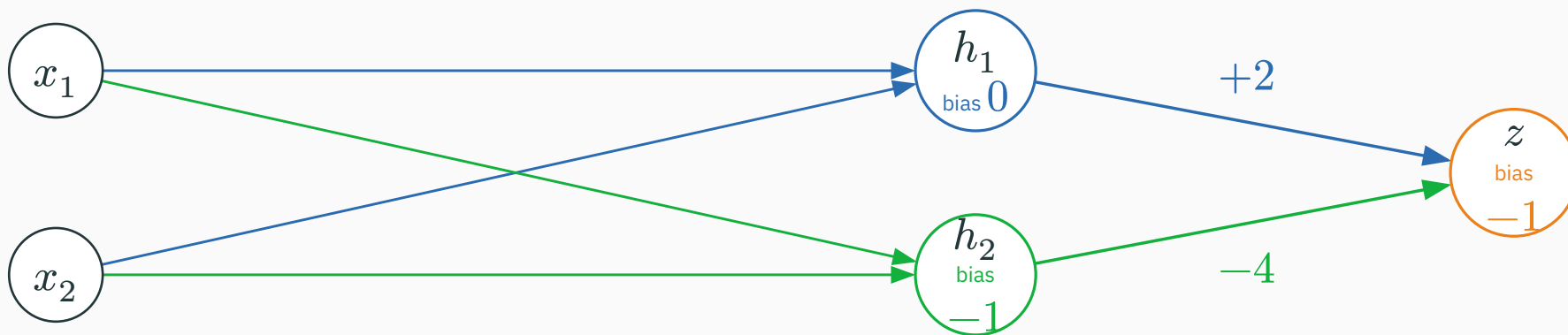
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all four input weights = $+1$

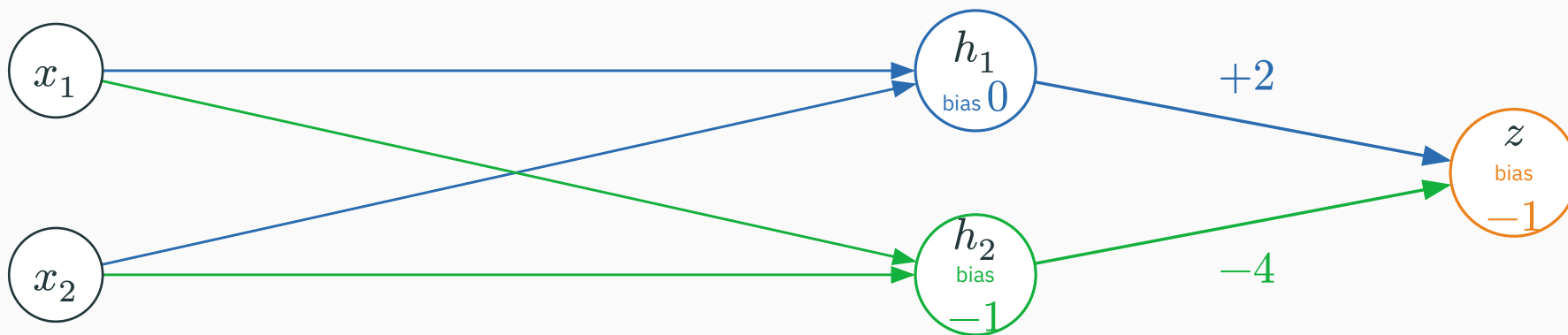


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The output node computes $z = 2h_1 - 4h_2 - 1$; we keep this exact network throughout the XOR example.

Verify the XOR network on all four inputs

Input	$u_1 =$ $x_1 + x_2$	h_1	$u_2 =$ $x_1 +$ $x_2 - 1$	h_2	$z =$ $2h_1 -$ $4h_2 - 1$	$\hat{y}$
(0, 0)	0	0	-1	0	-1	0
(1, 0)	1	1	0	0	1	1
(0, 1)	1	1	0	0	1	1
(1, 1)	2	2	1	1	-1	0

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$$\mathbf{W}_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{b}_1 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \mathbf{w}_{\text{out}} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}, \quad b_{\text{out}} = -1.$$

Verify the XOR network on all four inputs

Input	$u_1 =$ $x_1 + x_2$	h_1	$u_2 =$ $x_1 +$ $x_2 - 1$	h_2	$z =$ $2h_1 -$ $4h_2 - 1$	$\hat{y}$
(0, 0)	0	0	-1	0	-1	0
(1, 0)	1	1	0	0	1	1
(0, 1)	1	1	0	0	1	1
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Predict 1 when $z \geq 0$. One ordinary output node correctly classifies all four XOR inputs.

First derive the feature functions for a general input

Let $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$. The weights and bias define a **function** before we substitute any particular input.

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BLUE HIDDEN UNIT

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b_1 = 0$$

$$\begin{aligned} u_1(\mathbf{x}) &= \mathbf{w}_1^\top \mathbf{x} + b_1 \\ &= 1x_1 + 1x_2 + 0 = x_1 + x_2 \end{aligned}$$

$$\begin{aligned} h_1(\mathbf{x}) \\ &= \text{ReLU}(x_1 + x_2) \end{aligned}$$

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GREEN HIDDEN UNIT

$$\mathbf{w}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b_2 = -1$$

$$\begin{aligned} u_2(\mathbf{x}) &= \mathbf{w}_2^\top \mathbf{x} + b_2 \\ &= 1x_1 + 1x_2 - 1 = x_1 + x_2 - 1 \end{aligned}$$

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$$\begin{aligned} h_2(\mathbf{x}) \\ &= \text{ReLU}(x_1 + x_2 - 1) \end{aligned}$$

These are exactly the two XOR features already shown in the node diagram; now we evaluate them at one input.

Check both hidden calculations in matrix form

$$\begin{aligned} u &= \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1 \\ &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix}. \end{aligned}$$

$$u_1 = 1(1) + 1(0) + 0 = 1$$

$$u_2 = 1(1) + 1(0) - 1 = 0$$

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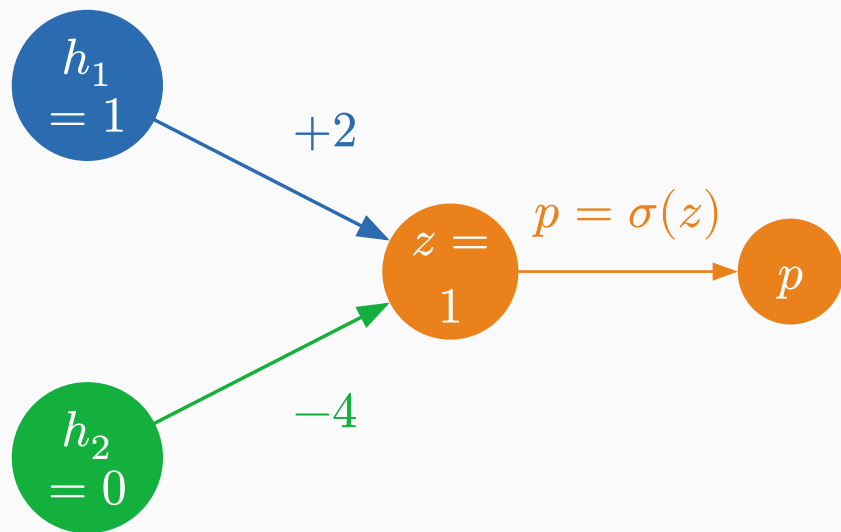
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Rows of W_1 are neurons: each row computes one pre-activation, then ReLU acts elementwise.

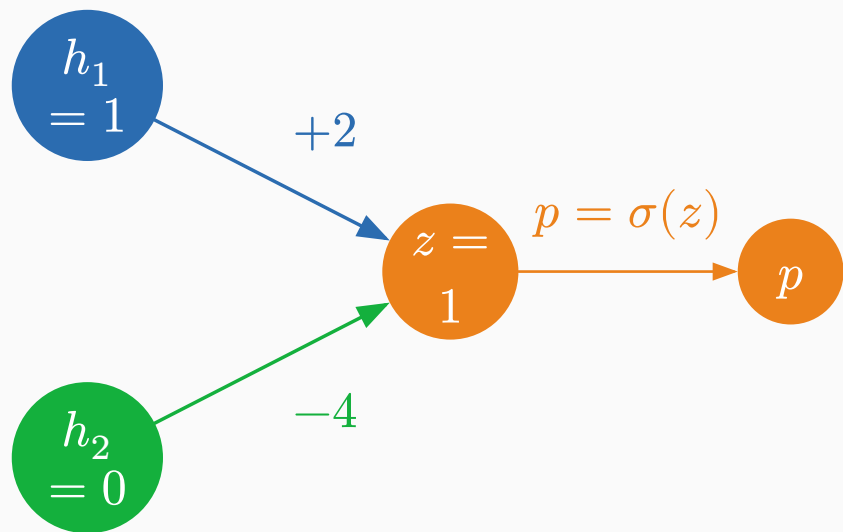
Combine the two features, then compute probability and loss



Output logit

$$\begin{aligned} z &= (2 \quad -4) \begin{pmatrix} 1 \\ 0 \end{pmatrix} - 1 \\ &= 2(1) - 4(0) - 1 = 1. \end{aligned}$$

Combine the two features, then compute probability and loss



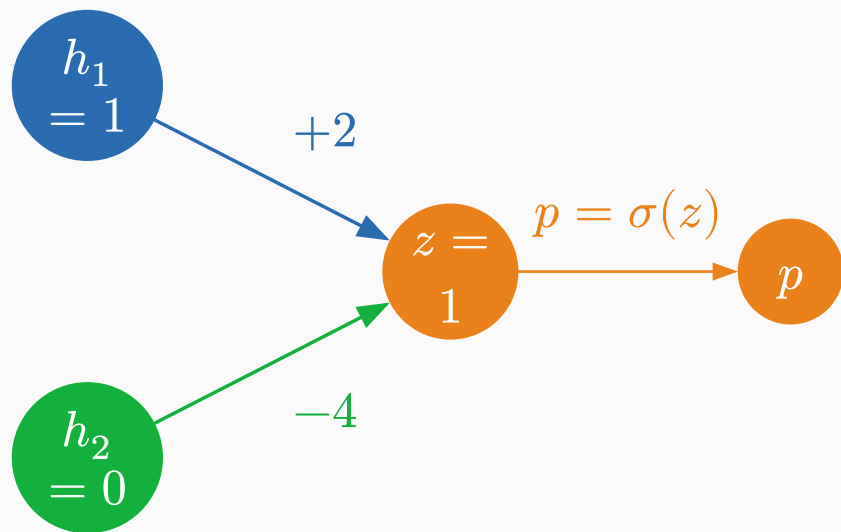
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Probability

$$p = \sigma(1) = \frac{1}{1 + \exp(-1)} \approx 0.731.$$

Combine the two features, then compute probability and loss



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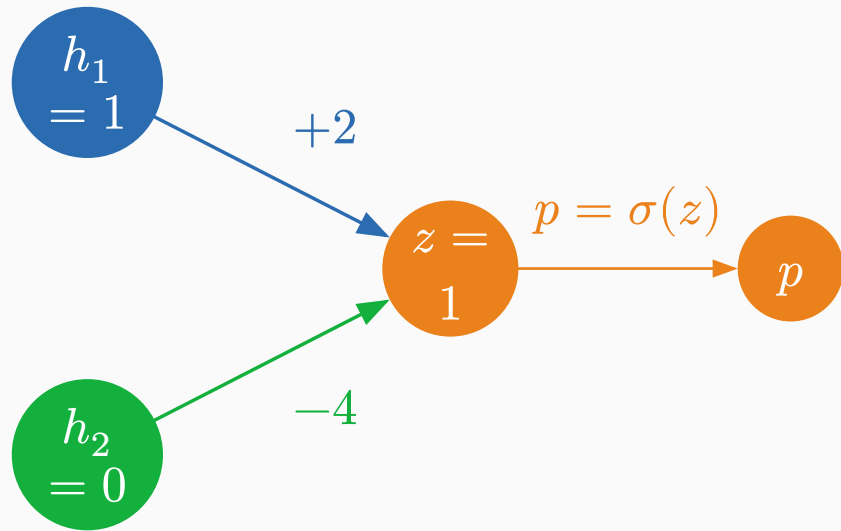
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Binary cross-entropy for $y = 1$

$$\mathcal{L} = -\log(0.731) \approx 0.313.$$



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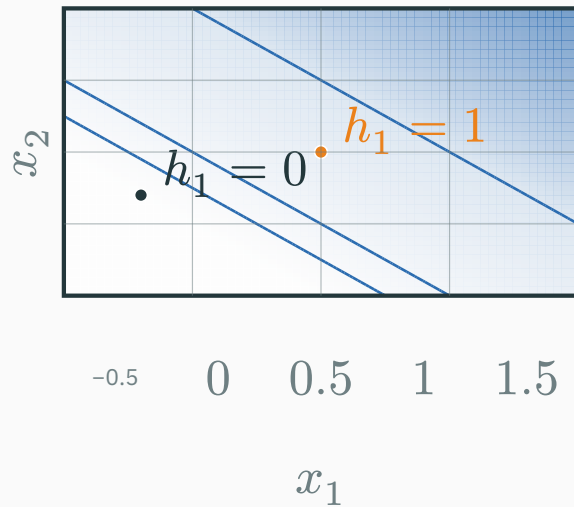
$$\mathcal{L} = -\log(0.731) \approx 0.313.$$

This is the same output node shown in the XOR network: weights $+2, -4$ and bias -1 . Sigmoid only turns z into p .

Each ReLU is flat on one side of a line and rises on the other

$$h_1(\mathbf{x}) = \text{ReLU}(x_1 + x_2)$$

active above $x_1 + x_2 = 0$



$h_j(\mathbf{x})$: 0  3

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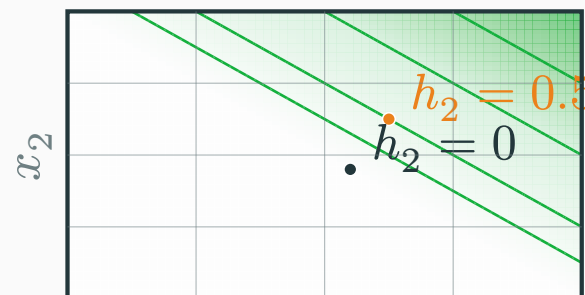
-0.5 0 0.5 1 1.5

x_1

$h_{j(x)} : 0$  3

$$h_2(x) = \text{ReLU}(x_1 + x_2 - 1)$$

active above $x_1 + x_2 = 1$



-0.5 0 0.5 1 1.5

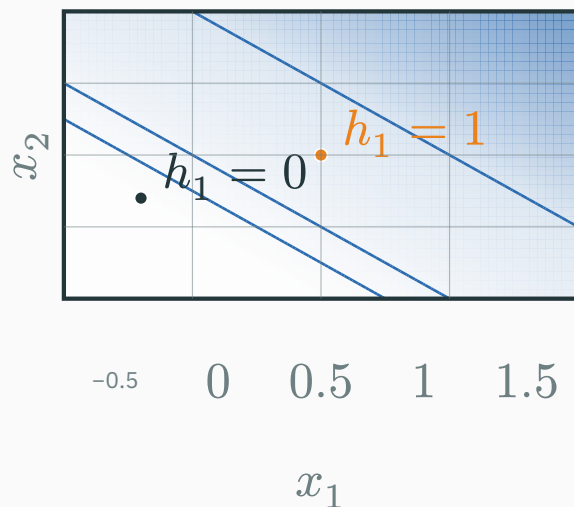
x_1

$h_{j(x)} : 0$  2

Each ReLU is flat on one side of a line and rises on the other

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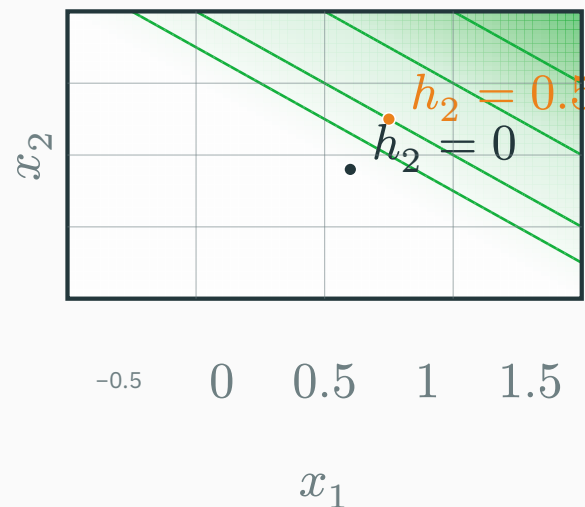
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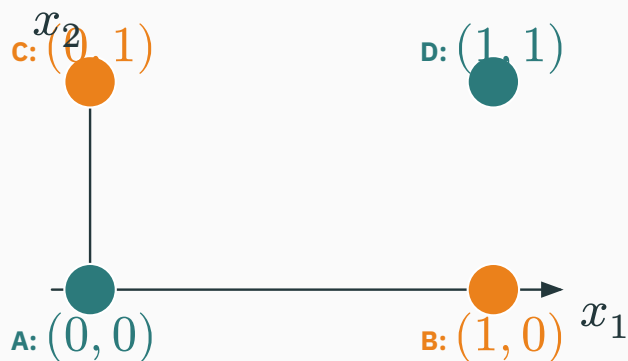
The grid is input space; each colour bar reports a hidden-unit activation value—not a probability.

Compute each XOR point's new hidden-space coordinates

Use the **same map** for every point: $T(x_1, x_2) = (h_1, h_2) = (\text{ReLU}(x_1 + x_2), \text{ReLU}(x_1 + x_2 - 1))$.

$$\mathbf{A}: (0, 0) \rightarrow (0, 0) \quad \mathbf{B}: (1, 0) \rightarrow (1, 0) \quad \mathbf{C}: (0, 1) \rightarrow (1, 0) \quad \mathbf{D}: (1, 1) \rightarrow (2, 1)$$

INPUT SPACE (x_1, x_2)

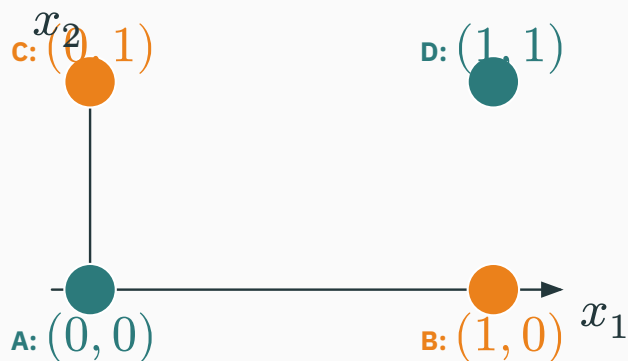


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APPLY T



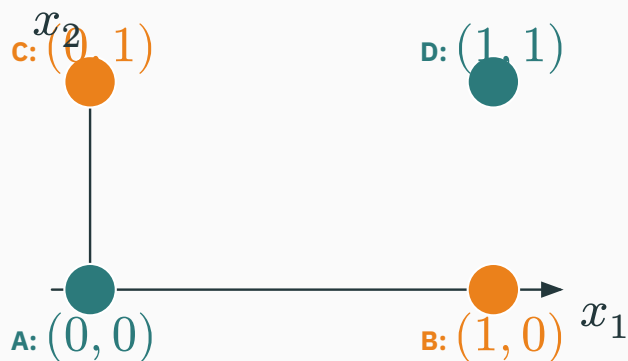
B and C
merge

Compute each XOR point's new hidden-space coordinates

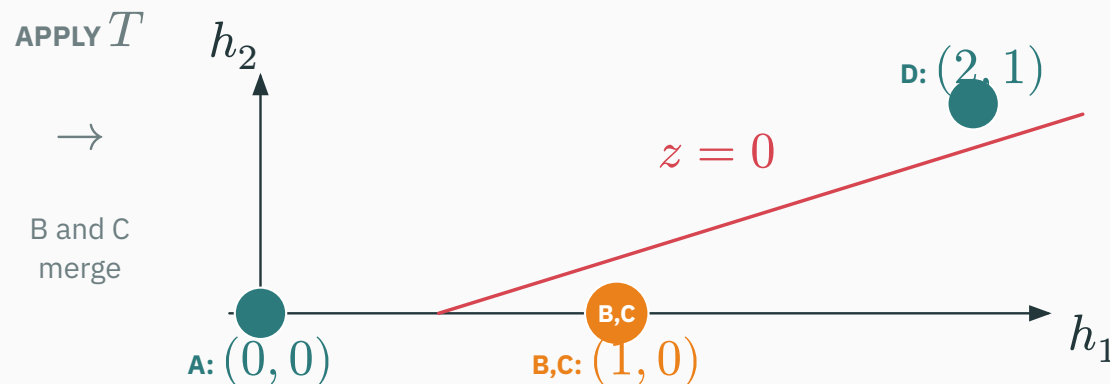
Use the **same map** for every point: $T(x_1, x_2) = (h_1, h_2) = (\text{ReLU}(x_1 + x_2), \text{ReLU}(x_1 + x_2 - 1))$.

A: $(0, 0) \rightarrow (0, 0)$ **B:** $(1, 0) \rightarrow (1, 0)$ **C:** $(0, 1) \rightarrow (1, 0)$ **D:** $(1, 1) \rightarrow (2, 1)$

INPUT SPACE (x_1, x_2)



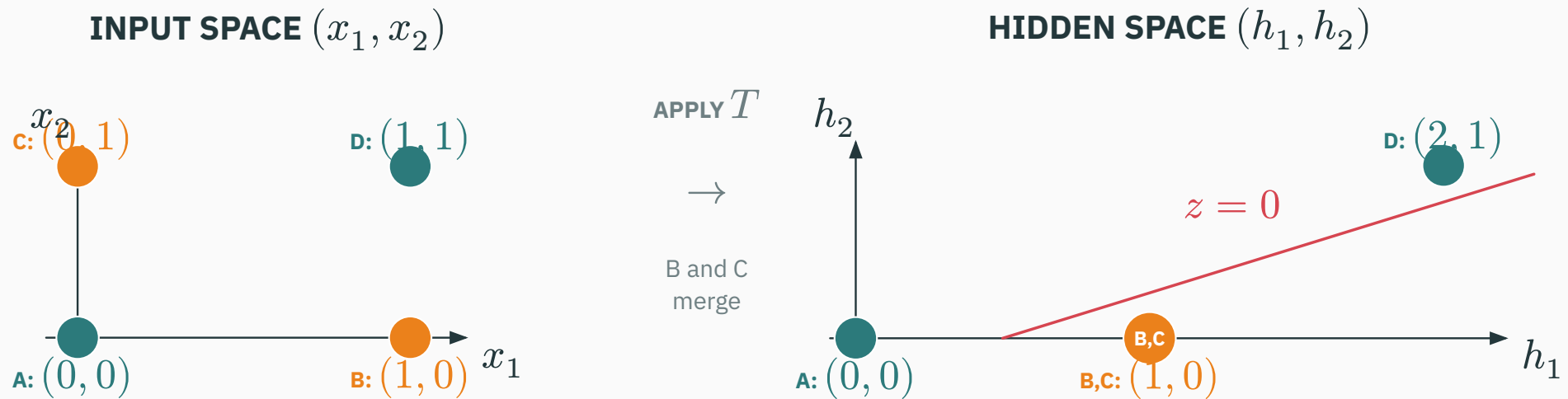
HIDDEN SPACE (h_1, h_2)



Compute each XOR point's new hidden-space coordinates

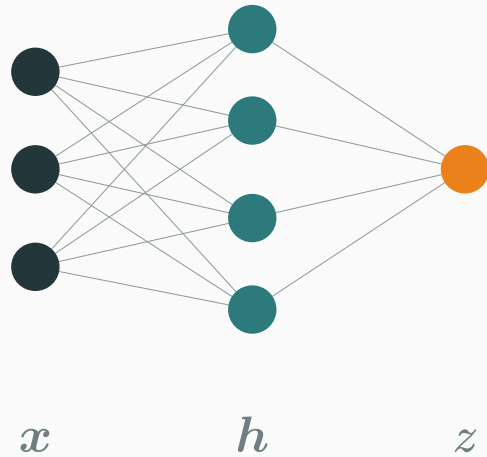
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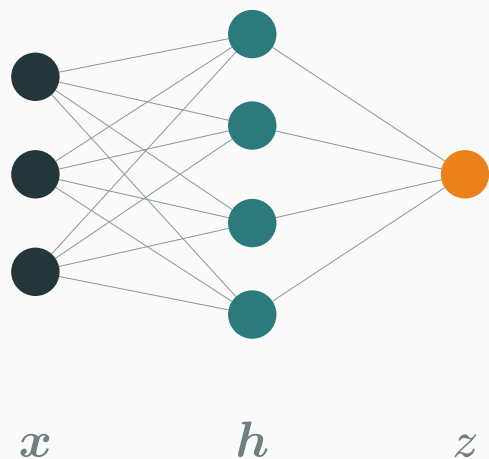


B and C both become $(1, 0)$. Since $z = 2h_1 - 4h_2 - 1$, the boundary $z = 0 \leftrightarrow h_1 - 2h_2 = 0.5$ separates them from A and D.

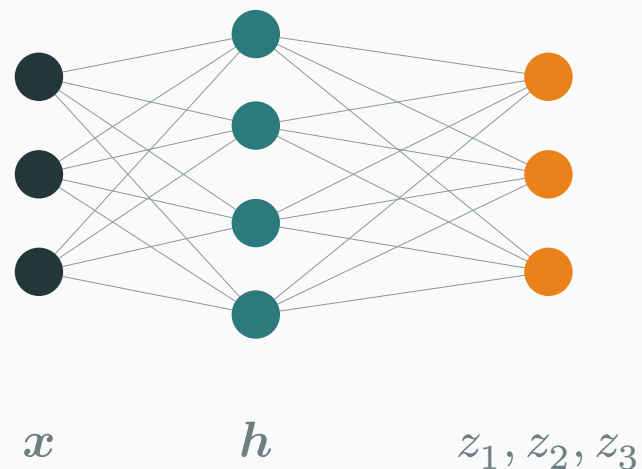
BINARY



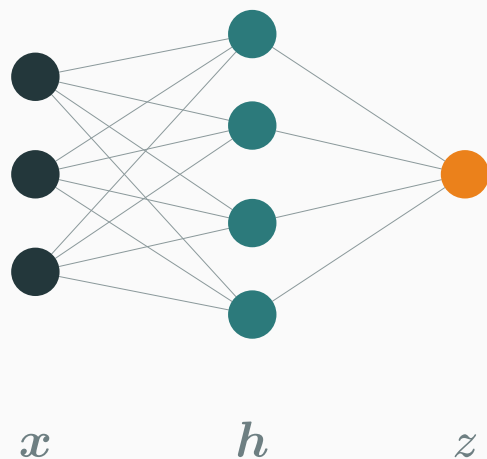
BINARY



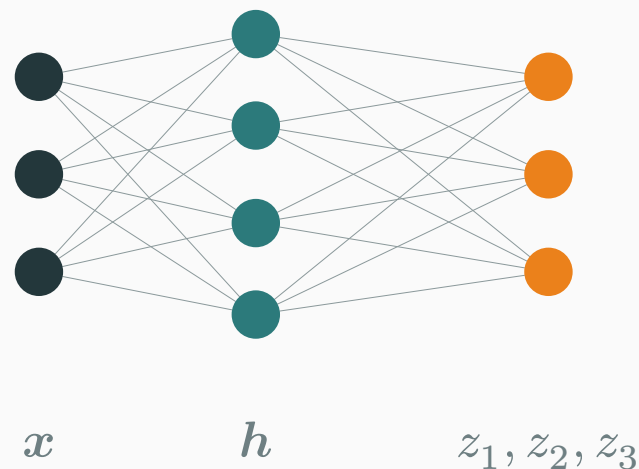
THREE CLASSES



BINARY

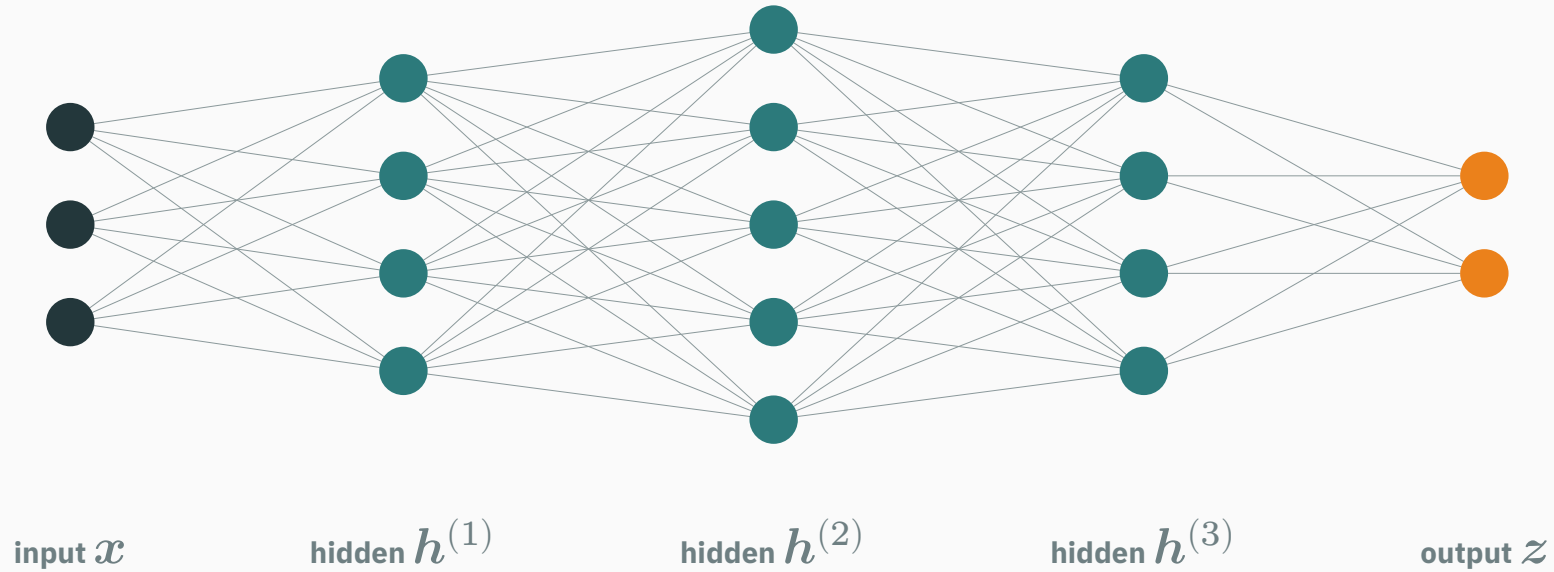


THREE CLASSES

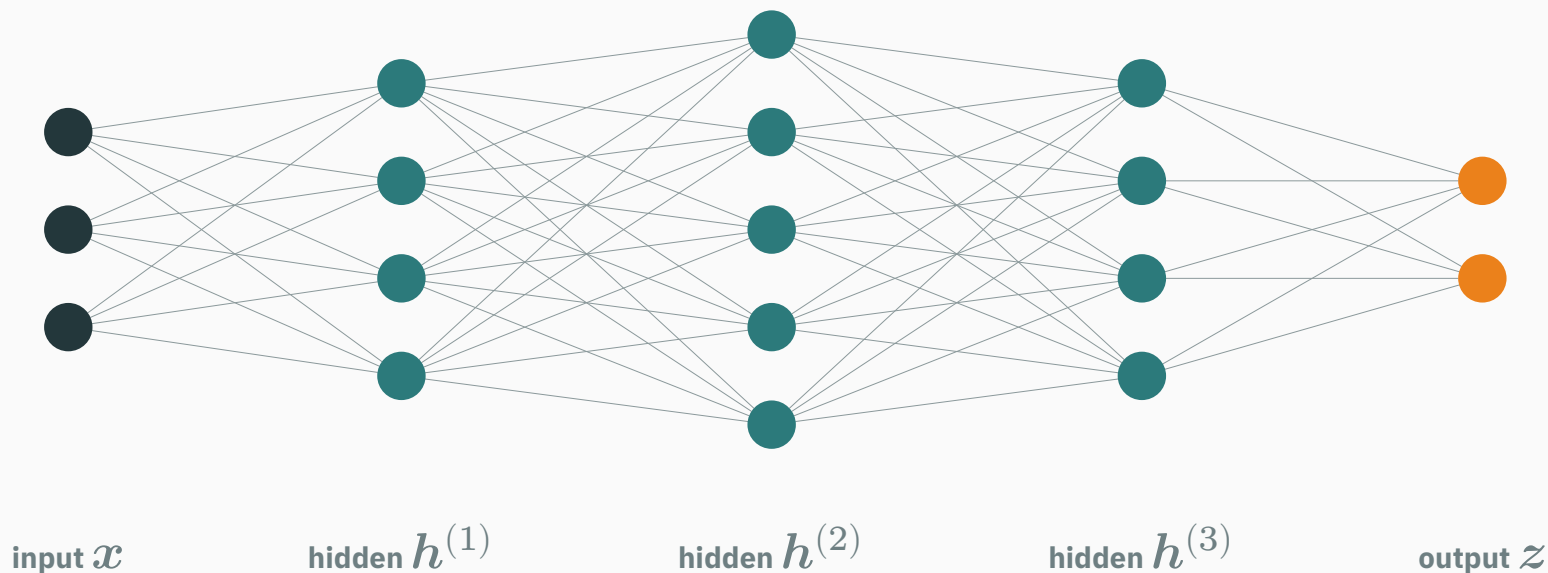


The hidden representation can stay the same; only the number and interpretation of output nodes changes.

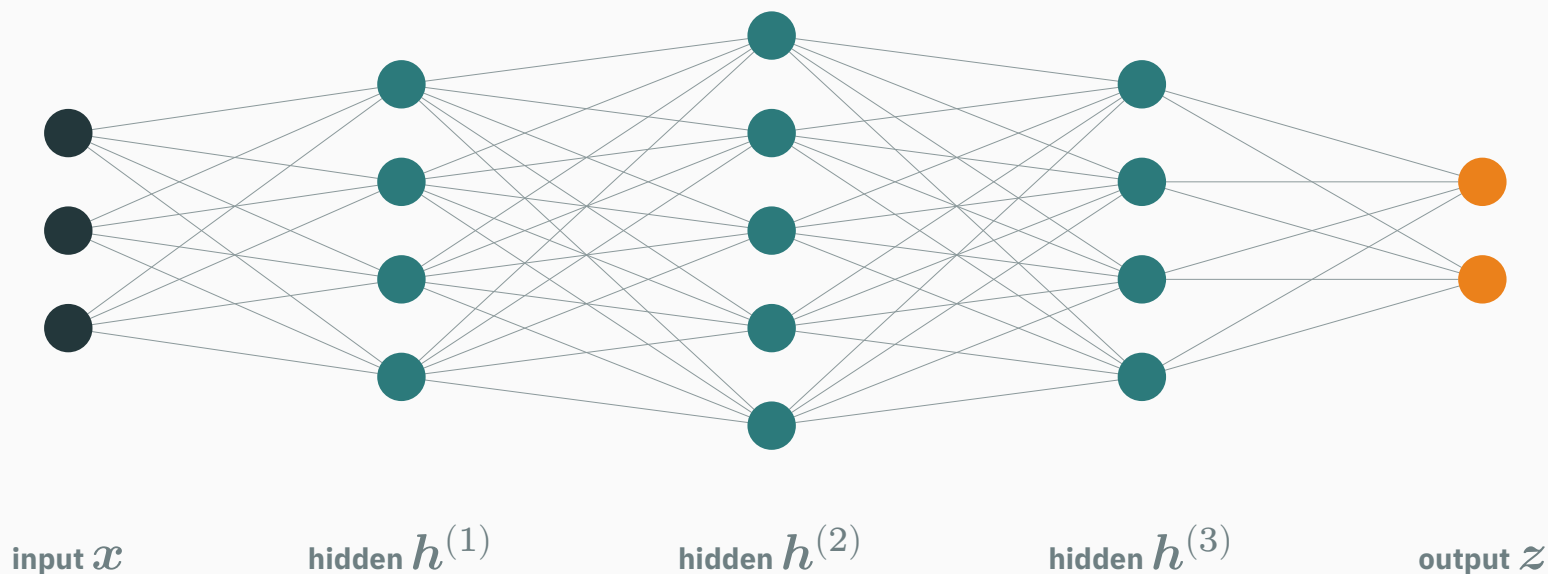
Node view: depth adds hidden columns



Node view: depth adds hidden columns



Every node connects only to the next column; information reaches the output through successive learned transformations.



Every node connects only to the next column; information reaches the output through successive learned transformations.

Depth is visible as the number of hidden columns, not the number of nodes within one column.

Universal approximation

Universal approximation asks one precise question

QUESTION

TARGET
continuous f $\rightarrow$ **REGION**
closed, bounded D $\rightarrow$ **TOLERANCE**
 $\varepsilon > 0$

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Can some finite network g make $|f(x) - g(x)| < \varepsilon$ for **every** $x \in D$?

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With a suitable nonlinear activation, the theorem says: yes—such a finite one-hidden-layer network exists.

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Can some finite network g make $|f(x) - g(x)| < \varepsilon$ for **every** $x \in D$?

With a suitable nonlinear activation, the theorem says: **yes—such a finite one-hidden-layer network exists.**

This is an existence claim about representational capacity; it does not promise that training will find the network.

A ReLU contributes one hinge

$$\varphi(z) = \max(0, z)$$

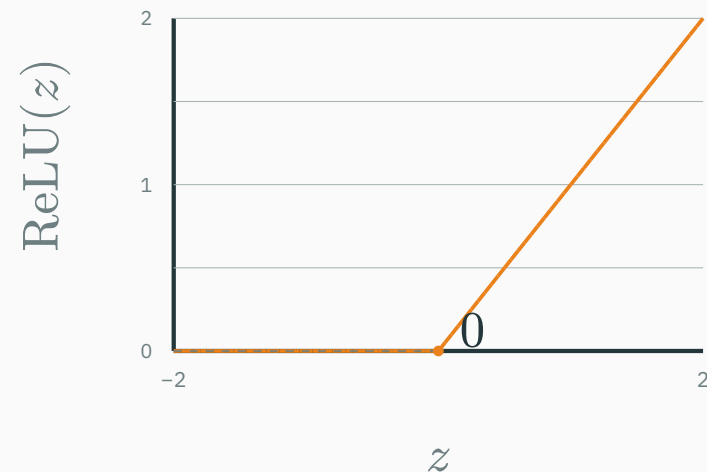
$$\varphi(z) = \max(0, z)$$

- $z < 0$: output is zero;
- $z > 0$: output grows linearly;
- at $z = 0$: the slope changes—a **hinge**.

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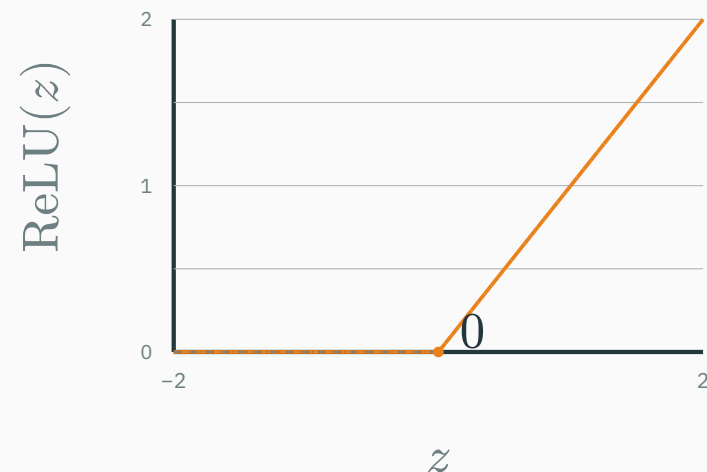
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- at $z = 0$: the slope changes—a **hinge**.



A hidden layer combines many shifted hinges into a piecewise-linear function.

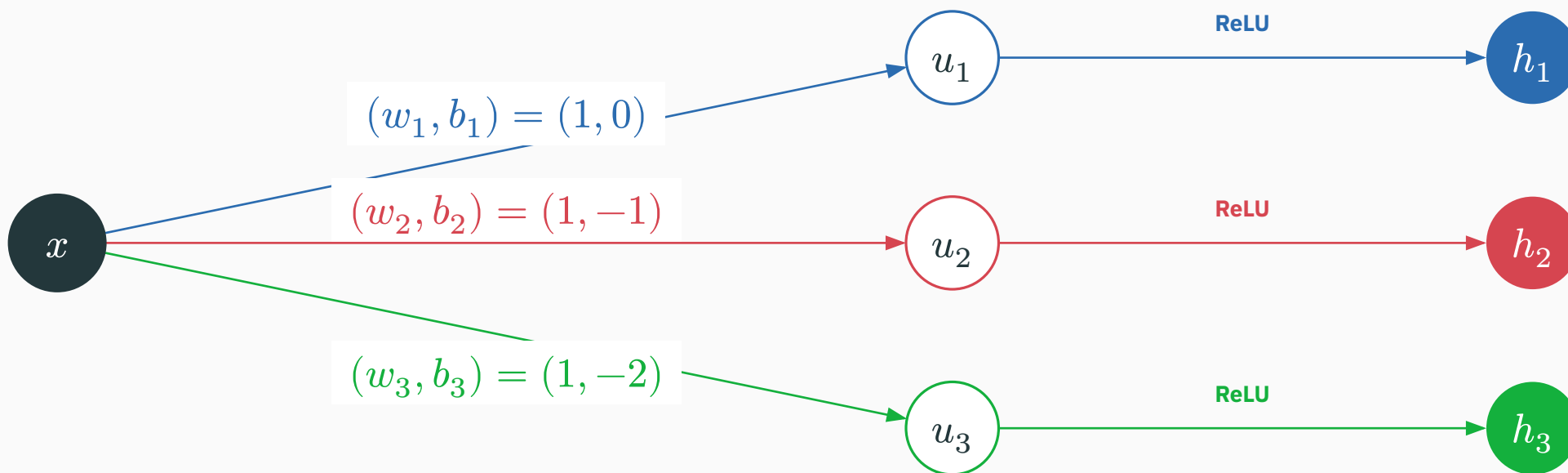
One input becomes three symbolic hinge features

Every unit first computes $u_i = w_i x + b_i$, then applies $h_i = \text{ReLU}(u_i)$.

ONE INPUT

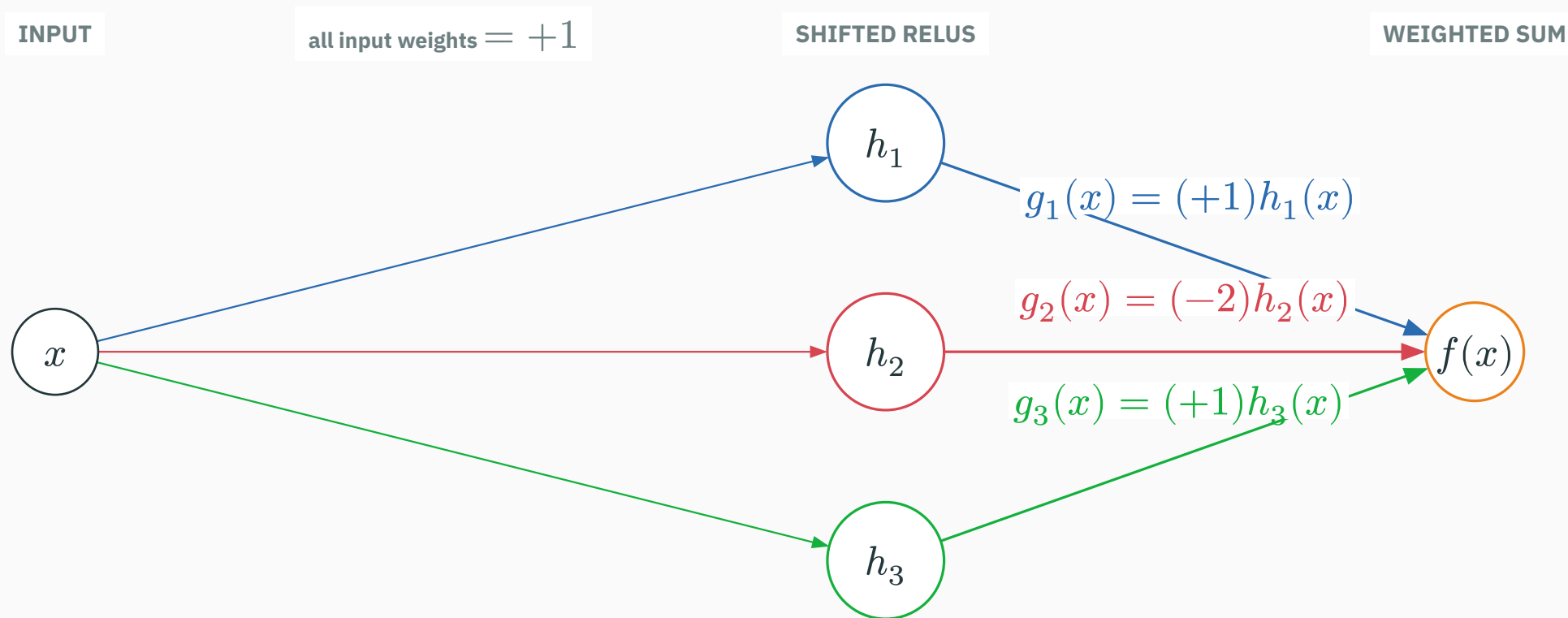
AFFINE SUMS

RELU FEATURES



The same x creates $h(x) = (\text{ReLU}(x), \text{ReLU}(x - 1), \text{ReLU}(x - 2))$; the hinges are at 0, 1, 2.

Output weights combine the three symbolic hinges

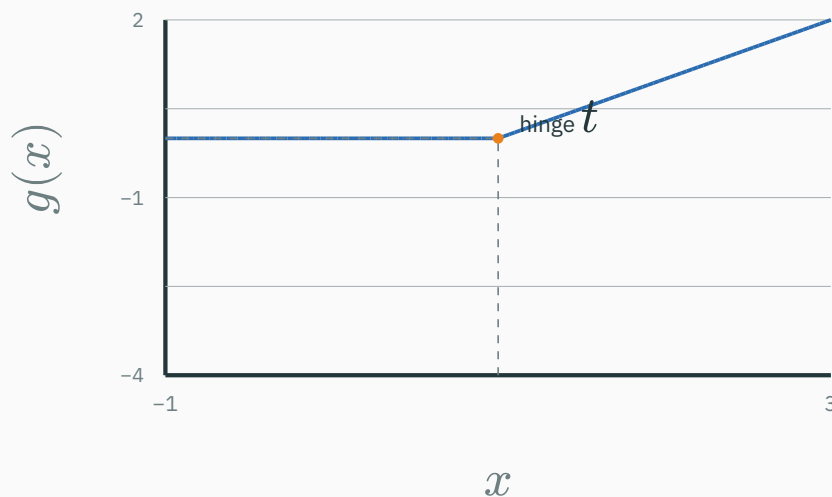


The output adds $f(x) = g_1(x) + g_2(x) + g_3(x)$ —the theorem’s “sum of hidden units” is literal.

Output weights choose each hinge's slope change

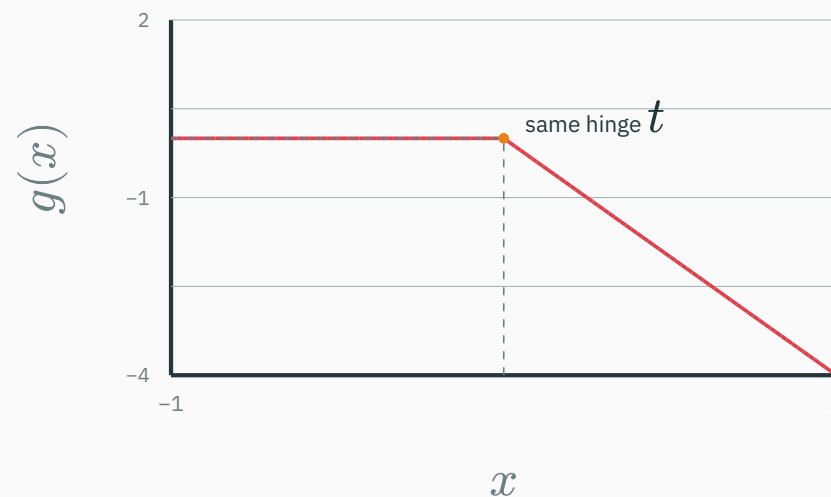
Start with $h(x) = \text{ReLU}(x - t)$; its output contribution is $g(x) = vh(x)$.

$$v = +1 \rightarrow g(x) = +h(x)$$



slope change $\Delta s = +1$

$$v = -2 \rightarrow g(x) = -2h(x)$$

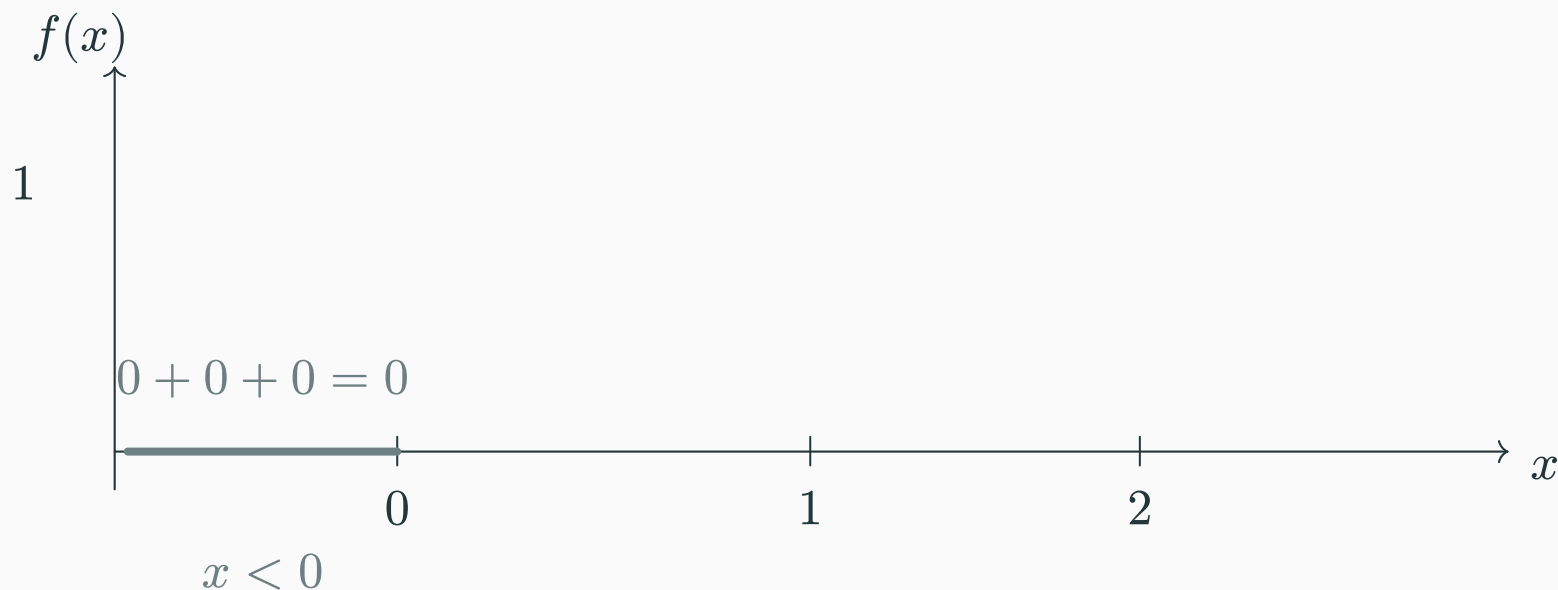


slope change $\Delta s = -2$

The hinge stays at t ; the sign of v chooses the bend, and $|v|$ sets its strength.

Three hinges assemble a tent interval by interval

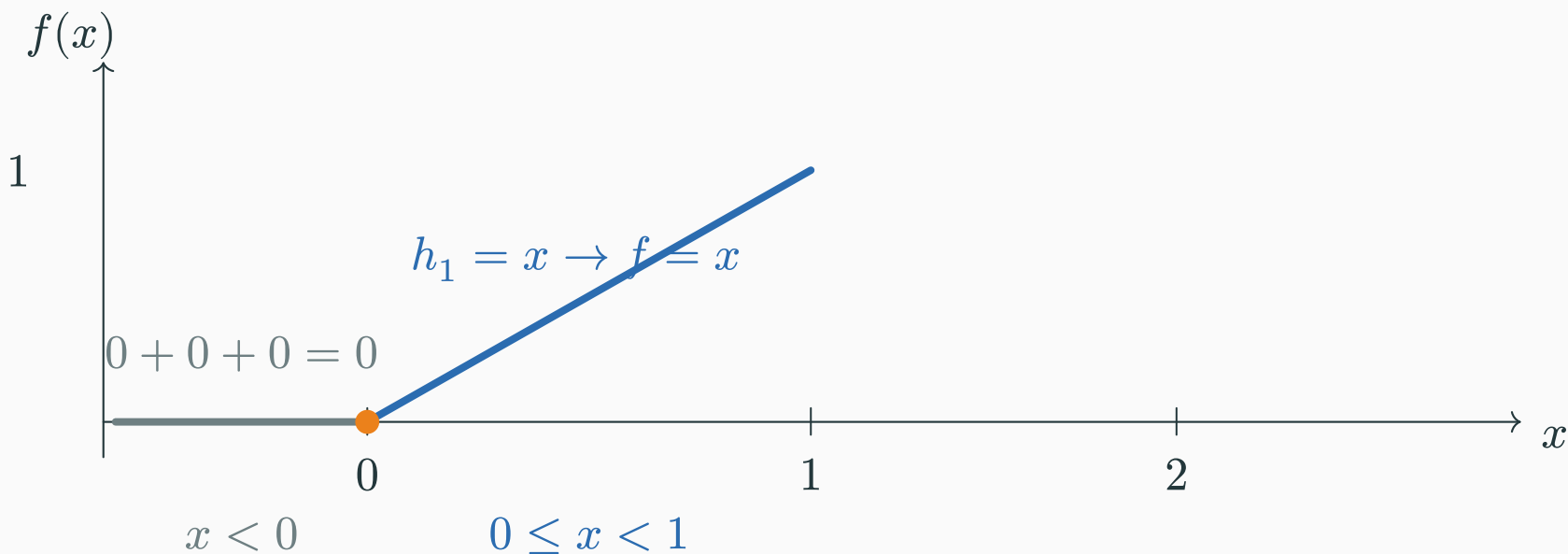
$$f(x) = h_1 - 2h_2 + h_3 = \text{ReLU}(x) - 2\text{ReLU}(x - 1) + \text{ReLU}(x - 2).$$



Region 1: every hidden unit is off, so $f(x) = 0$.

Three hinges assemble a tent interval by interval

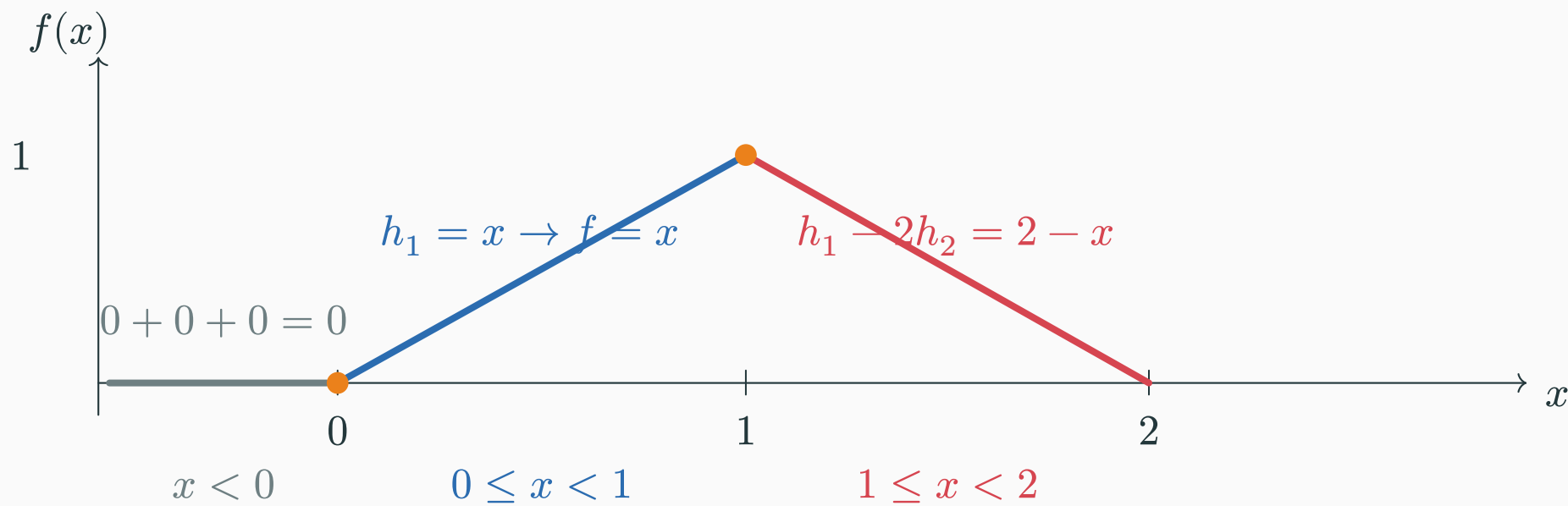
$$f(x) = h_1 - 2h_2 + h_3 = \text{ReLU}(x) - 2\text{ReLU}(x - 1) + \text{ReLU}(x - 2).$$



Region 2: $h_1 = x$ switches on, so the output rises with slope +1.

Three hinges assemble a tent interval by interval

$$f(x) = h_1 - 2h_2 + h_3 = \text{ReLU}(x) - 2\text{ReLU}(x - 1) + \text{ReLU}(x - 2).$$

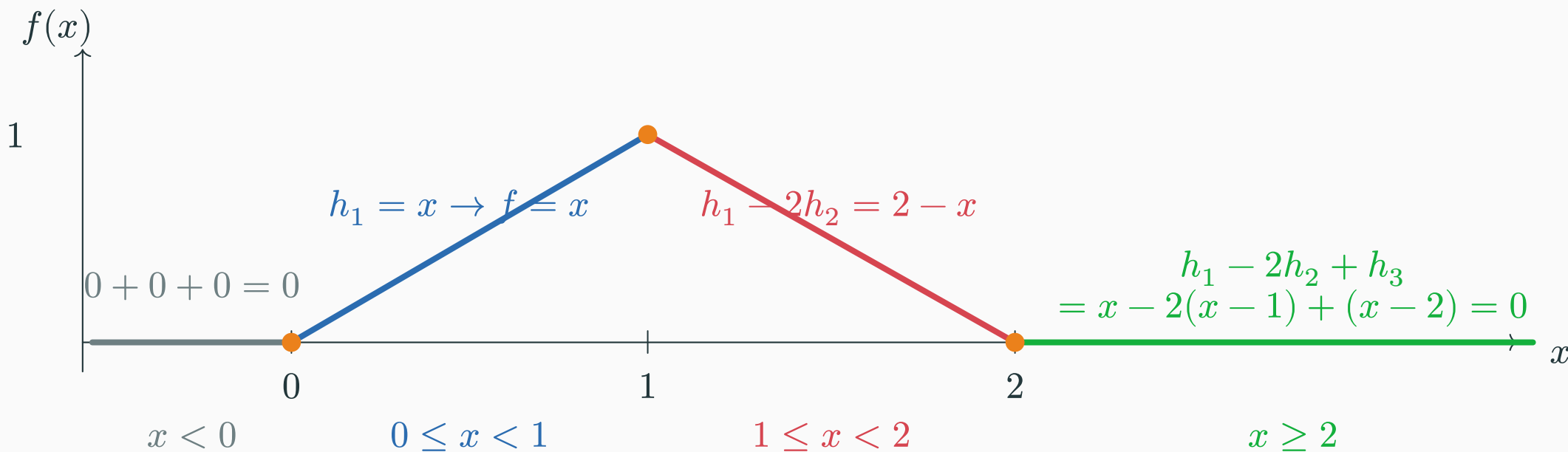


Region 3: $-2h_2$ switches on, changing the total slope from $+1$ to -1 .

Three hinges assemble a tent interval by interval

D DERIVATION

$$f(x) = h_1 - 2h_2 + h_3 = \text{ReLU}(x) - 2\text{ReLU}(x - 1) + \text{ReLU}(x - 2).$$



Region 4: h_3 switches on and cancels the remaining slope, so $f(x) = 0$.

Reading left to right: flat $\rightarrow$ rise $\rightarrow$ fall $\rightarrow$ flat—the weighted sum is a tent.

Learned ReLU units are the theorem's building blocks

D DERIVATION

For a one-dimensional input, a trained hidden unit has the form

$$h_{j(x)} = \text{ReLU}(w_j x + b_j), \quad g(x) = c + \sum_j v_j h_{j(x)}.$$

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j	hinge t_j	w_j	b_j	output v_j	signed contribution $g_{j(x)}$
1	0	1	0	+1	+ReLU(x)
2	1	1	-1	-2	-2ReLU($x - 1$)
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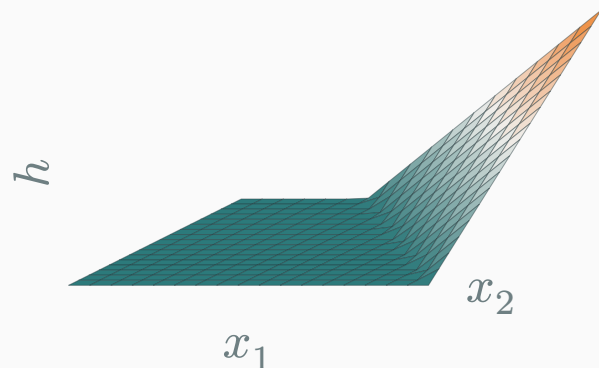
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Training learns hinge locations and signed contributions; universal approximation says enough such units can make the remaining gap arbitrarily small.

A ReLU unit adds one fold to a multidimensional function

$$h(x) = \text{ReLU}(w^T x + b)$$



computed ReLU surface for a two-dimensional input

THE FOLD

$$w^T x + b = 0$$

is a line in 2-D and a hyperplane
in d -D

THE TWO SIDES

one side is flat at $h = 0$; the
other rises linearly

A weighted sum of ReLU units combines folds in different positions and directions into a piecewise-linear surface.

Classification: one score per class, then pick the largest

FOR EACH INPUT

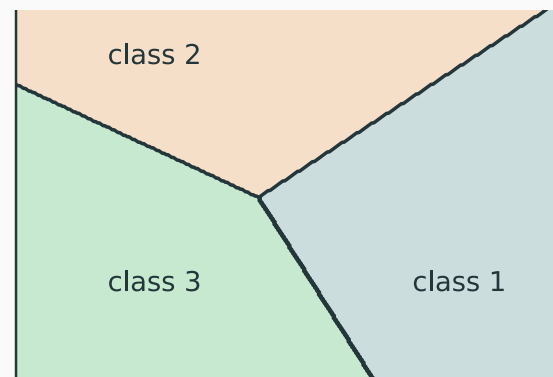
$$x \rightarrow h \rightarrow (z_1, z_2, z_3)$$

input $\rightarrow$ ReLU features $\rightarrow$ one score per class

PICK THE WINNING SCORE

$$\hat{y} = \arg \max_k z_k(x)$$

IN HIDDEN-FEATURE SPACE (h_1, h_2)



colour shows which output score is largest

ReLU features reshape the input; the largest output score determines the class, and its winner regions define the boundaries.

INTERACTIVE

↗ nipunbatra.github.io/dl-teaching/interactives/relu-function-builder.html

Follow the same sequence as the slides: **one hinge** → **tent** → **smooth target** → **binary classifier**.

$$g_{j(x)} = v_j \text{ReLU}(x - t_j)$$

WHERE IS THE BEND?

t_j

moves the hinge left or right

WHAT DOES THE BEND DO?

v_j

sets its direction and strength

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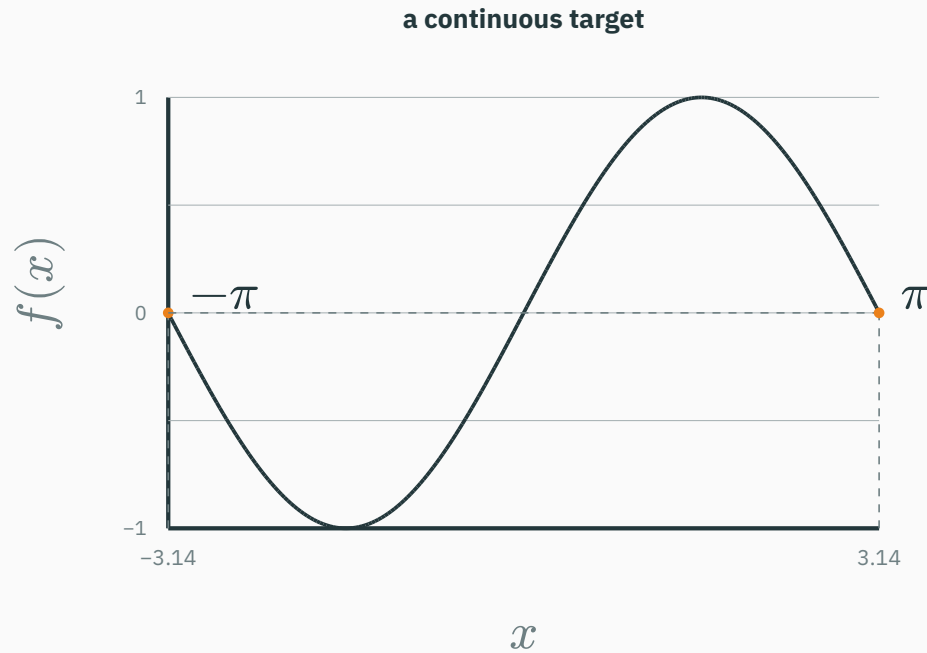
v_j

sets its direction and strength

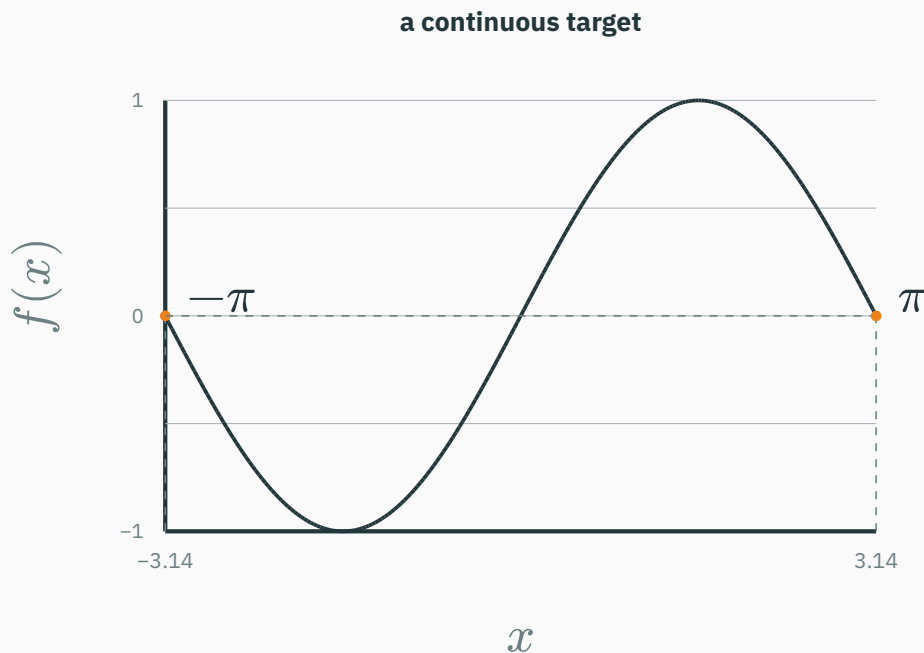
Add bends to trace a target; when their sum is a logit $z(x)$, the predicted class changes where $z = 0$.

First fix the target and the input region

Continuous means that the target has no jumps.



First fix the target and the input region

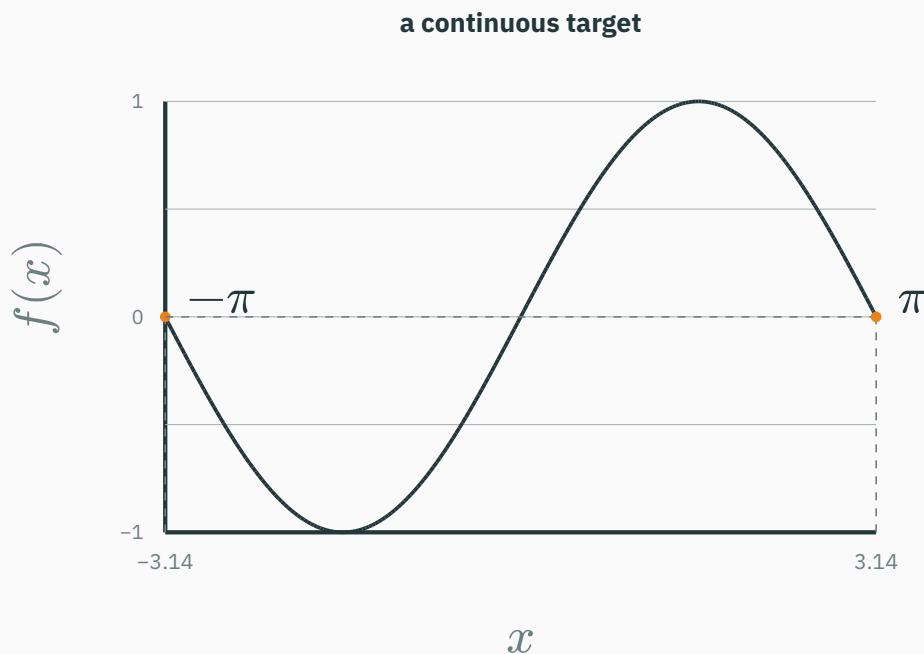


Continuous means that the target has no jumps.

In one dimension, a **compact domain** can be a closed, bounded interval such as

$$D = [-\pi, \pi].$$

First fix the target and the input region



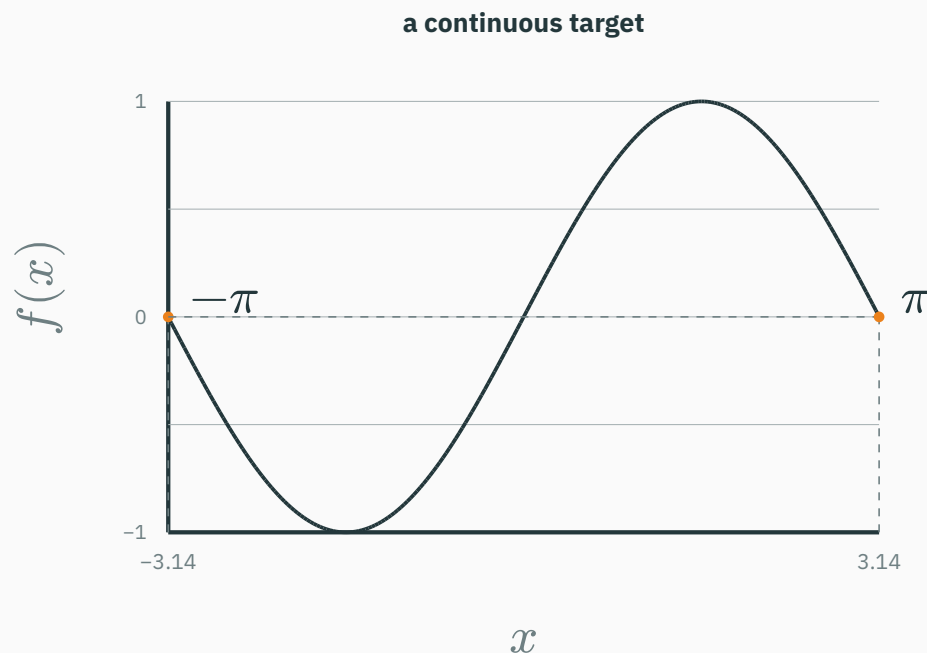
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We ask the network to match f **on** D ; the theorem makes no claim outside that chosen region.

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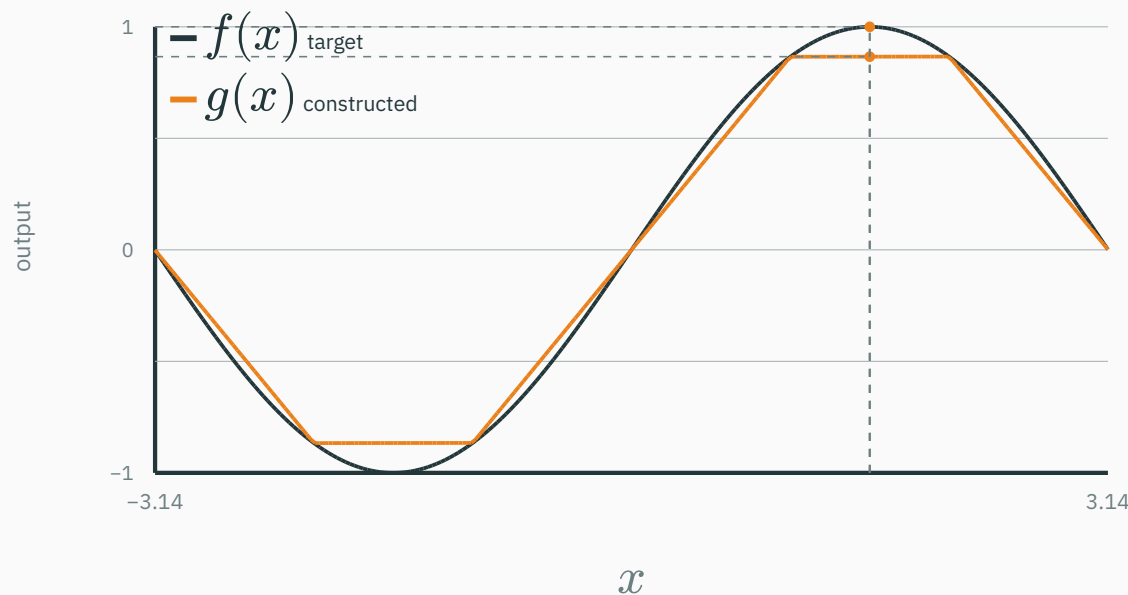
$$D = [-\pi, \pi].$$

We ask the network to match f **on** D ; the theorem makes no claim outside that chosen region.

Approximation begins with one target function on one specified input region.

$$e(x_0) = |f(x_0) - g(x_0)|$$

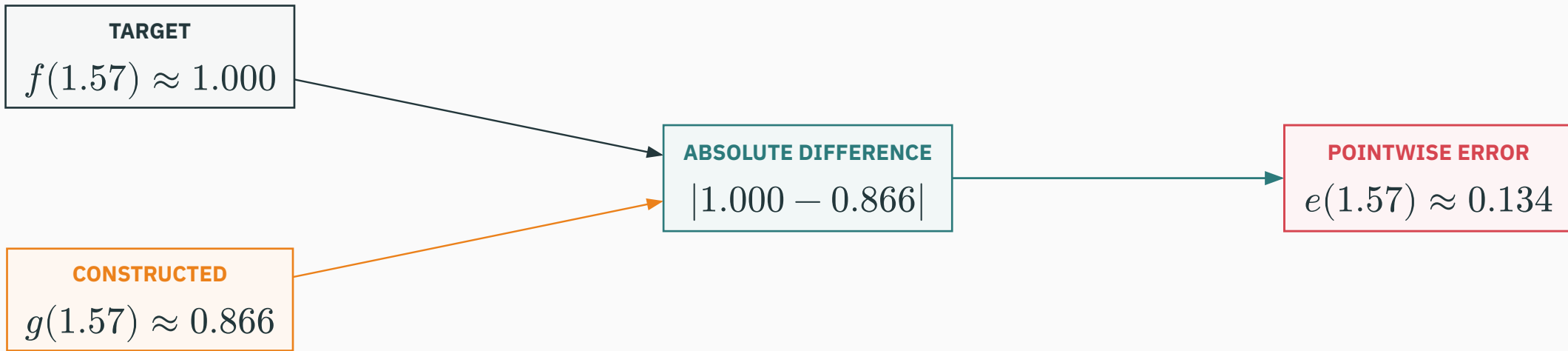
vertical slice at $x_0 = 1.57$



Pointwise error is the vertical distance between the target and its constructed approximation at the same input.

Calculate the gap at $x_0 = 1.57$

$$e(x_0) = |f(x_0) - g(x_0)|$$



One point is not enough: universal approximation asks for the largest gap over the whole region.

“sup” means the worst gap over the whole region

$$\sup_{x \in D} |f(x) - g(x)|$$

sup means **supremum**: the smallest number
at least as large as every pointwise error.

“sup” means the worst gap over the whole region

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Here the error is continuous and $D = [-\pi, \pi]$ is closed and bounded, so a highest error exists. In this plot, read **sup** as **maximum**.

sup $< \epsilon$: no input in D crosses the tolerance.

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The entire orange error curve stays below $\epsilon = 0.05$, so the uniform guarantee is satisfied.

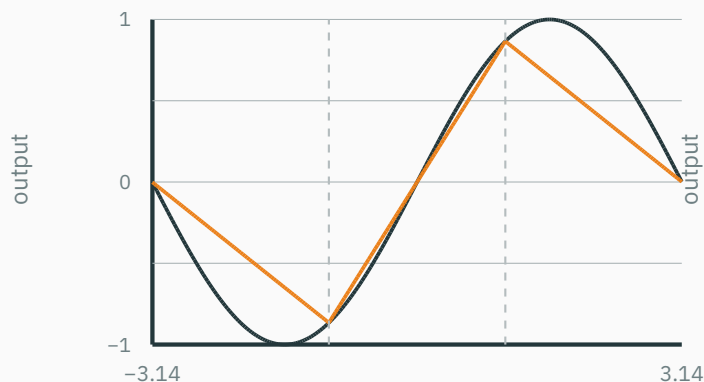
More hinges reduce the largest approximation error

CONSTRUCTED INTERPOLANTS · FIXED KNOTS · NOT TRAINED solid: target $\sin x$ dashed: $g_m(x)$

Each g_m matches $\sin x$ at fixed knots and is linear between adjacent knots.

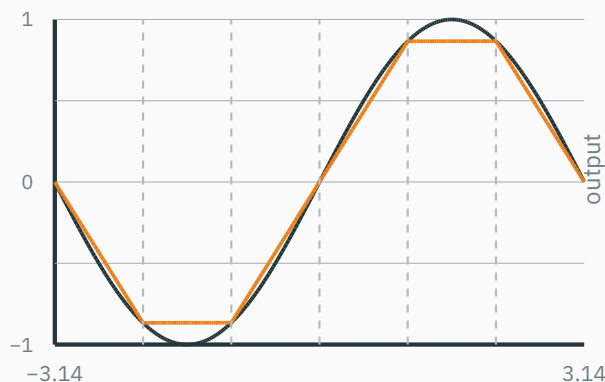
2 hinges

$$\varepsilon_m \approx 0.437$$



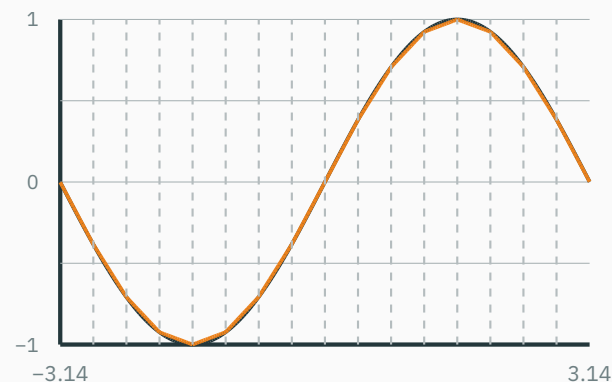
5 hinges

$$\varepsilon_m \approx 0.134$$



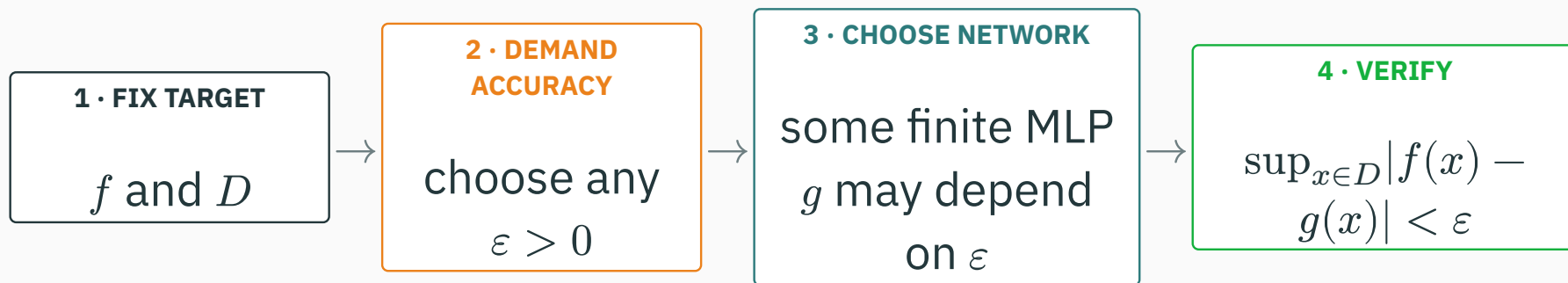
15 hinges

$$\varepsilon_m \approx 0.019$$

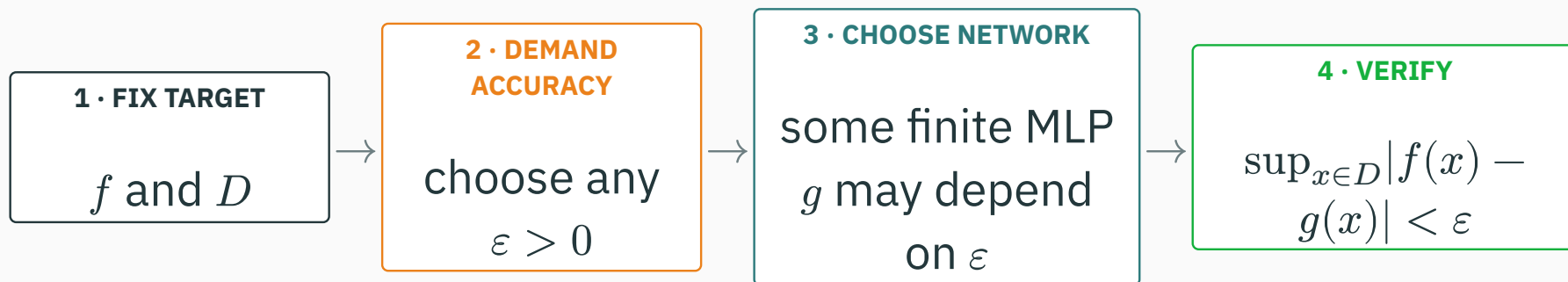


More hinges shorten each linear segment, so the worst gap falls from 0.437 to 0.019.

Read the universal-approximation claim in order



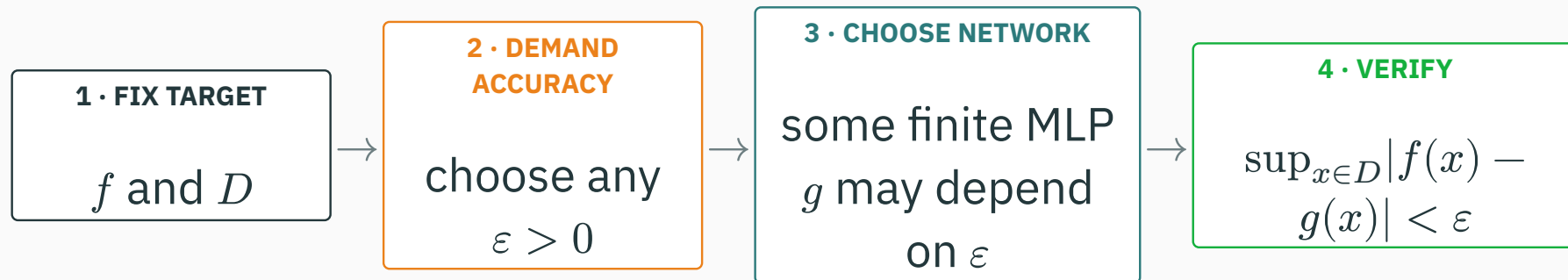
Read the universal-approximation claim in order



$\forall \varepsilon > 0, \exists \text{ finite MLP } g \text{ such that } \sup_{x \in D} |f(x) - g(x)| < \varepsilon.$

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D DERIVATION



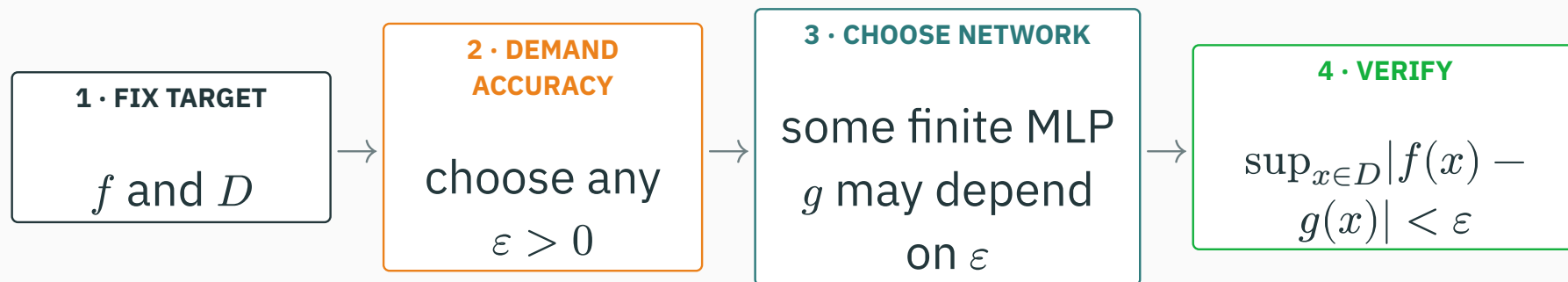
$\forall \varepsilon > 0, \exists \text{ finite MLP } g \text{ such that } \sup_{x \in D} |f(x) - g(x)| < \varepsilon.$

For every ε : we may request any positive tolerance.

There exists g : the network may change when the tolerance changes.

Read the universal-approximation claim in order

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$\forall \varepsilon > 0, \exists \text{ finite MLP } g \text{ such that } \sup_{x \in D} |f(x) - g(x)| < \varepsilon.$

For every ε : we may request any positive tolerance.

There exists g : the network may change when the tolerance changes.

This is a statement about representational capacity—not yet about how to find the network.

Existence does not guarantee learnability

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- that training will **find** that function;
- that **finite data** is enough;

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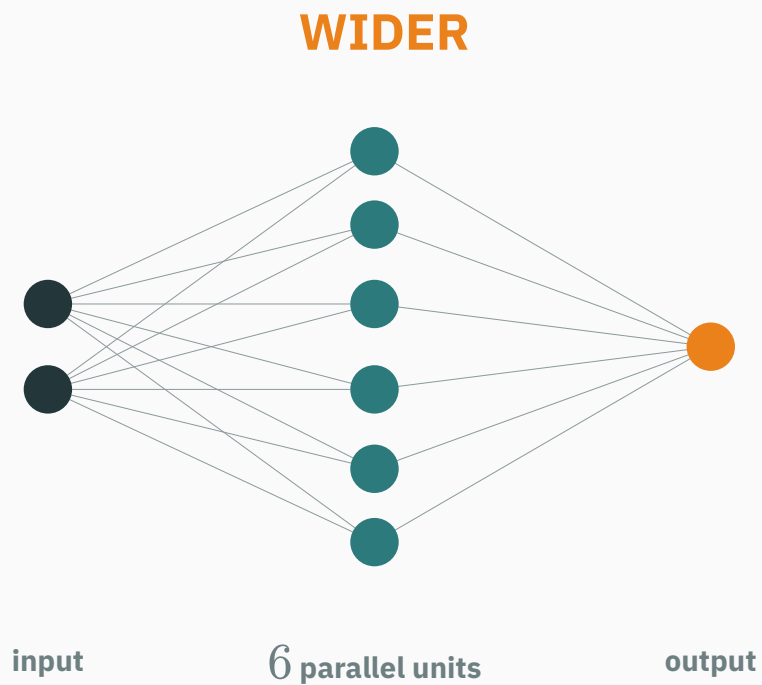
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expressivity $\neq$ trainability $\neq$ generalization

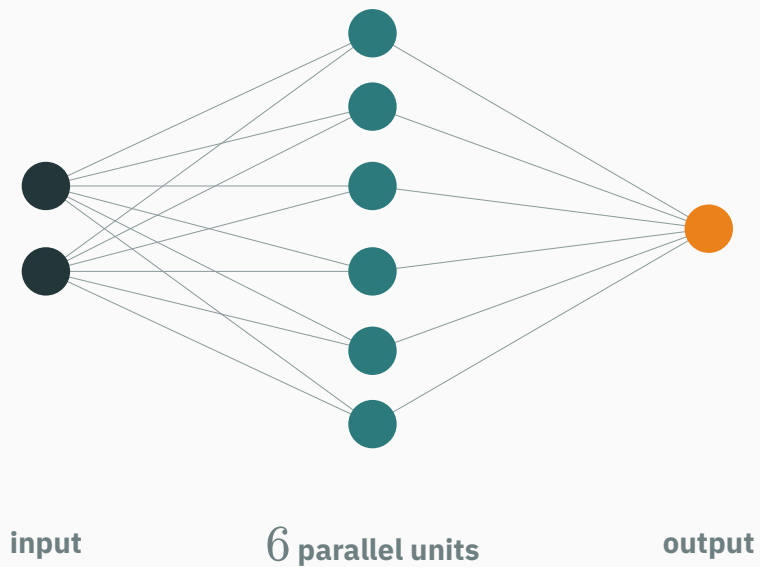
Depth versus width

Node view: width adds rows; depth adds columns

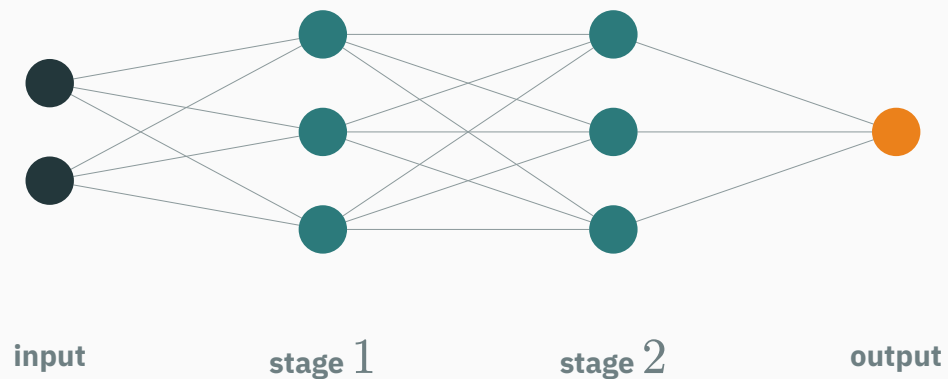


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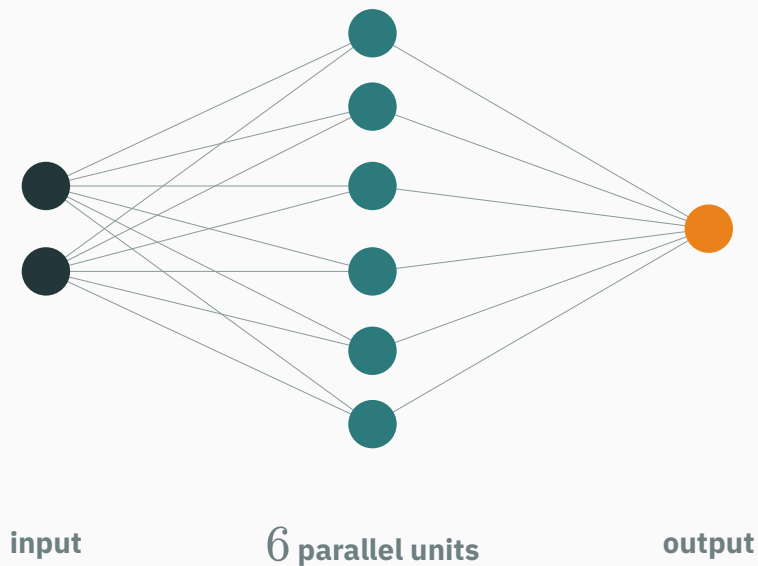
WIDER



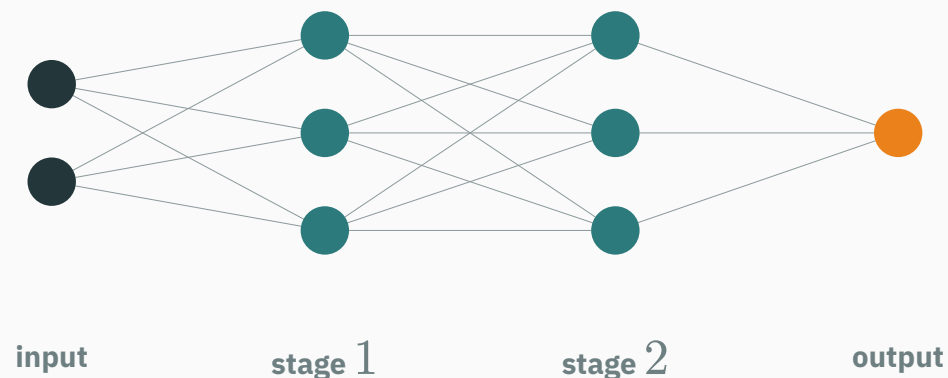
DEEPER



WIDER

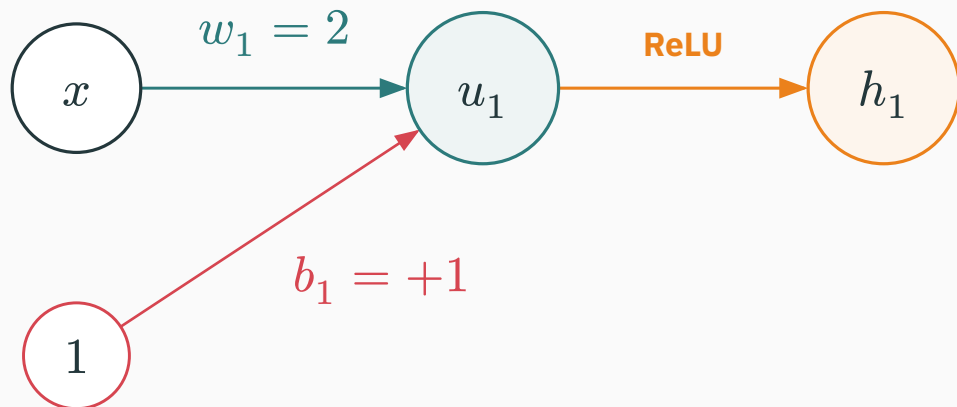


DEEPER



Count nodes vertically to read width; count hidden columns horizontally to read depth.

Layer 1 creates a hinge at $x = -0.5$

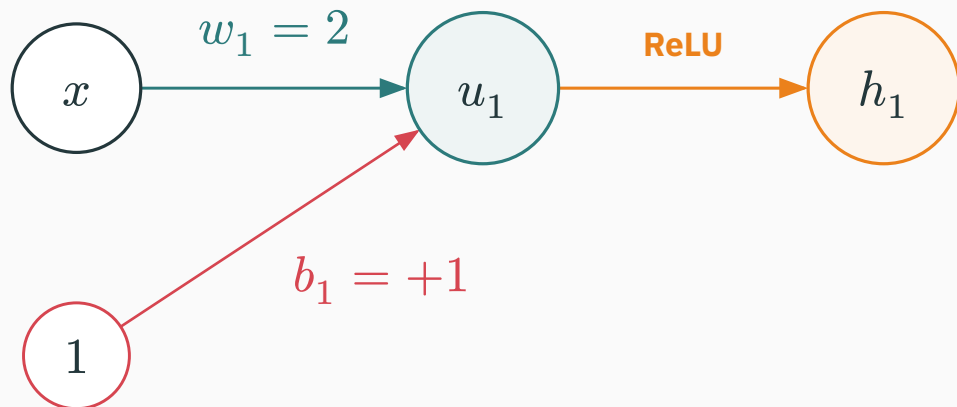


$$u_1 = 2x + 1$$
$$h_1 = f_1(x) = \text{ReLU}(u_1)$$

At $x = 0.25$:

$$u_1 = 1.5 \rightarrow h_1 = 1.5$$

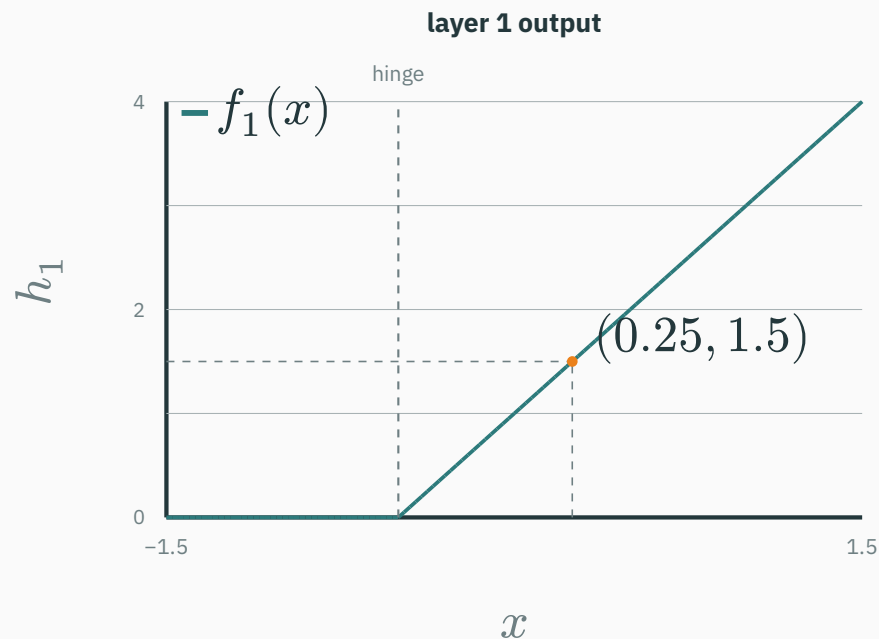
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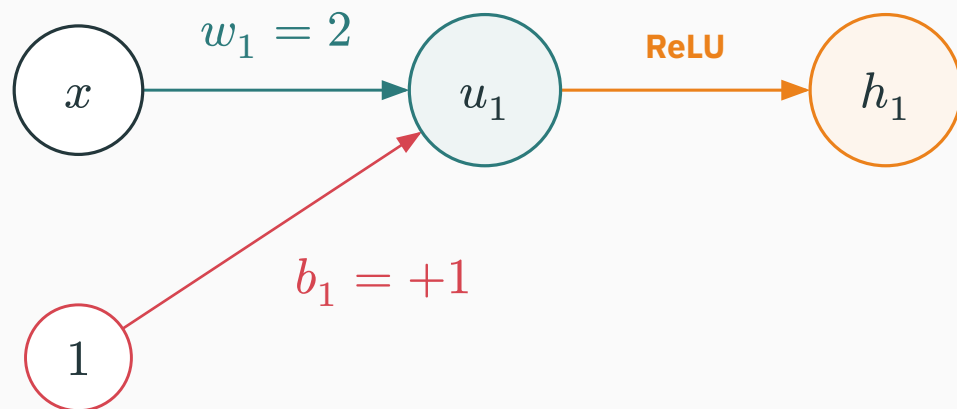
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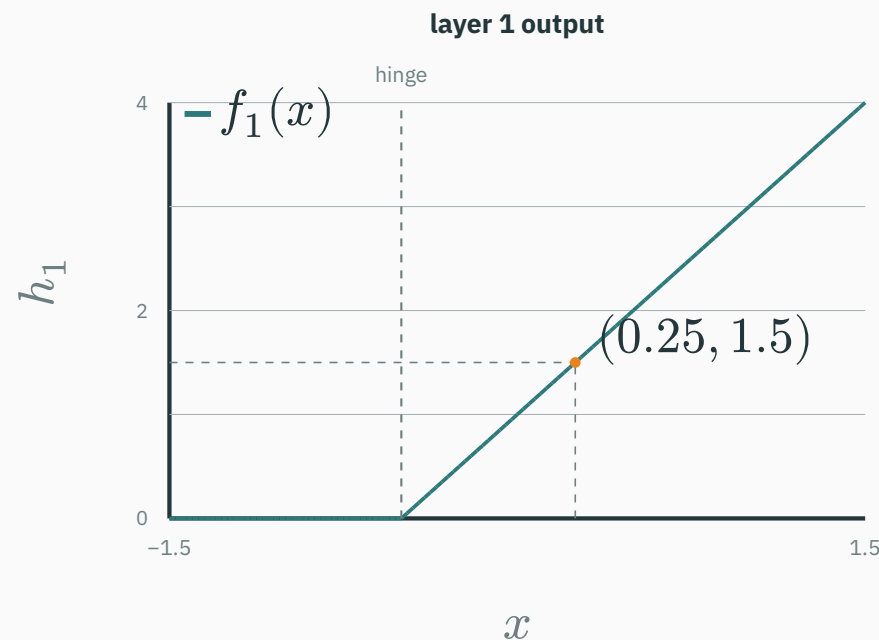
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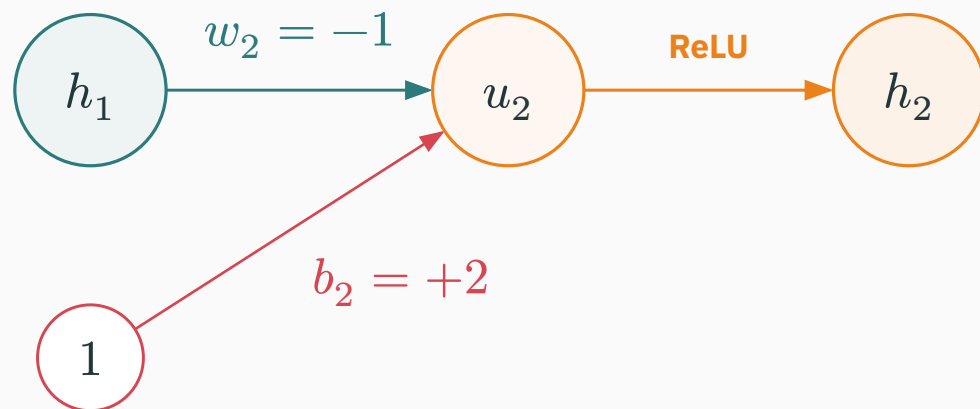
At $x = 0.25$:

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Layer 1 is flat until $2x + 1$ becomes positive; then its learned feature grows linearly.

The second layer transforms the learned feature

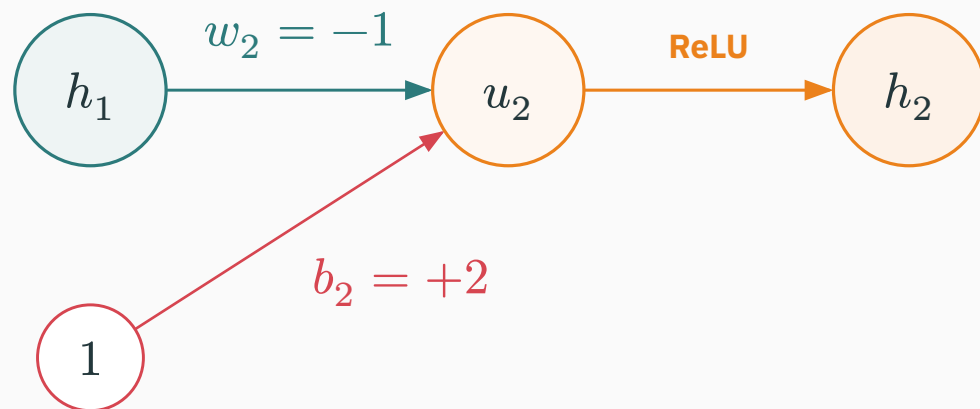


$$u_2 = 2 - h_1$$
$$h_2 = f_2(h_1) = \text{ReLU}(u_2)$$

For $h_1 = 1.5$:

$$u_2 = 0.5 \rightarrow h_2 = 0.5$$

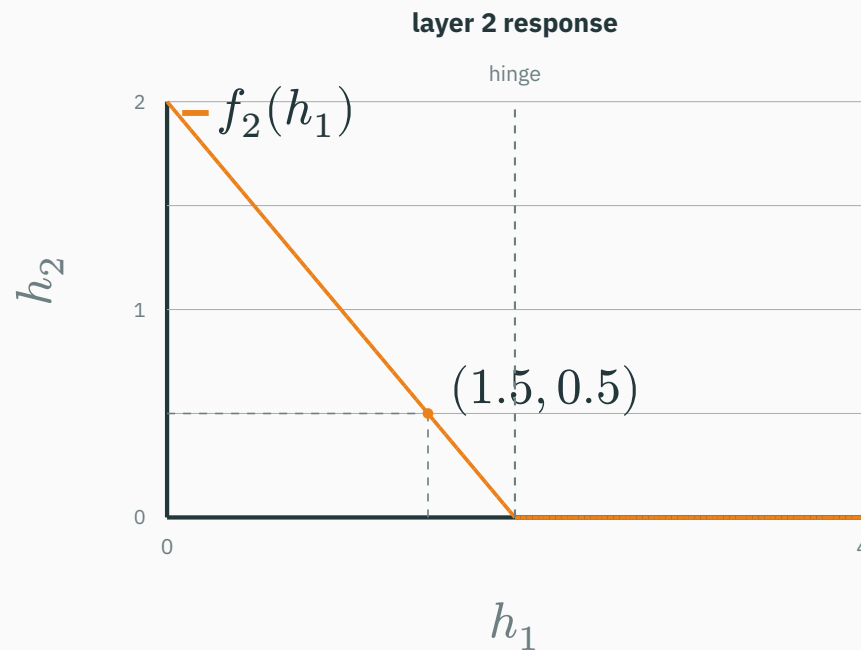
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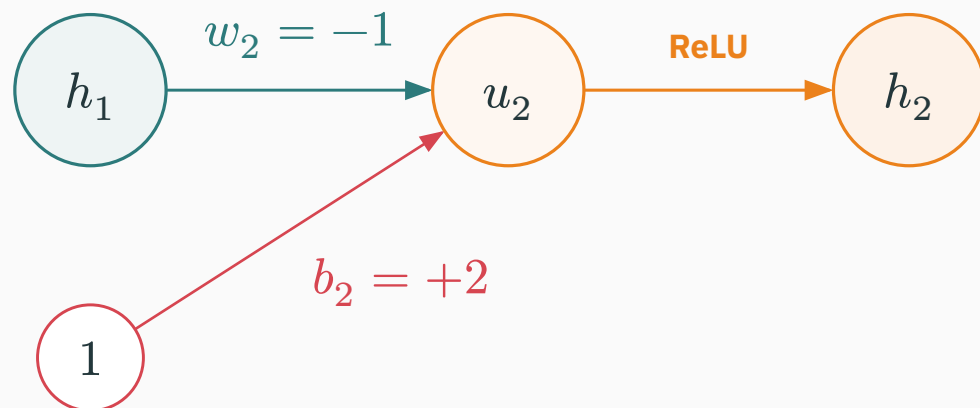
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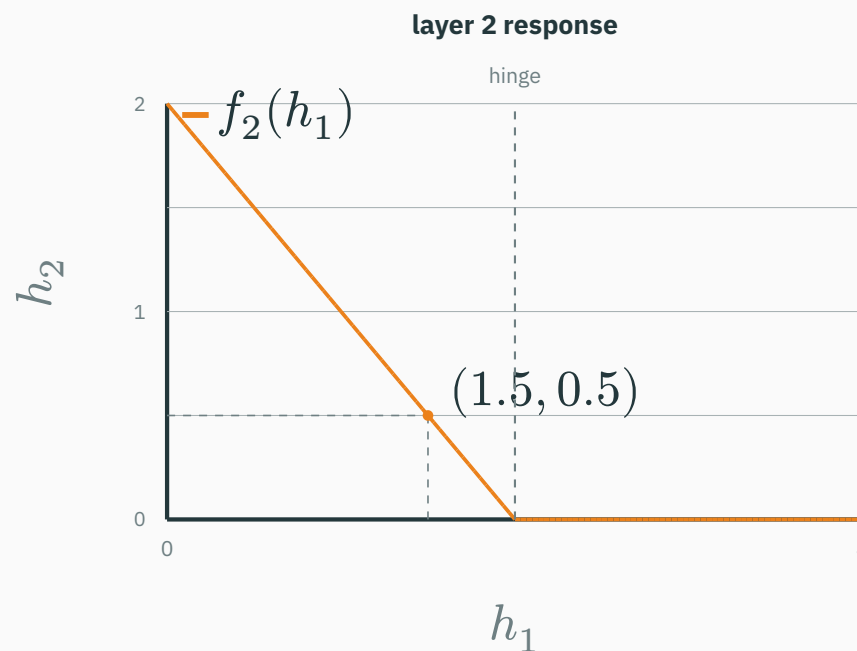
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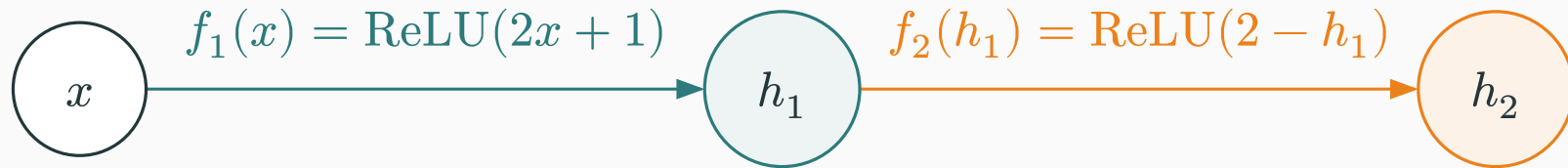
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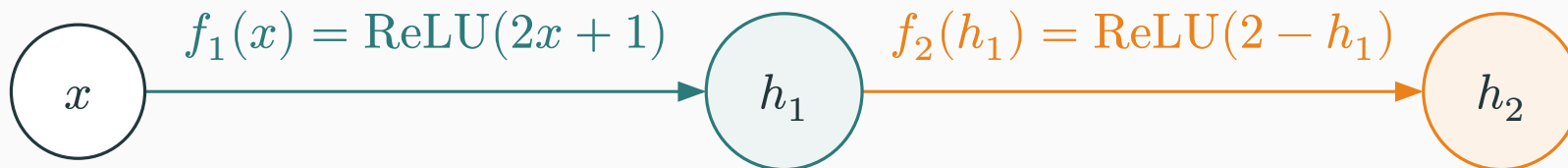
Layer 2 decreases as h_1 grows, then ReLU clips every negative result to zero.

Two layers create a clipped nonlinear ramp

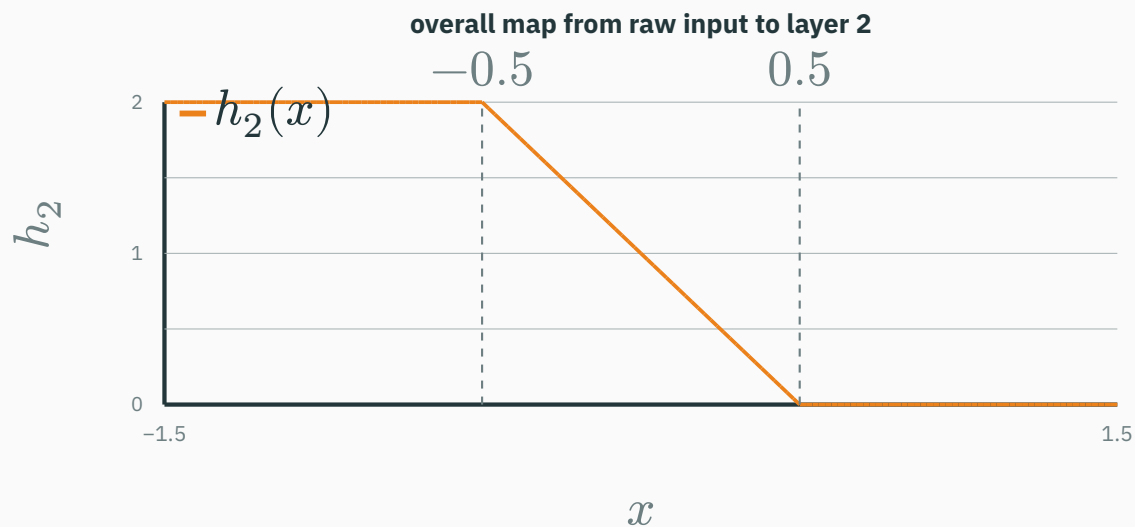


$$h_2(x) = f_2(f_1(x)) = \text{ReLU}(2 - \text{ReLU}(2x + 1))$$

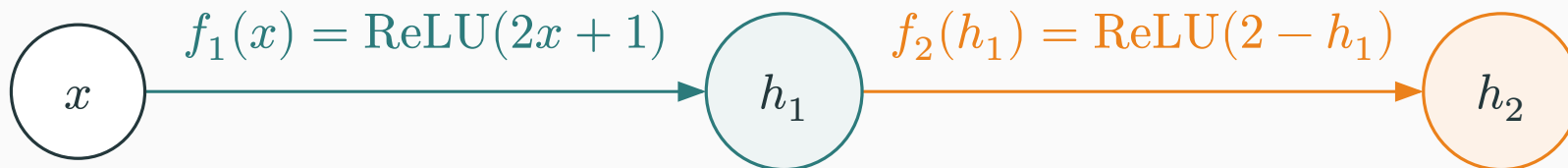
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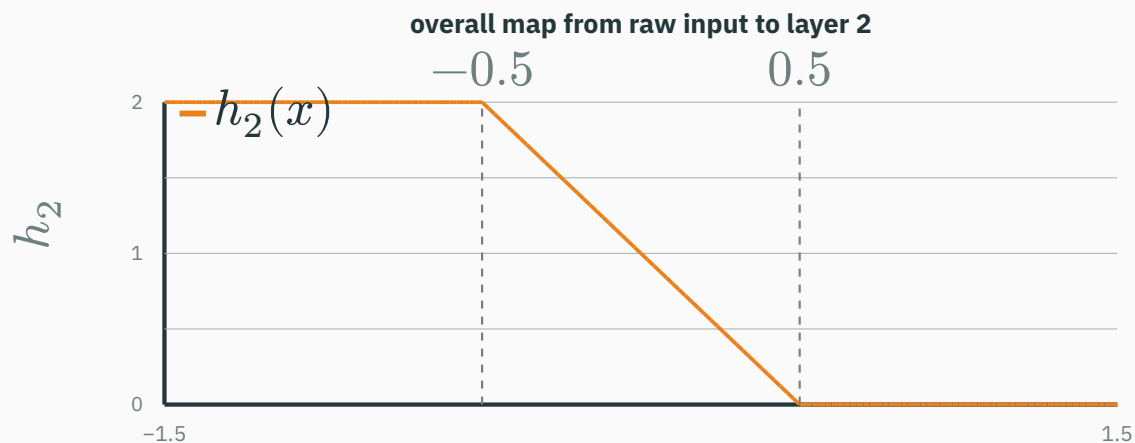
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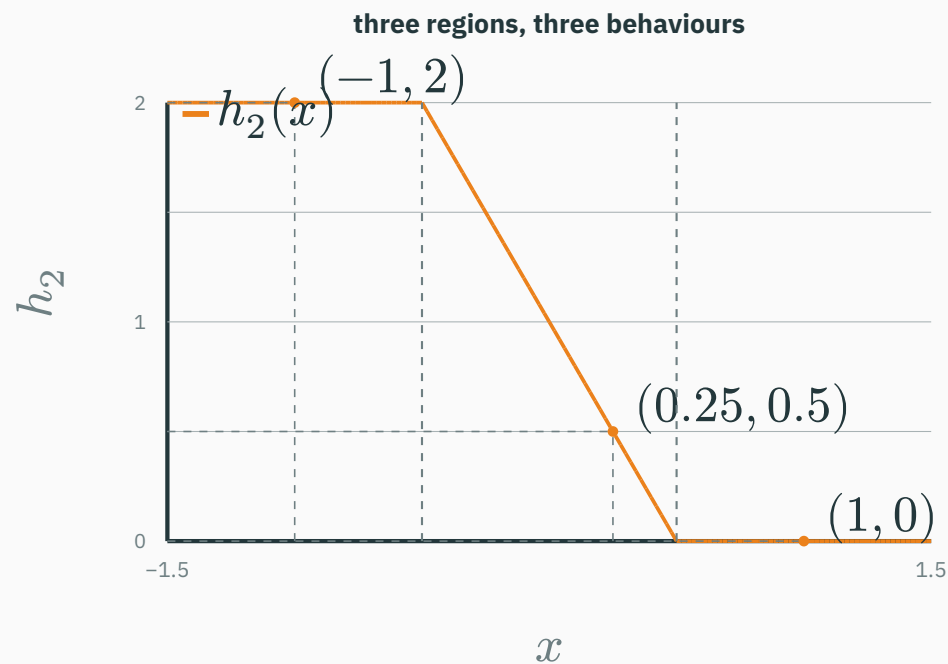


$$h_2(x) = f_2(f_1(x)) = \text{ReLU}(2 - \text{ReLU}(2x + 1))$$



The first hinge turns on at $x = -0.5$; the second layer creates another bend at $x = 0.5$.

Three inputs reveal the complete composition



$$x = -1 \quad u_1 = -1 \rightarrow h_1 = 0$$

$$u_2 = 2 \rightarrow h_2 = 2$$

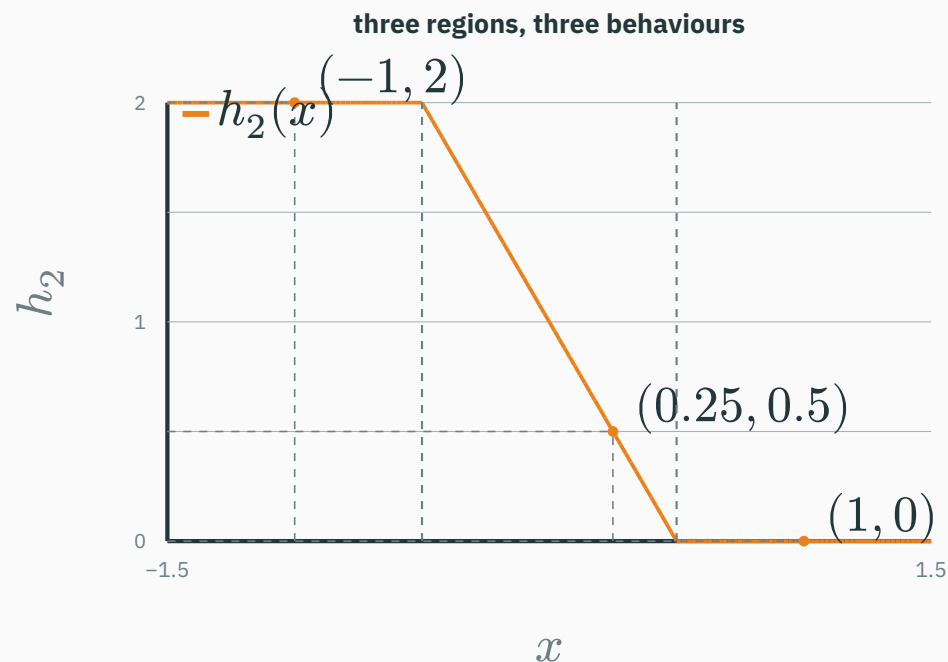
$$x = 0.25 \quad u_1 = 1.5 \rightarrow h_1 = 1.5$$

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$$x = 1 \quad u_1 = 3 \rightarrow h_1 = 3$$

$$u_2 = -1 \rightarrow h_2 = 0$$

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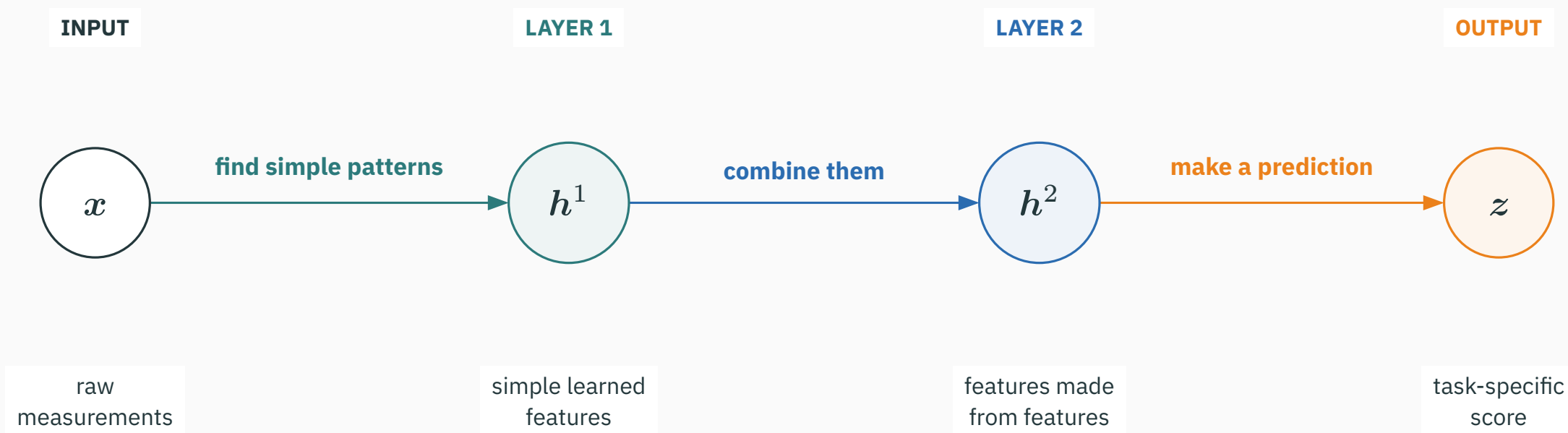
$$u_2 = -1 \rightarrow h_2 = 0$$

Depth means sequential reuse: layer 2 receives the learned value h_1 and transforms it again.

Depth builds features one stage at a time

Each layer receives the representation produced immediately before it.

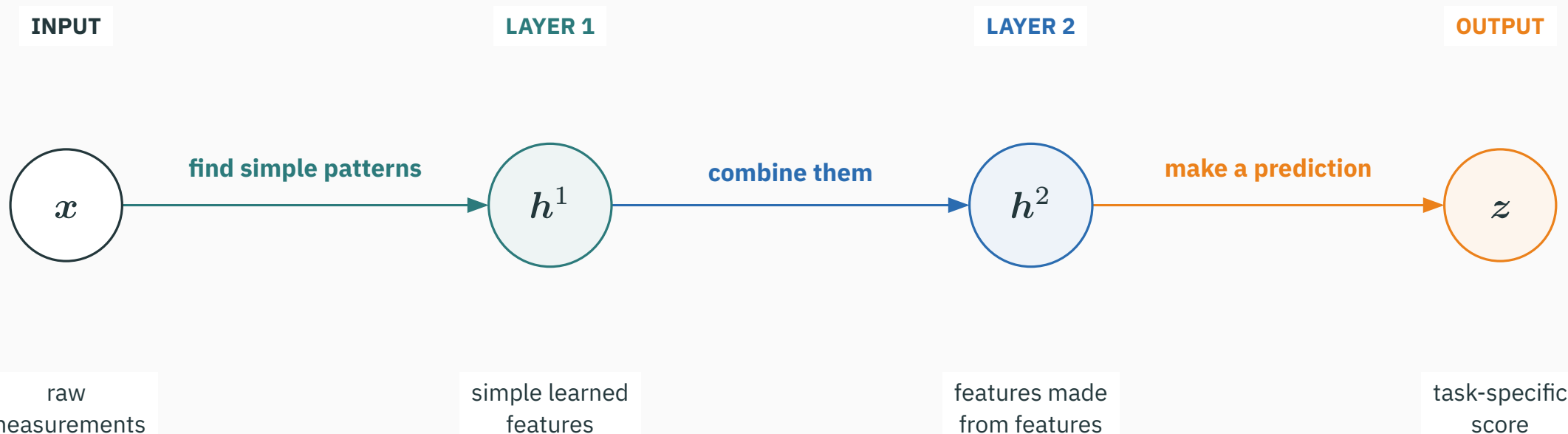
pixels → **short strokes** → **closed loops** → **digit score**



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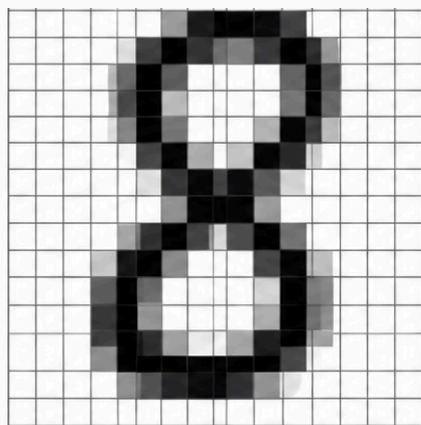
pixels → **short strokes** → **closed loops** → **digit score**



Later layers build on earlier features; they do not solve the whole task from raw input again.

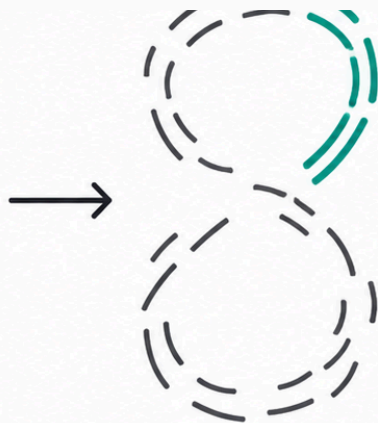
Vision: depth builds strokes, then loops

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



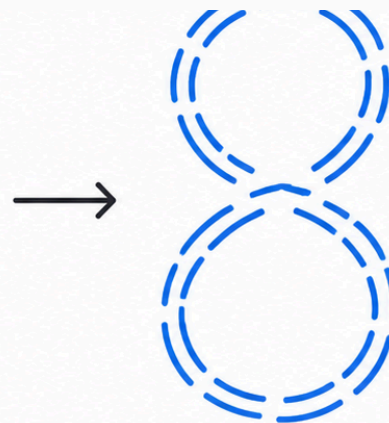
INPUT

pixel intensities



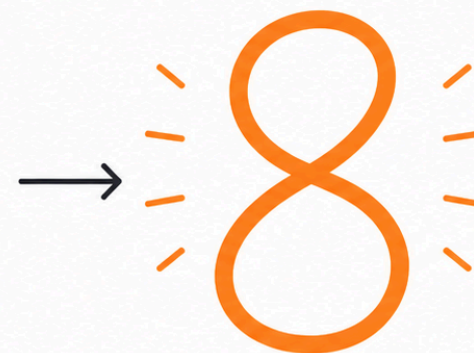
LAYER 1

curved stroke



LAYER 2

two closed loops

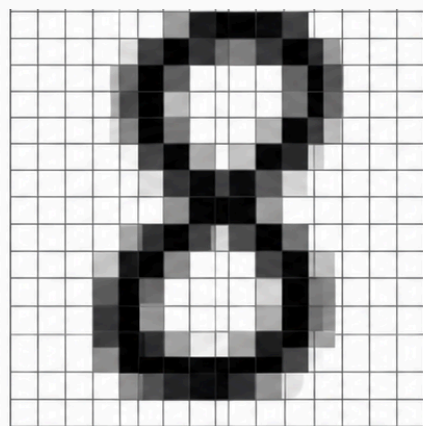


OUTPUT

evidence for 8

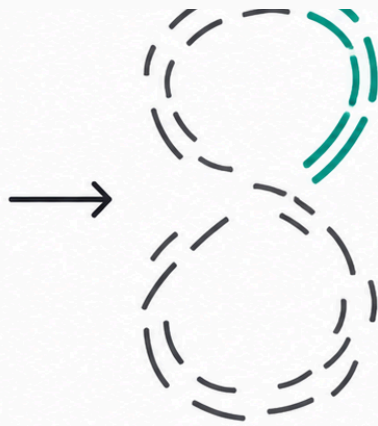
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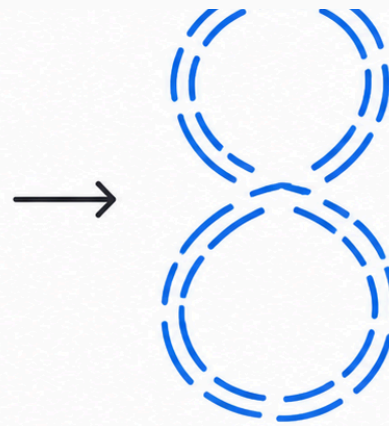
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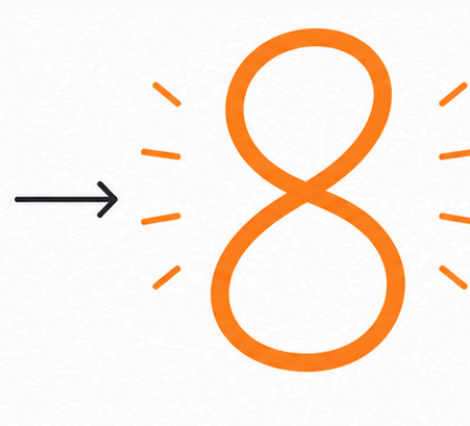
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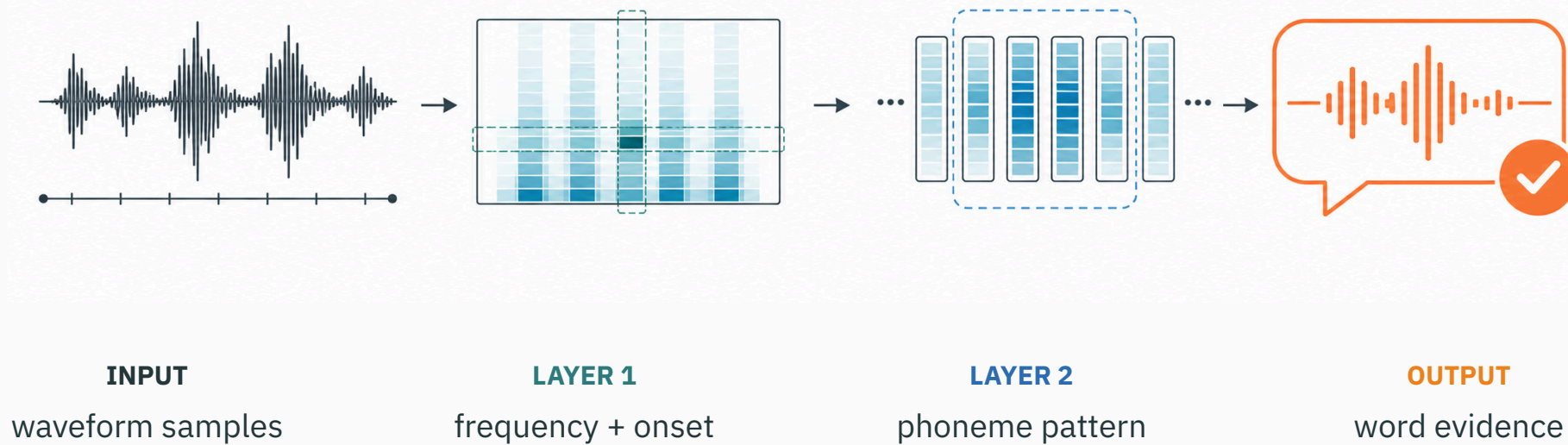
OUTPUT

evidence for 8

Early units respond to local edges; later units combine them into shapes that support the final class.

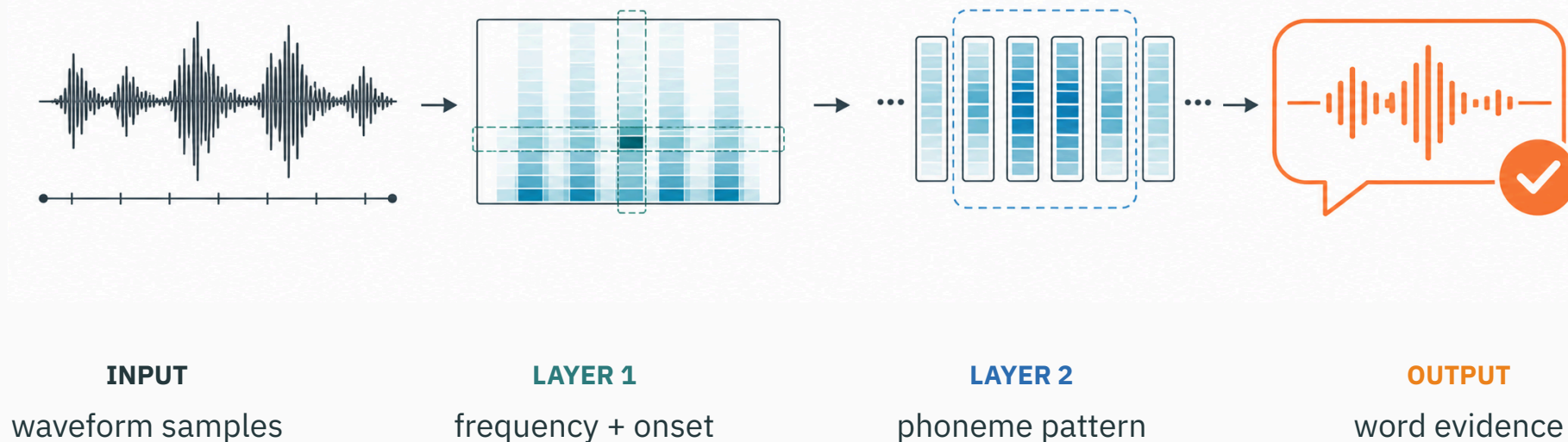
Audio: depth builds tones, then phonemes

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



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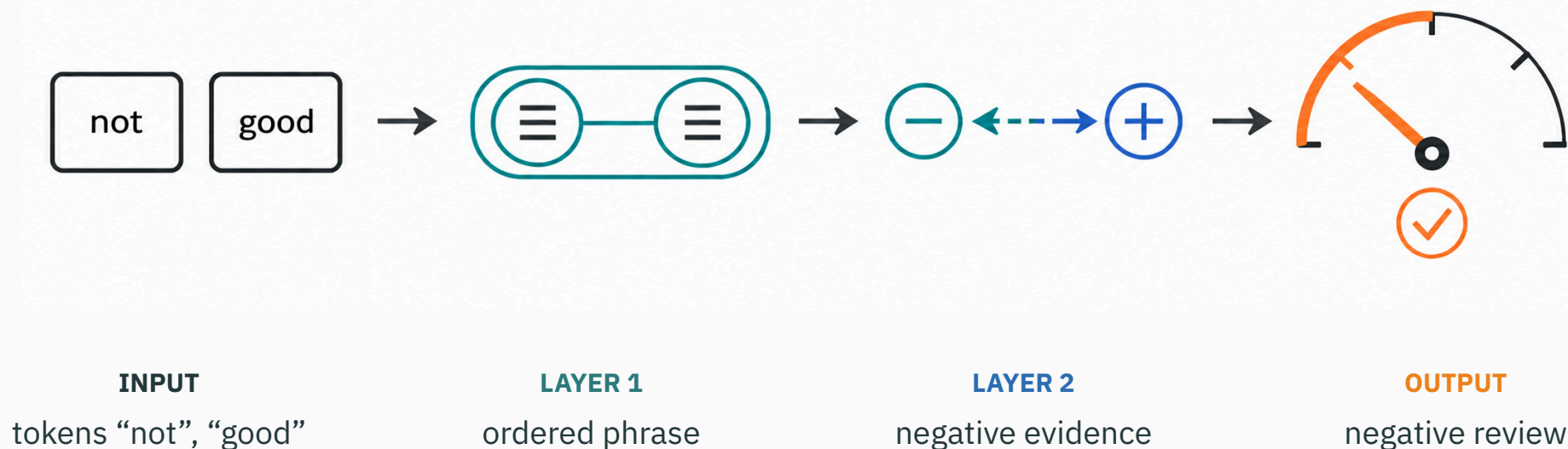
CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



Early layers detect local frequency and timing cues; later layers combine them over longer time windows.

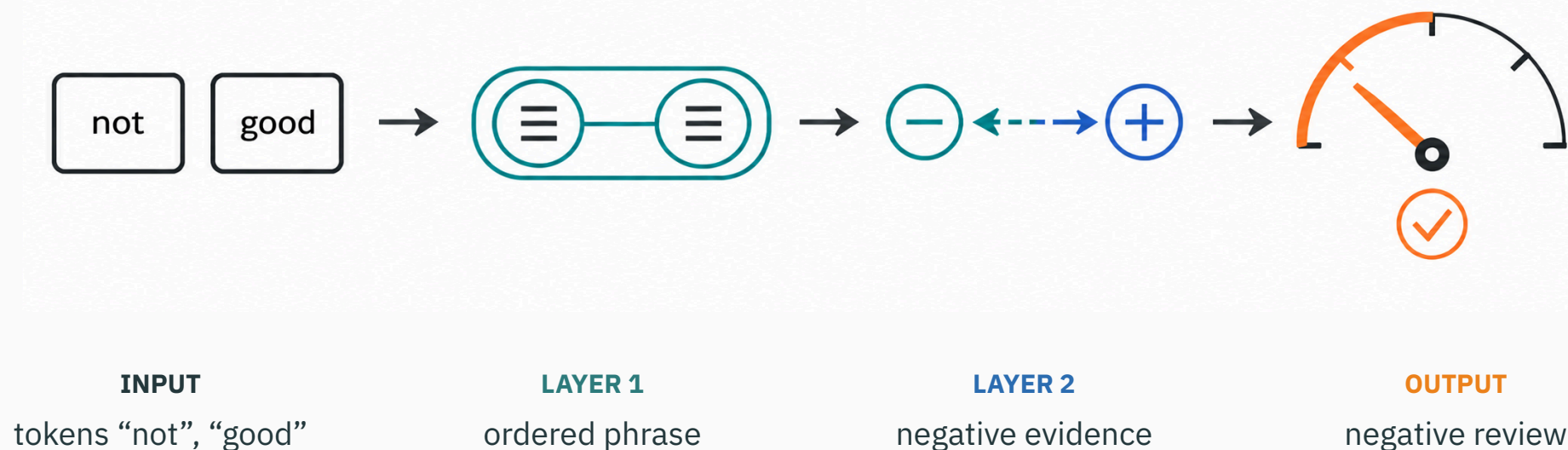
Text: depth combines words into phrase meaning

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



Text: depth combines words into phrase meaning

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



A later unit can reverse the evidence from "good" after combining it with the preceding token "not".

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- adds more features **in parallel**;
- helps when many patterns live at the same abstraction level;
- costs wider activations and weight matrices.

Example: add hinge locations to follow a detailed curve.

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There is no universally best shape. Compare models under a meaningful parameter, memory, and compute budget.

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Example: strokes → loops → digit identity.

Modern networks usually need both: enough width per stage and enough depth to compose stages.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/mlp-decision-boundary

Train a linear classifier or a small MLP on **XOR / circles / spirals / moons**.

Compare one versus two hidden layers, then adjust width while watching the learned boundary. Controls: dataset · linear/MLP · hidden width · one/two layers · activation · train/reset/resample.

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Hold the dataset fixed. First add width within one layer; then redistribute a similar number of hidden units across two layers. Which boundary appears sooner and which trains more reliably?

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The comparison is empirical: width and depth change both representational structure and optimization behaviour.

Connecting to deep-learning practice

Forward pass, step 1: flatten the image into a vector

Start with a 2×2 toy image so that every number stays visible.

PIXEL GRID

$$\mathbf{X}_{\text{image}} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

four pixel intensities

read row by row $\rightarrow$

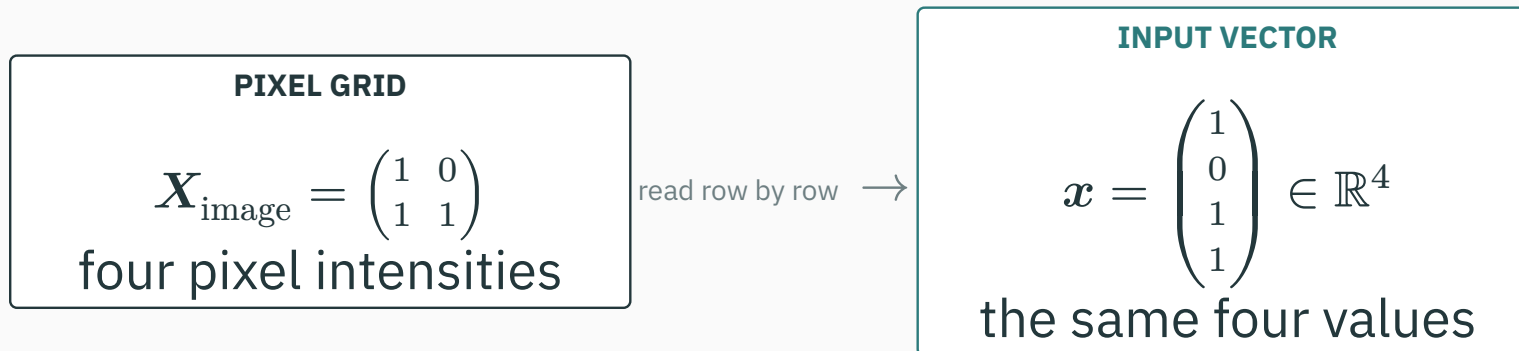
INPUT VECTOR

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \in \mathbb{R}^4$$

the same four values

Forward pass, step 1: flatten the image into a vector

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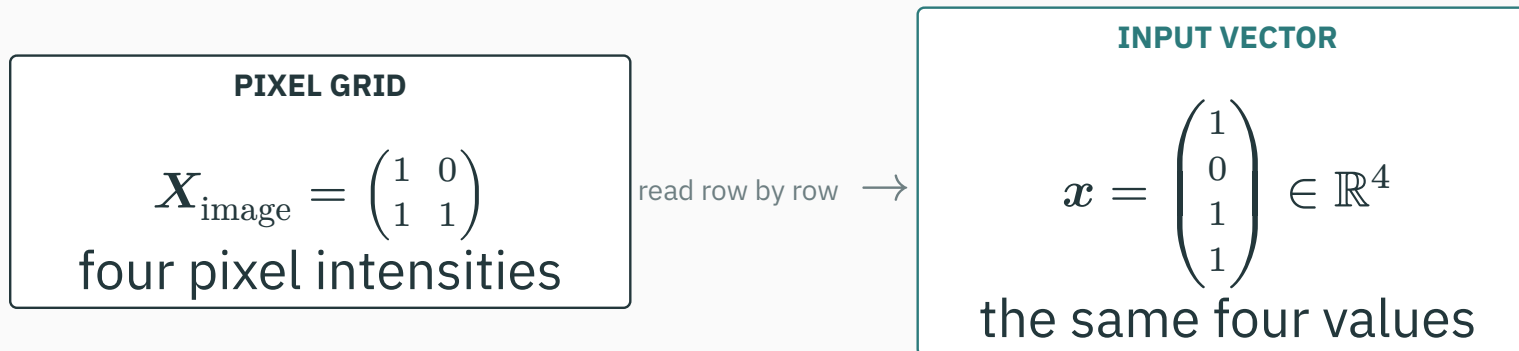


REAL IMAGE

a 28×28 image becomes $x \in \mathbb{R}^{784}$ because
 $28(28) = 784$

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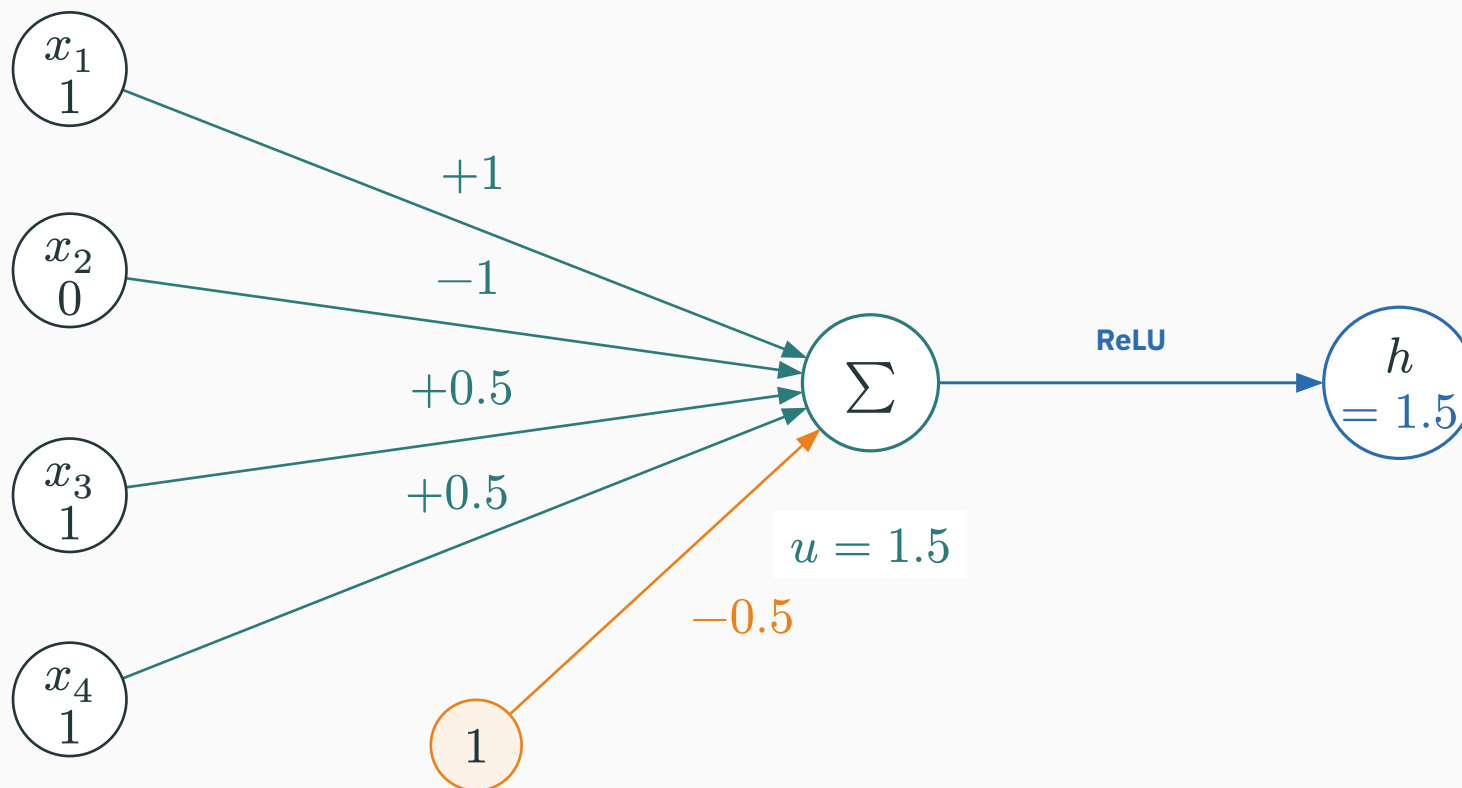
REAL IMAGE

a 28×28 image becomes $x \in \mathbb{R}^{784}$ because
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Flattening changes the arrangement of the pixels—not their values or meaning.

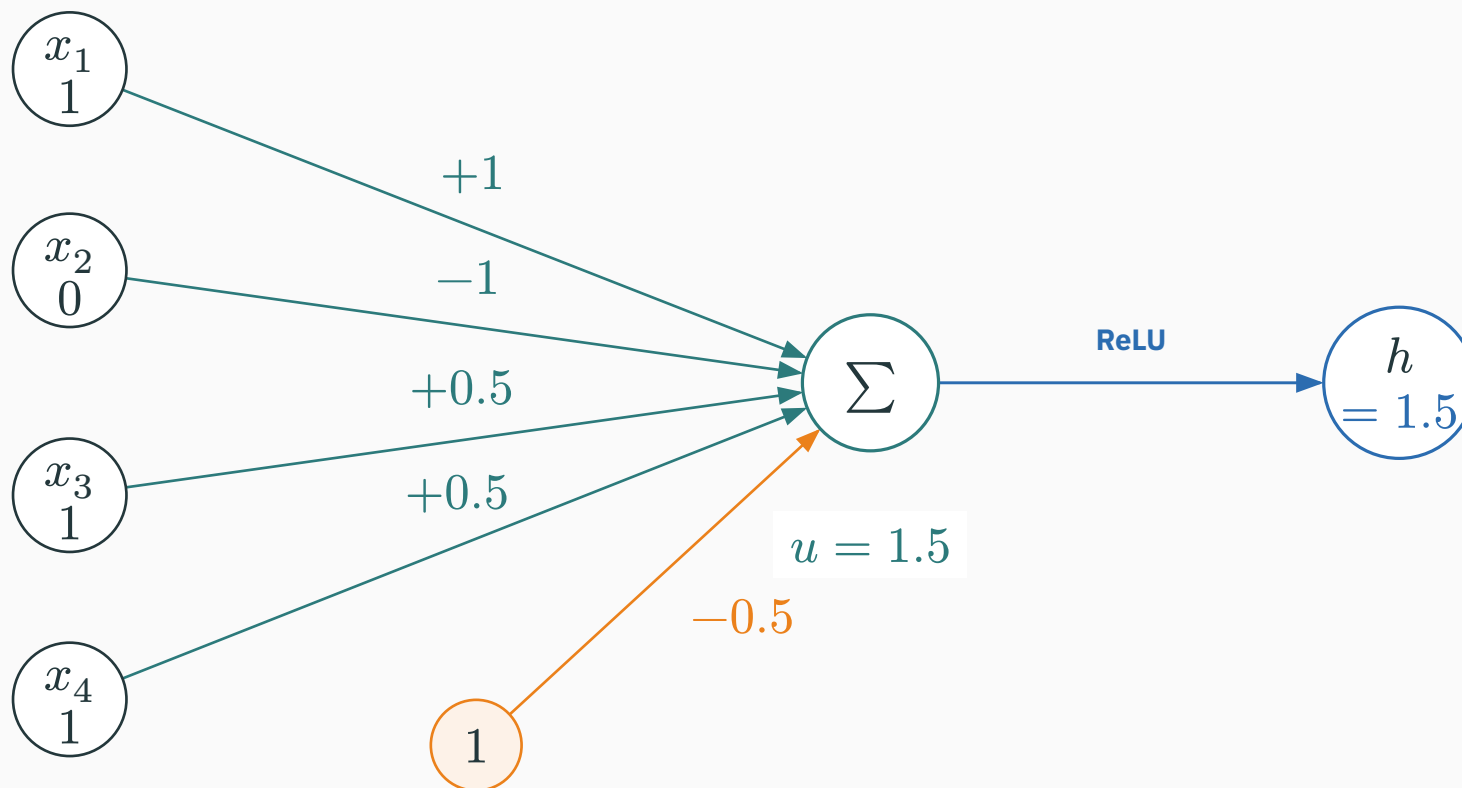
Forward pass, step 2: one hidden unit tests one pattern

D DERIVATION



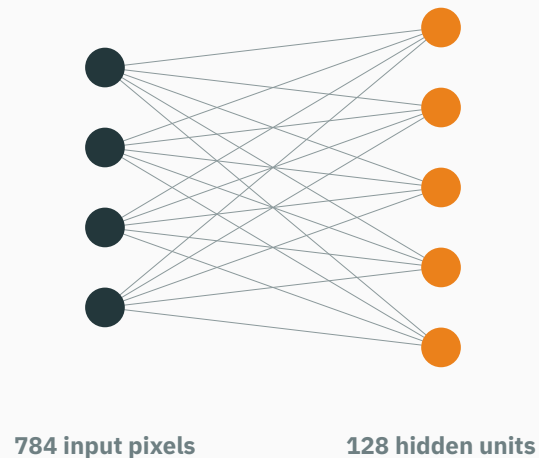
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D DERIVATION



$u = 1(1) - 1(0) + 0.5(1) + 0.5(1) - 0.5 = 1.5$, then $h = \max(0, u) = 1.5$. The edge weights define the pixel pattern this feature detects.

Forward pass, step 3: 128 units test 128 patterns



Each receiving unit has its own row of weights:

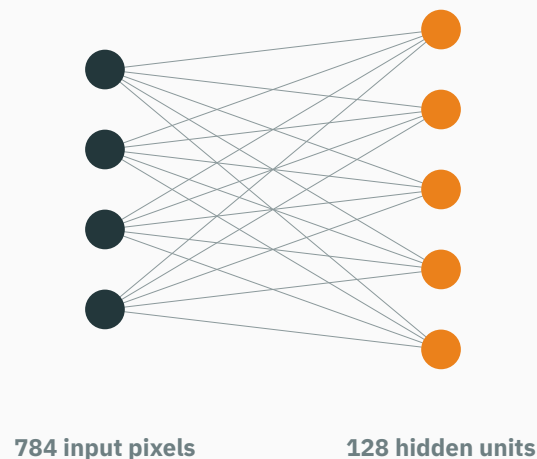
$$u_j = \mathbf{w}_j^\top \mathbf{x} + b_j$$
$$h_j = \max(0, u_j)$$

$$\mathbf{u} = \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1 \in \mathbb{R}^{128}$$

$$\mathbf{h} = \text{ReLU}(\mathbf{u}) \in \mathbb{R}^{128}$$

schematic: only a few nodes are drawn

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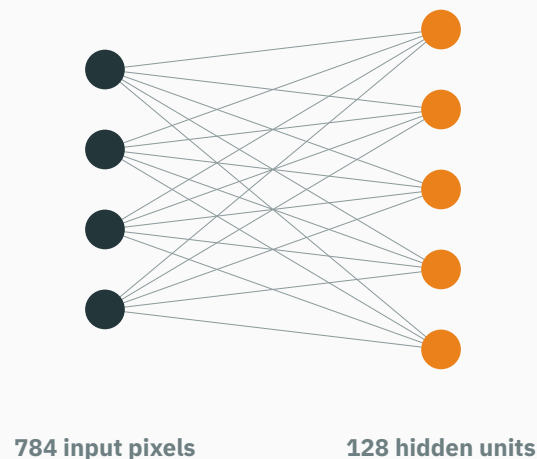
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For one trained layer: $h_{17} = 2.1 \rightarrow$ curved-stroke unit **on**; $h_{83} = 0 \rightarrow$ another unit **off**.

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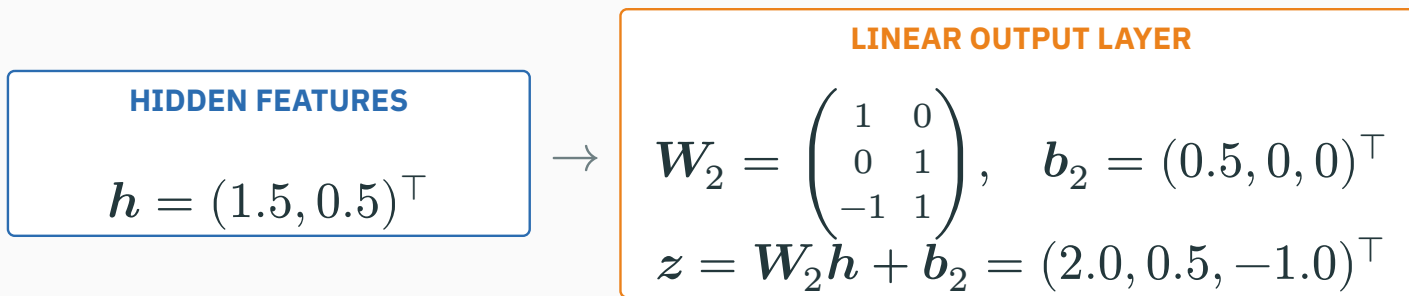
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The hidden vector is a list of 128 learned measurements of the same image.

Forward pass, step 4: hidden features become class scores

Use a three-class toy output so that every logit can be computed explicitly.



Forward pass, step 4: hidden features become class scores

Use a three-class toy output so that every logit can be computed explicitly.

HIDDEN FEATURES

$$\mathbf{h} = (1.5, 0.5)^\top$$

$\rightarrow$

LINEAR OUTPUT LAYER

$$\mathbf{W}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{b}_2 = (0.5, 0, 0)^\top$$
$$\mathbf{z} = \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2 = (2.0, 0.5, -1.0)^\top$$

$$z_A = 1(1.5) + 0(0.5) + 0.5 = 2.0$$

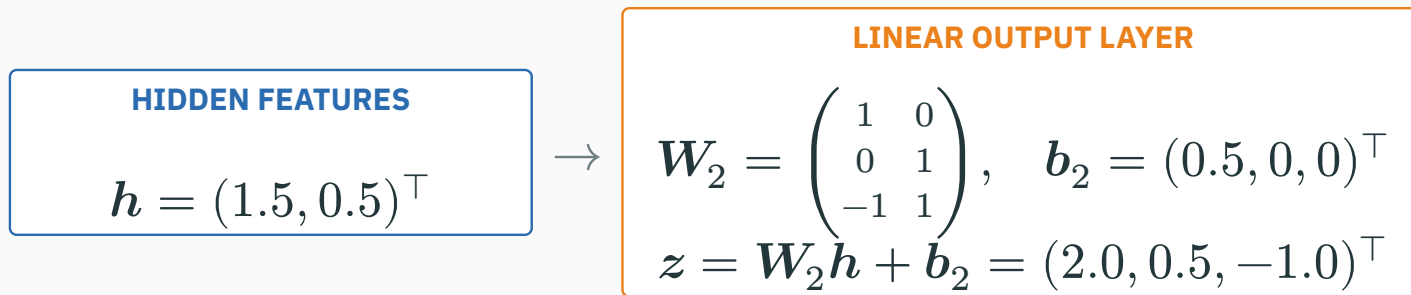
$$z_B = 0(1.5) + 1(0.5) + 0 = 0.5$$

$$z_C = -1(1.5) + 1(0.5) + 0 = -1.0$$

Forward pass, step 4: hidden features become class scores

D DERIVATION

Use a three-class toy output so that every logit can be computed explicitly.



$$z_A = 1(1.5) + 0(0.5) + 0.5 = 2.0$$

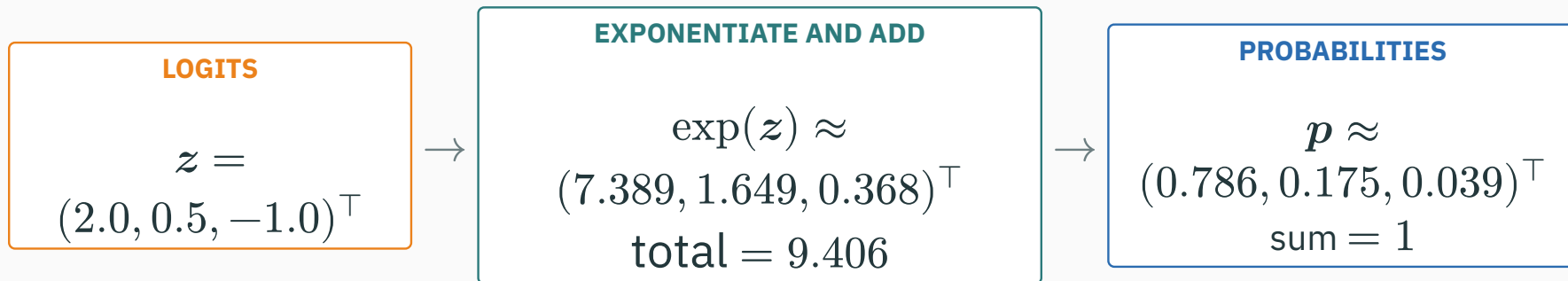
$$z_B = 0(1.5) + 1(0.5) + 0 = 0.5$$

$$z_C = -1(1.5) + 1(0.5) + 0 = -1.0$$

Logits are unrestricted class scores: larger means preferred, but they are not probabilities yet.

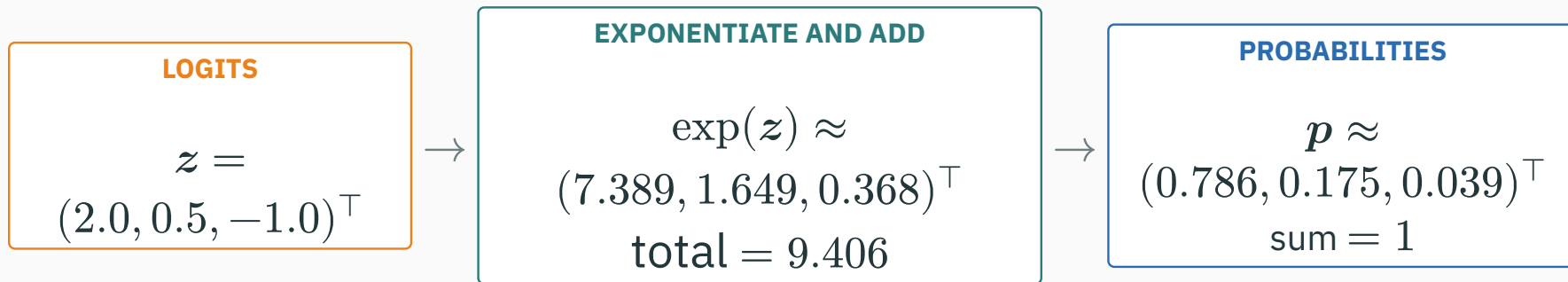
Forward pass, step 5: softmax compares every score

D DERIVATION



Forward pass, step 5: softmax compares every score

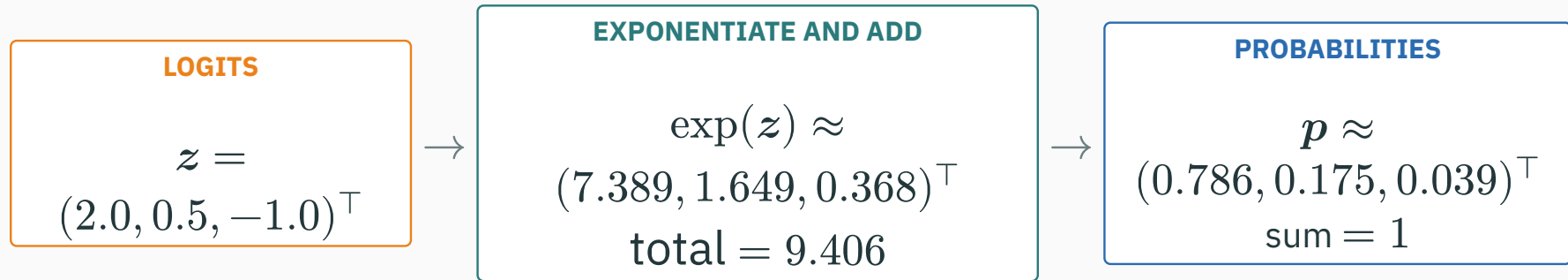
D DERIVATION



$$p_A = \frac{\exp(2.0)}{\exp(2.0) + \exp(0.5) + \exp(-1.0)} = \frac{7.389}{9.406} \approx 0.786$$

Forward pass, step 5: softmax compares every score

D DERIVATION



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Softmax preserves the score ordering while turning all three scores into one probability distribution.

Forward pass, step 6: the label selects the loss

Cross-entropy reads the probability assigned to the true class: $\mathcal{L} = -\log p_y$.

IF $Y = A$

$$\mathcal{L} = -\log(0.786) \approx 0.241$$

IF $Y = B$

$$\mathcal{L} = -\log(0.175) \approx 1.743$$

IF $Y = C$

$$\mathcal{L} = -\log(0.039) \approx 3.244$$

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Confident probability on the wrong class gives a large loss.

The forward pass ends with one scalar loss; it has evaluated the weights but has not updated them.

A minibatch stacks examples as rows

Apply the same hidden unit from step 2 to three inputs at once.

THREE INPUT ROWS

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

shape 3×4

→

ONE FEATURE ACROSS THE BATCH

$$\begin{aligned} \mathbf{u} &= \mathbf{X}\mathbf{w} + b\mathbf{1} = (1.5, -1, -0.5)^\top \\ \mathbf{h} &= \text{ReLU}(\mathbf{u}) = (1.5, 0, 0)^\top \end{aligned}$$

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The same $\mathbf{w}$ and b are reused for row 1, row 2, and row 3.

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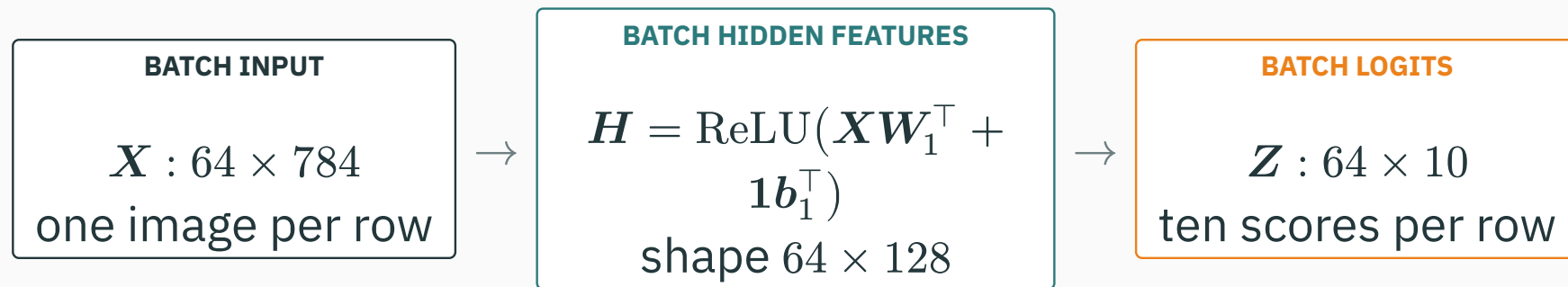
THREE INPUT ROWS $\mathbf{X} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$ shape 3×4	$\rightarrow$	ONE FEATURE ACROSS THE BATCH $\mathbf{u} = \mathbf{X}\mathbf{w} + b\mathbf{1} = (1.5, -1, -0.5)^\top$ $\mathbf{h} = \text{ReLU}(\mathbf{u}) = (1.5, 0, 0)^\top$
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The same $\mathbf{w}$ and b are reused for row 1, row 2, and row 3.

Rows are different examples; columns are the features computed for those examples.

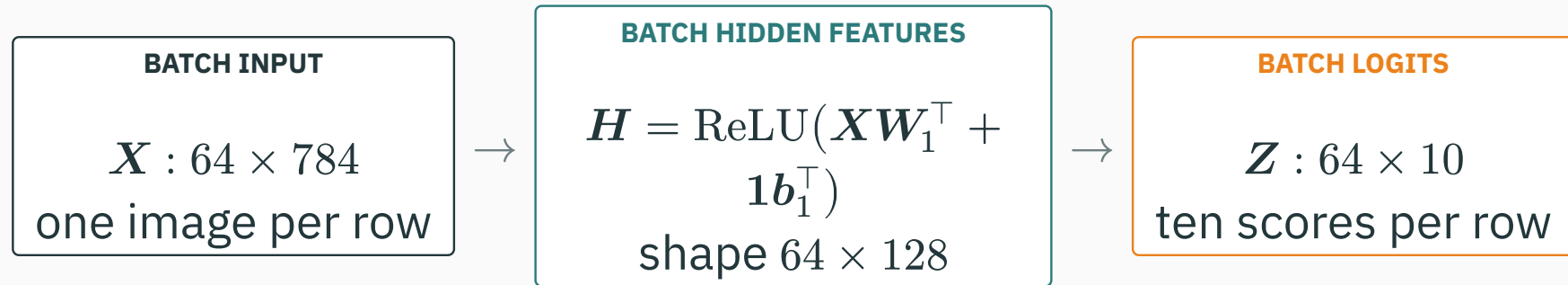
The full minibatch keeps the same network

For $n = 64$ images, all 128 hidden units and all 10 logits are computed together:



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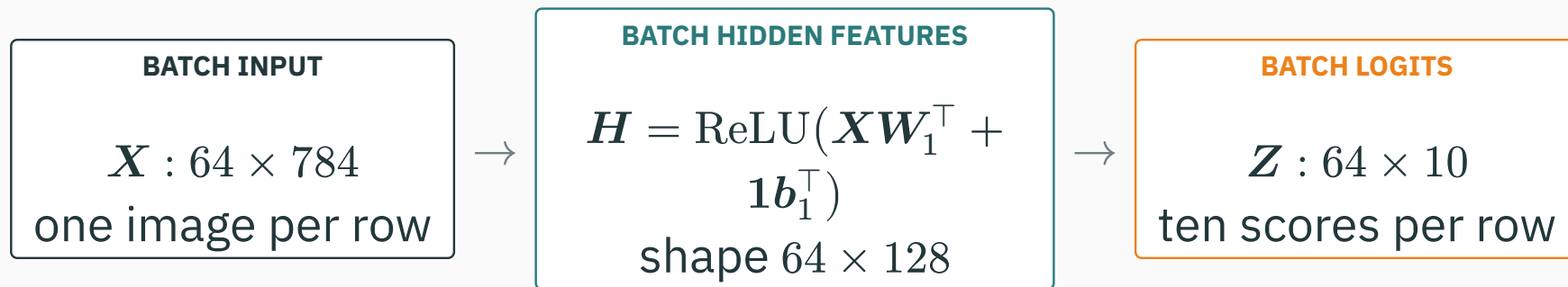


Shared across rows: weights and biases.

Recomputed for this batch: activations, probabilities, and losses.

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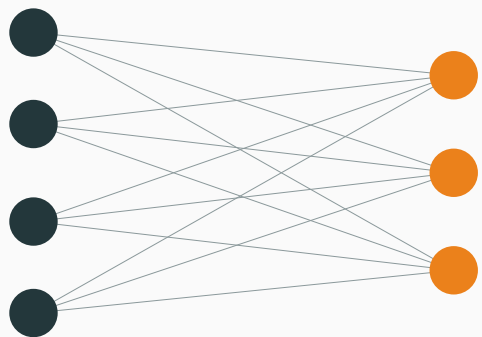


Shared across rows: weights and biases.

Recomputed for this batch: activations, probabilities, and losses.

Batching changes the array shapes, not the prediction rule: every row follows the same network.

A tiny layer reveals the parameter-count rule



4 incoming values

3 receiving units

every connecting edge carries one weight

Weights

$4 \text{ inputs} \times 3 \text{ units} = 12$

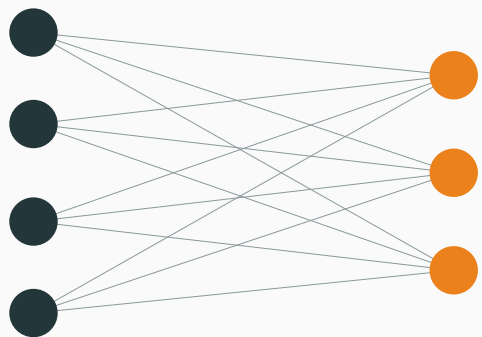
Biases

one per receiving unit = 3

Total

$12 + 3 = 15$ parameters

A tiny layer reveals the parameter-count rule



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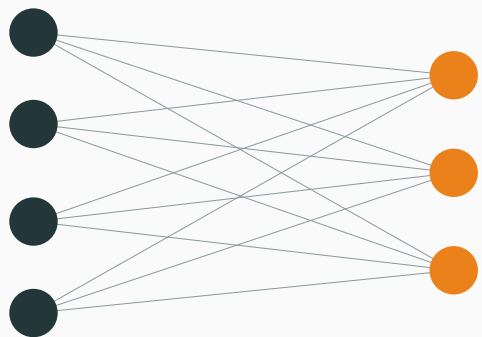
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GENERAL RULE

$$(\text{output width} \times \text{input width}) + \text{output width}$$

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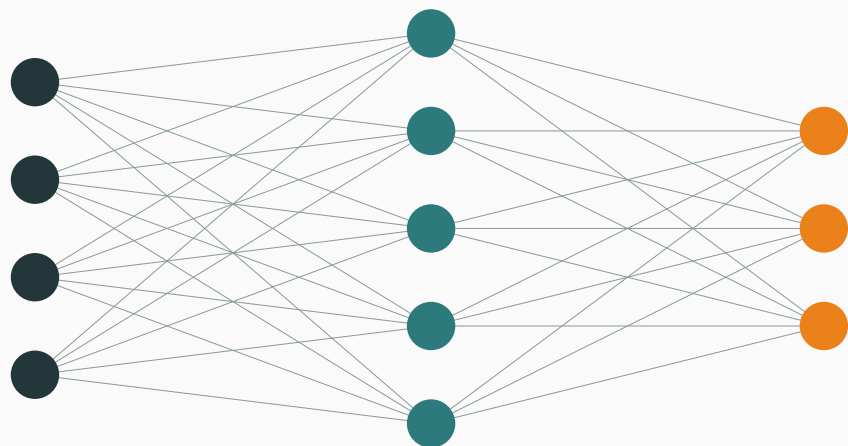
$$12 + 3 = 15 \text{ parameters}$$

GENERAL RULE

$$(\text{output width} \times \text{input width}) + \text{output width}$$

Count the arrows, then add one bias for every receiving node.

Apply the same counting rule to both affine layers



784 inputs

128 hidden units

10 outputs

The drawing is schematic; the labels give the actual widths.

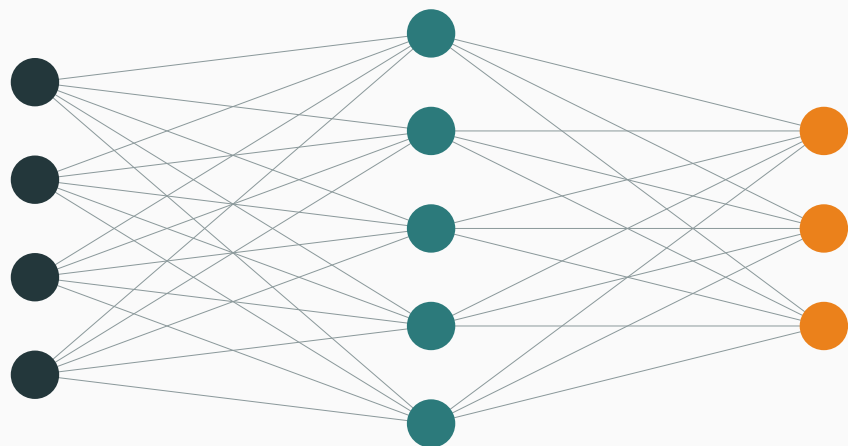
INPUT $\rightarrow$ HIDDEN

$$W_1 : 128 \times 784 \rightarrow 128(784) = 100,352$$

$$b_1 \in \mathbb{R}^{128} \rightarrow 128$$

Subtotal: $100,352 + 128 = 100,480$

Apply the same counting rule to both affine layers



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HIDDEN → OUTPUT

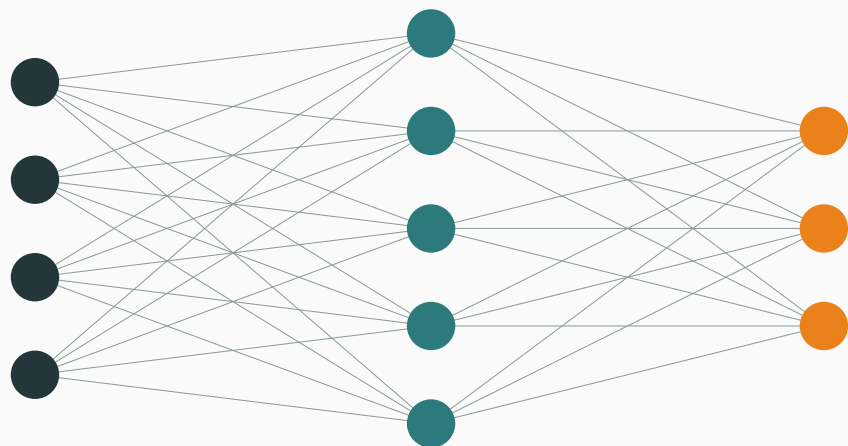
$$W_2 : 10 \times 128 \rightarrow 10(128) = 1,280$$

$$b_2 \in \mathbb{R}^{10} \rightarrow 10$$

$$\text{Subtotal: } 1,280 + 10 = 1,290$$

Apply the same counting rule to both affine layers

D DERIVATION



784 inputs

128 hidden units

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The drawing is schematic; the labels give the actual widths.

INPUT $\rightarrow$ HIDDEN

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HIDDEN $\rightarrow$ OUTPUT

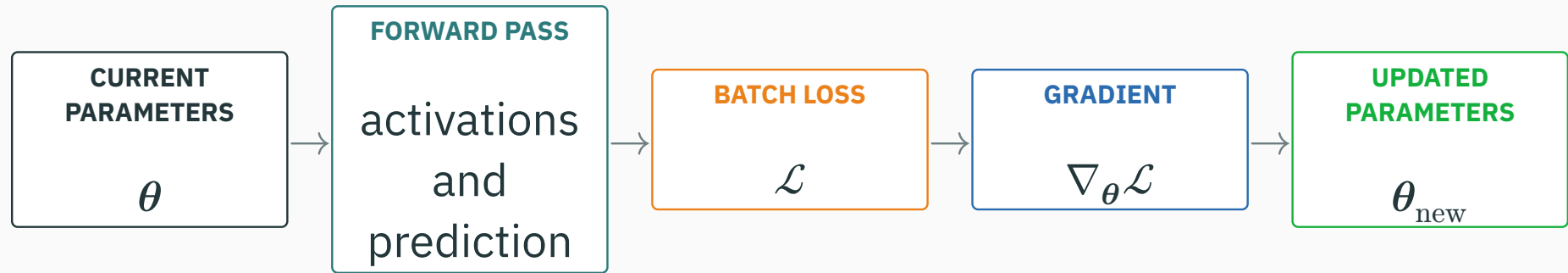
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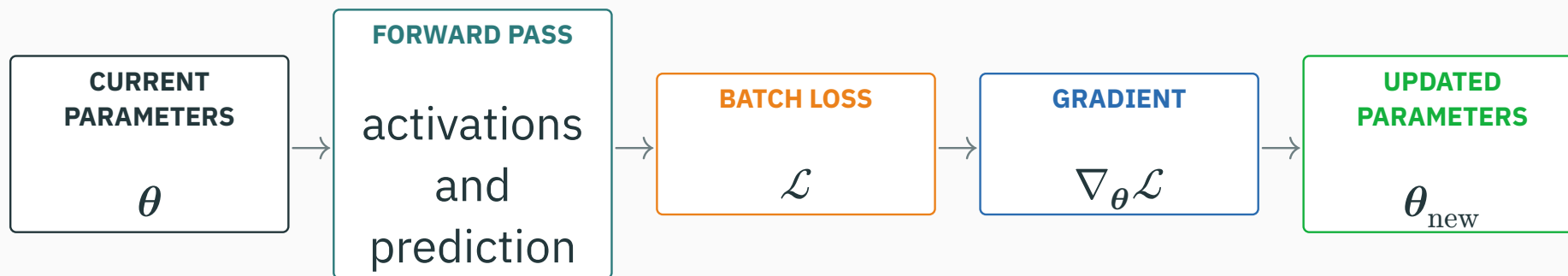
$$\text{Subtotal: } 1,280 + 10 = 1,290$$

Total trainable parameters: $100,480 + 1,290 = 101,770$.

Training changes parameters, not the architecture

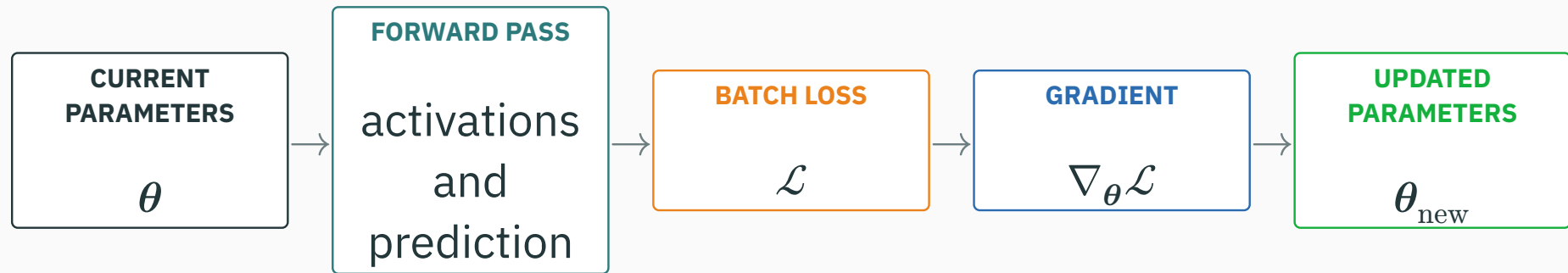


Training changes parameters, not the architecture



$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_{\theta} \mathcal{L}$$

Training changes parameters, not the architecture

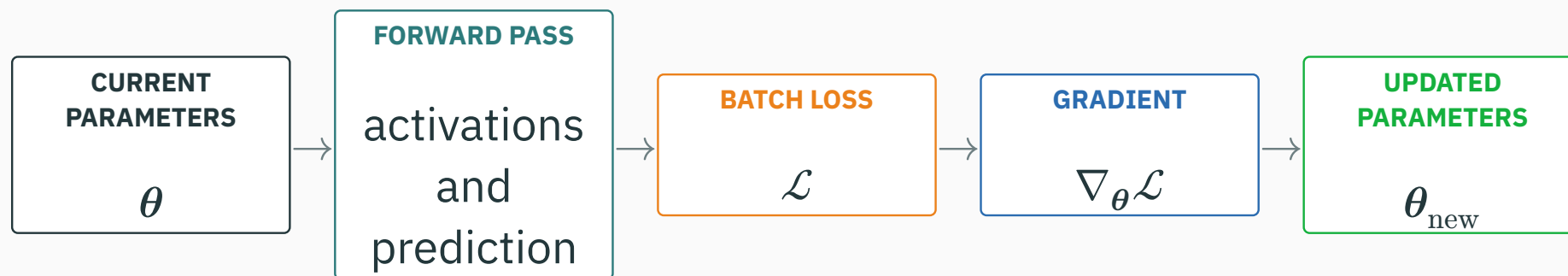


$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_{\theta} \mathcal{L}$$

Fixed: layer sizes, connectivity pattern, and activation choices.

Changed: the entries of weight matrices and bias vectors.

Training changes parameters, not the architecture



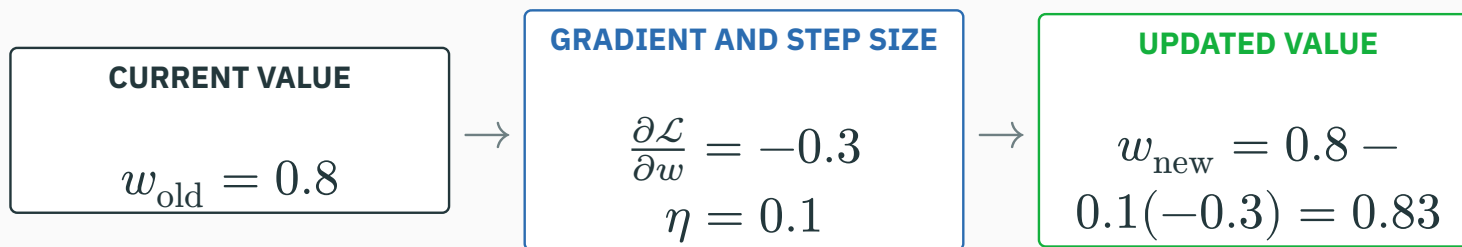
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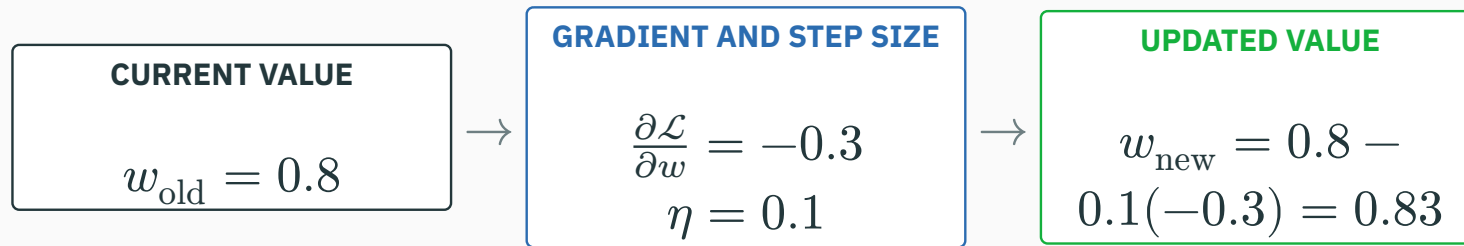
Changed: the entries of weight matrices and bias vectors.

Forward propagation produces activations and loss; learning uses that loss to update the shared parameters.

One numerical parameter update

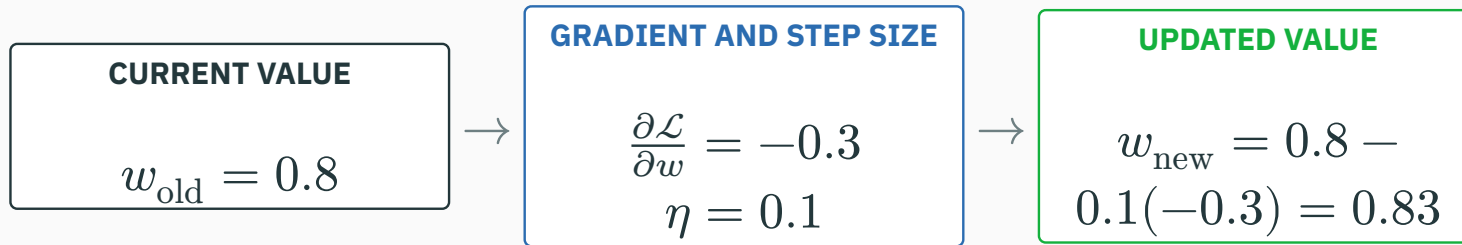


One numerical parameter update



The negative gradient says increasing w locally decreases the loss, so the update moves w upward.

One numerical parameter update

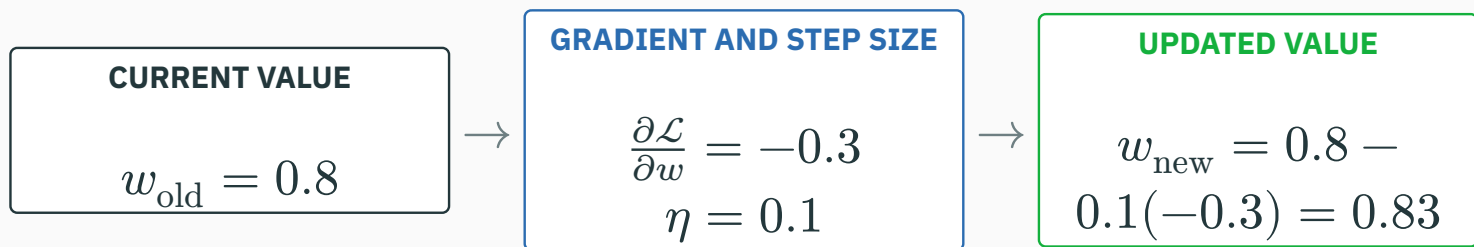


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Lecture 3 will explain how backpropagation computes every component of $\nabla_{\theta} \mathcal{L}$ efficiently.

One numerical parameter update

D DERIVATION

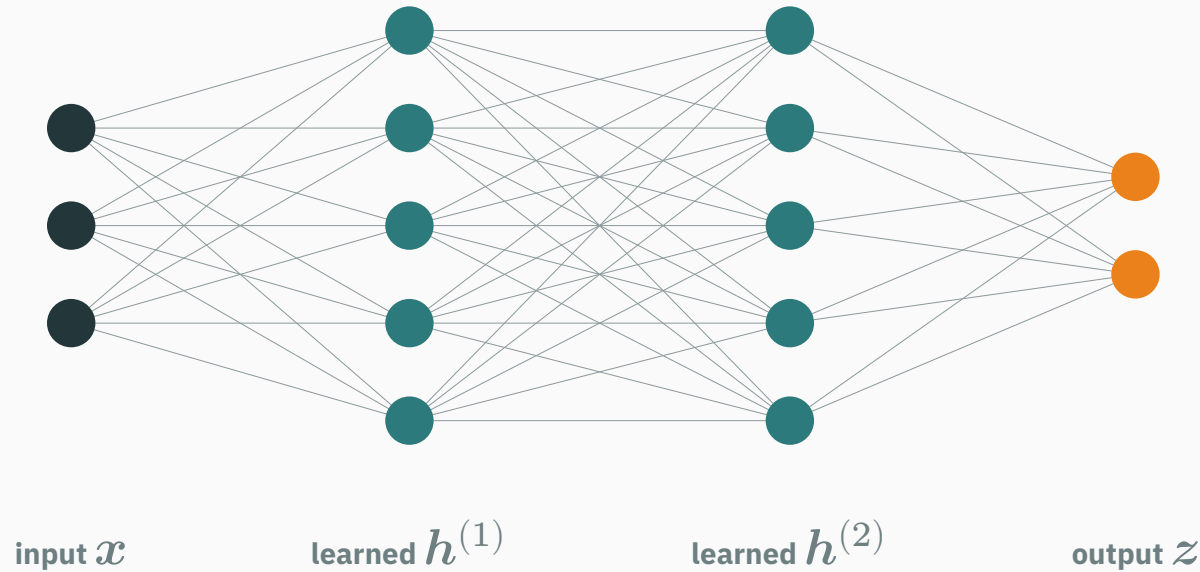


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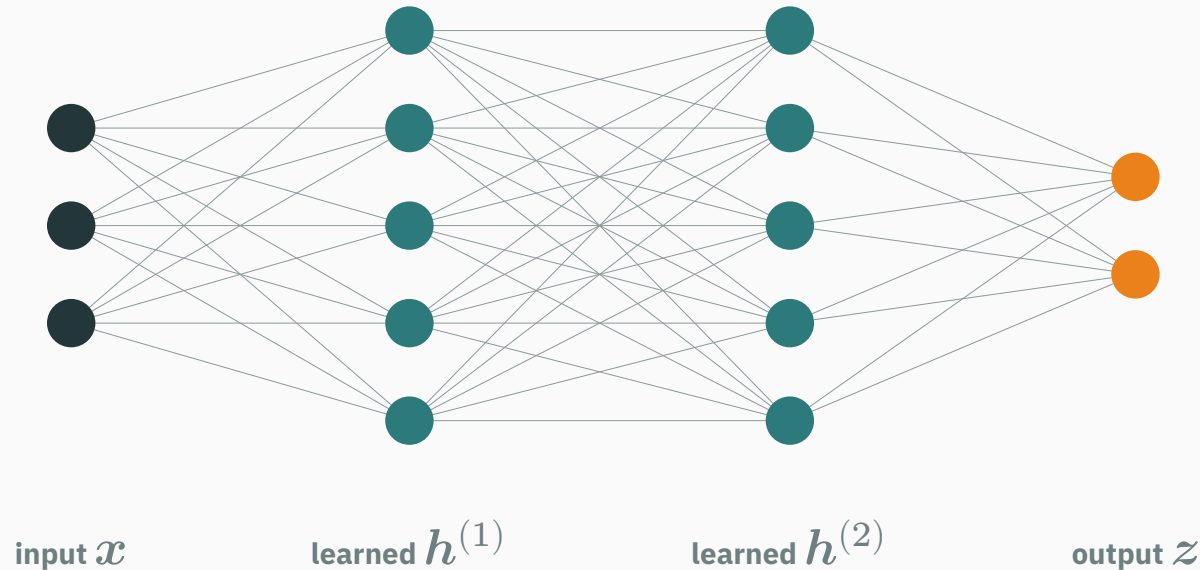
The forward pass evaluates the current network; the update changes the network for the next pass.

The decisive change is the learned representation



The final scores still feed the familiar Lecture 1 output model and loss.

The decisive change is the learned representation



The final scores still feed the familiar Lecture 1 output model and loss.

The output loss can stay the same while hidden layers make the representation nonlinear.

We now have $x \rightarrow f_{\theta}(x) \rightarrow \mathcal{L}$.

Lecture 3: how do we compute $\nabla_{\theta}\mathcal{L}$ efficiently?

computation graphs · the chain rule · backpropagation · automatic differentiation